TRIPLE QUANTUM SCHUBERT CALCULUS

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Abstract. We prove a triple quantum Pieri formula: the product of a quantum double Schubert polynomial $\widetilde{\mathfrak{S}}_u(x;y)$ by a quantum factorial elementary polynomial $\widetilde{E}_{p,k}(x;z)$ in a second set of variables z is a sum over chains in the quantum Bruhat graph, indexed by a quantum analogue of Sottile's relation, whose coefficients are monomials in the quantum parameters times polynomials in the differences $y_i - z_j$ with nonnegative integer coefficients, and which is Graham-positive at $z = y$. At $y = z = 0$ it is the quantum Pieri rule of Ciocan-Fontanine and Postnikov; at $z = y$ it is an equivariant quantum Pieri rule for the complete flag variety, indexed by chains in the quantum Bruhat graph rather than, as in the rule of Huang and Li, by the data of a reduction to classical cohomology, and proved directly. A doubled Pieri formula, positive in the same senses, treats multiplication by the quantization of a product of two elementary polynomials in the same variables, which is not the product of the quantizations: it is two iterated Pieri steps with the incompatible pairs of chains omitted. More generally, the triple quantum Schubert structure constants $c_{u,v}^{w,d}(y;z)$, the coefficients of $\widetilde{\mathfrak{S}}_u(x;y)\,\widetilde{\mathfrak{S}}_v(x;z)$ in the basis $\{\widetilde{\mathfrak{S}}_w(x;y)\}$, are polynomials in the $y_i - z_j$ with nonnegative integer coefficients; the proof of this appears as an appendix. We extend the Peterson–Woodward comparison formula and the quantum-to-classical reduction of Leung–Li and Huang–Li to the triple constants, and we describe and prove correct the algorithms of `schubmult`, which compute products of arbitrary quantum double Schubert polynomials well beyond the reach of previous methods.

Keywords: quantum double Schubert polynomial, structure constant, Littlewood-Richardson rule, Pieri formula, quantum Bruhat graph, Peterson–Woodward comparison formula

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1. Introduction

Let $\mathfrak{S}_w(x;y)$, $w \in S_\infty$, denote the double Schubert polynomials of Lascoux and Schützenberger, which represent the torus-equivariant Schubert classes of the complete flag variety, and let $\widetilde{\mathfrak{S}}_w(x;y)$ denote the quantum double Schubert polynomials of Kirillov and Maeno [20], which represent the Schubert classes in torus-equivariant quantum cohomology [21]. The structure constants of the equivariant quantum cohomology ring are the coefficients in

$$\widetilde{\mathfrak{S}}_u(x;y)\,\widetilde{\mathfrak{S}}_v(x;y) = \sum_{w,d} q^d\, c_{u,v}^{w,d}(y;y)\ \widetilde{\mathfrak{S}}_w(x;y),$$

and Mihalcea [25] proved that they are polynomials in the negative simple roots $y_{i+1} - y_i$ with nonnegative integer coefficients.

In this paper we study the coefficients obtained when the two factors are polynomials in different sets of variables, y and z:

$$\widetilde{\mathfrak{S}}_u(x;y)\,\widetilde{\mathfrak{S}}_v(x;z) = \sum_{w,d} q^d\, c_{u,v}^{w,d}(y;z)\ \widetilde{\mathfrak{S}}_w(x;y).$$

We call the $c_{u,v}^{w,d}(y;z)$ the *triple quantum Schubert structure constants*. At $z = y$ they recover the equivariant quantum structure constants, and at $q = 0$ they recover the triple Schubert structure constants $c_{u,v}^{w}(y;z)$ of [33], which were recently shown by Gao and Xiong [15] to be polynomials in the differences $y_i - z_j$ with nonnegative integer coefficients. The triple constants carry strictly more information than the equivariant ones: the equivariant quantum constant $c_{u,v}^{w,d}(y;y)$ is a specialization of $c_{u,v}^{w,d}(y;z)$, while conversely $c_{u,v}^{w,d}(y;z)$ is recovered from the constants $c_{u,v_2}^{w,d}(y;y)$, v_2 ranging over the right factors of the reduced factorizations $v = v_1 \cdot v_2$, by the quantum Cauchy identity (Lemma 2.1).

The main results of this paper are the following; the positivity theorem is due to the authors of Appendix A, the rest are ours.

(1) *A triple quantum Pieri formula* (Theorem 3.4; general degree in Theorem 8.1). Let $\widetilde{E}_{k,k}(x;z) = \widetilde{\mathfrak{S}}_{s_1 s_2 \cdots s_k}(x;z)$ be the quantum factorial elementary polynomial. We prove

$$\widetilde{\mathfrak{S}}_u(x;y)\widetilde{E}_{k,k}(x;z) = \sum_{u \overset{k}{\rightsquigarrow} w} q_k(u,w) \prod_{a \in \mathrm{fix}_k(u,w)} (y_a - z_1)\ \widetilde{\mathfrak{S}}_w(x;y),$$

where $\overset{k}{\rightsquigarrow}$ is a quantum analogue of the relation introduced by Sottile [35] for the classical Pieri rule, $q_k(u,w)$ is a monomial in the q_i recording the length-decreasing steps of a chain from u to w in the quantum Bruhat graph, and $\mathrm{fix}_k(u,w)$ is the set of values fixed by the chain in the first k positions. The general-degree version expands $\widetilde{\mathfrak{S}}_u(x;y)\widetilde{E}_{p,k}(x;z)$ for every $p \le k$ over the same chains, the coefficient of $\widetilde{\mathfrak{S}}_w(x;y)$ being $q_k(u,w)$ times a factorial elementary polynomial in the variables y_a, $a \in \mathrm{fix}_k(u,w)$, and z. In either form the coefficient of each q-monomial is a polynomial in the differences $y_i - z_j$ with nonnegative integer coefficients. This is the main result of the paper.

The word triple is essential. Mixed-variable formulas of this kind go back to the Littlewood–Richardson rule of Molev and Sagan for factorial Schur functions [27], and the classical triple Pieri formula [33, Theorem 7.1], to which the present one reduces at $q = 0$, has already shown what such a formula is good for: it is the engine of the multiplication algorithm of `schubmult` for double Schubert polynomials. The reason is that a formula with $z \neq y$ composes: an arbitrary $\widetilde{\mathfrak{S}}_v(x;z)$ expands into products of factorial elementary polynomials in the z-variables alone (Theorems 4.6 and 4.7), and the triple Pieri formula then multiplies each factor against $\widetilde{\mathfrak{S}}_u(x;y)$ without any further change of variables. An equivariant formula, with $z = y$, cannot be iterated in this way. Conversely the triple formula determines the equivariant one by specialization, while recovering it from equivariant data requires the quantum Cauchy identity and a sum over factorizations (Lemma 2.1).

The specializations locate the formula among known results. At $q = 0$ it is the triple Pieri formula of [33], and at $q = 0$, $y = z = 0$ it is Sottile's rule. At $y = z = 0$ it is the quantum Pieri formula for complete flag manifolds, due to Ciocan-Fontanine [11] and reformulated by Postnikov [28] in terms of what is now called the quantum Bruhat graph; the relation $\overset{k}{\rightsquigarrow}$ is the factorial version of Postnikov's. At $p = 1$ and $z = y$ it is the equivariant quantum Chevalley (Monk) formula of Mihalcea [26]. At $z = y$ and general p it is an equivariant quantum Pieri rule for the complete flag variety. Such a rule was first obtained by Huang and Li [17, Theorem 3.10], for all partial flag varieties of type A, by applying the quantum-to-classical reduction (a principle going back to Buch, Kresch and Tamvakis [8]) repeatedly until no quantum parameters remain and then invoking Robinson's equivariant Pieri rule [31]; its terms are indexed by the data of Ciocan-Fontanine's rule, namely a unimodal degree d, the permutations τ_d and ϕ_d it determines, and classical chains from $u\tau_d$ to $w\phi_d$, and its coefficients are restrictions of special Schubert classes of Grassmannians to torus-fixed points. Our formula differs from theirs in three respects. It has independent sets of variables y and z, which is what makes it composable. Its terms are indexed intrinsically, by chains in the quantum Bruhat graph starting at u, with the quantum degree $q_k(u,w)$ and the weight $\mathrm{fix}_k(u,w)$ read off from the chain. This is closer in spirit to Postnikov's reformulation of Ciocan-Fontanine's rule than to the reduction-based formula of Huang and Li. And its proof is a direct analysis of the quantum Bruhat graph rather than a reduction.

It is worth separating the kinds of positivity in play. Positivity of the coefficients follows from Mihalcea's theorem at $z = y$ and from Theorem 2.3 in general, without any formula; triple positivity does not by itself imply Graham positivity at $z = y$, but the Pieri coefficients are specializations of factorial elementary polynomials and are manifestly Graham-positive there (Corollary 3.5), exactly as in [33]. The quantum-to-classical reduction of Leung–Li [22] and Huang–Li [17] gives more: whenever one factor is Grassmannian, each structure constant equals an explicitly determined classical one, a statement we show holds verbatim for the triple constants (Proposition 5.4), and combining this with a positive classical rule gives a positive expression for each coefficient separately, which is how Huang and Li obtain their formula. A rule indexed by chains from u is a different kind of statement: it describes the support of the product, the degrees d and the permutations w that occur, together with all coefficients, without reference to a target (w,d), and it is this that the multiplication algorithm of Section 4 iterates. The proof occupies Section 6; it proceeds by induction on k via the single-variable Monk formula, and rests on an intrinsic description of the chains (every nontrivial cycle of $u^{-1}w$

has exactly one element exceeding k, and a single length identity) which makes the term-by-term cancellations tractable.

(2) *A doubled quantum Pieri formula* (Theorem 8.3). The quantization map of Fomin, Gelfand and Postnikov is not multiplicative on products of elementary polynomials in the same number of variables, so the quantum Schubert polynomial of a $k \times 2$ rectangle is not the product of two quantum factorial elementary polynomials. We show that the product $\widetilde{\mathfrak{S}}_u(x;y)\,\widetilde{\mathfrak{S}}_{d_2[k]}(x;z)$, where $d_2[k]$ is the dominant permutation whose diagram is the $k \times 2$ rectangle, is nevertheless given by two iterated Pieri steps, provided one omits the pairs of chains that are not *compatible* in a precise cycle-theoretic sense (Definition 8.2). The doubled rule is equivalent to a termwise identity between pairs of chains (Theorem 8.10) which telescopes along explicit blocks of pairs and is then established by an injection built from a single local lemma, a generalization of the up/down pairing used in the proof of the Pieri formula (Theorem 8.23). By quantum Giambelli the same product is an alternating sum of iterated Pieri steps; the doubled rule is the cancellation-free form of that sum, and it is the "paired step" of the multiplication algorithm.

(3) *Positivity* (Theorem 2.3). The constants $c_{u,v}^{w,d}(y;z)$ lie in $\mathbb{Z}_{\geq 0}[y_i - z_j]$. This theorem and its proof are due to the authors of Appendix A; the proof combines the geometric argument of Mihalcea with the Graham-positivity theorem of Gao and Xiong for B^--stable effective cycles. We include it because it settles what would otherwise be a conjecture about the constants studied here, of which the Pieri formulas above are explicit combinatorial instances.

(4) *Parabolic and quantum-to-classical reductions* (Proposition 2.9, Theorem 5.3). The Peterson–Woodward comparison formula [37], in the form proved by Huang and Li [17] for equivariant quantum cohomology, extends verbatim to the triple constants. Consequently the triple structure constants of the parabolic quantum double Schubert polynomials of Lam and Shimozono are equal to triple constants of the complete flag variety, and in particular they are also Graham-positive (Corollary 2.10). The quantum-to-classical reduction of Leung–Li and Huang–Li extends to the triple constants as well. For Grassmannians, considerably more is known at $z = y$: Mihalcea [24] gave an equivariant quantum Pieri rule and a recursive characterization of the equivariant quantum structure constants, and Buch's mutation algorithm for puzzles [6], combined with the equivariant "quantum equals classical" theorem of Buch and Mihalcea [9], provides a Graham-positive combinatorial Littlewood–Richardson rule for $QH_T^*(\mathrm{Gr}(k,n))$. No such rule is known to us for the triple constants with $z \neq y$, even in the Grassmannian case.

(5) *Algorithms* (Section 4). Computing products of Schubert classes in $QH^*(Fl_n)$, even non-equivariantly, is difficult in practice: the general methods available, namely reduction of quantum Giambelli determinants modulo the quantum ideal and iteration of the quantum Monk or Chevalley formula with a recursive reduction, become infeasible well before n reaches the range in which interesting phenomena appear. We describe the algorithms implemented in the package `schubmult` [34], which compute products of arbitrary (quantum) double Schubert polynomials, with independent parameters y and z, by combining the Pieri formulas on one factor with a signed expansion of the other factor into products of factorial elementary polynomials. In our experience this is by a wide margin the fastest method available, and it is the only one we know of that handles the triple constants directly. We prove the expansion, in its classical form (Theorem 4.5) and in two quantum forms, over a strict dominant shape (Theorem 4.6) and over the "medium" shape actually used by the implementation, in which a pair of equal columns is allowed when it is preceded by a drop of at least two and contributes a single doubled factor (Theorem 4.7); the key step is a formula for the skew divided differences, in the sense of [32, 33], of a product $\prod_{j \leq k}(x_j - z_i)$ with respect to the z-variables (Theorem 7.6), in which the terms of the iterated Monk rule that revisit a position cancel by a rotation argument (Lemma 7.5). The doubled Pieri rule is exactly what is needed to process the equal columns of the medium shape.

The paper is organized as follows. Section 2 defines the triple constants, states the positivity theorem and treats the parabolic case. Section 3 states the Pieri formula, whose proof is given in Section 6. Section 4 states the expansion formulas for (quantum) double Schubert polynomials and describes the multiplication algorithms built on them; the expansion formulas are proved in Section 7. Section 5 treats the quantum-to-classical reduction, Section 8 the general-degree and doubled Pieri formulas, and Section 9 proves the

parabolic reduction via the Peterson–Woodward comparison formula. Appendix A contains the proof of the positivity theorem.

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2. Triple quantum Schubert structure constants

2.1. **Quantum double Schubert polynomials and the triple constants.** We fix $n \geq 1$ and write

$$x = (x_1, \ldots, x_n), \qquad y = (y_1, \ldots, y_n), \qquad q = (q_1, \ldots, q_{n-1}).$$

For $1 \leq a < b \leq n$, set

$$q_{ab} := q_a q_{a+1} \cdots q_{b-1},$$

and use the cohomological grading $\deg q_i = 2$ (equivalently $\deg q_{ab} = 2(b-a)$).

Let s_i act on the y-variables by swapping y_i, y_{i+1}, and define the divided difference operators

$$\partial_i^{(y)} f := \frac{f - s_i f}{y_i - y_{i+1}} \qquad (1 \leq i \leq n-1).$$

For $w \in S_n$, if $w = s_{i_1} \cdots s_{i_r}$ is any reduced word, set

$$\partial_w^{(y)} := \partial_{i_1}^{(y)} \cdots \partial_{i_r}^{(y)}.$$

Following Kirillov–Maeno [20], define the quantum elementary polynomials $e_j^{(q)}(x_1, \ldots, x_k)$ by

$$\Delta_k(t \mid x_1, \ldots, x_k) := \sum_{j=0}^{k} t^{k-j} e_j^{(q)}(x_1, \ldots, x_k),$$

where

$$\Delta_k(t \mid x_1, \ldots, x_k) = \det \begin{pmatrix} x_1 + t & q_1 & 0 & \cdots & 0 \\ -1 & x_2 + t & q_2 & \ddots & \vdots \\ 0 & -1 & x_3 + t & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & q_{k-1} \\ 0 & \cdots & 0 & -1 & x_k + t \end{pmatrix}.$$

Let $w_0 \in S_n$ be the longest permutation. The top quantum double Schubert polynomial is

$$\widetilde{\mathfrak{S}}_{w_0}(x; y) := \prod_{i=1}^{n-1} \Delta_i(-y_{n-i} \mid x_1, \ldots, x_i),$$

and for $w \in S_n$ we define

$$\widetilde{\mathfrak{S}}_w(x; y) := (-1)^{\ell(w_0) - \ell(w)} \partial_{w w_0}^{(y)} \widetilde{\mathfrak{S}}_{w_0}(x; y).$$

Its specialization

$$\widetilde{\mathfrak{S}}_w(x) := \widetilde{\mathfrak{S}}_w(x; 0)$$

is the quantum Schubert polynomial. In particular, at $q = 0$ these recover the ordinary double and single Schubert polynomials.

The stable setting. The polynomials $\widetilde{\mathfrak{S}}_w(x; y)$ are stable [20]: for $w \in S_n$, $\widetilde{\mathfrak{S}}_w(x; y)$ involves only $x_1, \ldots, x_{n-1}$, $y_1, \ldots, y_{n-1}$ and $q_1, \ldots, q_{n-2}$, and it does not change when w is regarded as an element of S_{n+1} under the standard inclusion $S_n \subset S_{n+1}$. (This is also immediate from the description $\widetilde{\mathfrak{S}}_w(x; y) = \mathcal{Q}(\mathfrak{S}_w(x; y))$ in the proof of Lemma 2.1 below, since the quantization map $\mathcal{Q}$ does not depend on n.) From now on we therefore let the number of variables be infinite,

$$x = (x_1, x_2, \ldots), \qquad y = (y_1, y_2, \ldots), \qquad z = (z_1, z_2, \ldots), \qquad q = (q_1, q_2, \ldots),$$

index the polynomials by $S_\infty = \bigcup_n S_n$, and work in the polynomial ring $\mathbb{Z}[q][x, y, z]$ in these (countably many) variables, where $\mathbb{Z}[q] = \mathbb{Z}[q_1, q_2, \ldots]$. A *degree* is an element $d = (d_1, d_2, \ldots)$ of $\bigoplus_{i \geq 1} \mathbb{Z}_{\geq 0}$, i.e. a weak composition with finitely many nonzero parts, and $q^d = \prod_i q_i^{d_i}$; in Section 2.3 degrees will be regarded as effective elements of the coroot lattice. As a module over $\mathbb{Z}[q][y]$, the quantum double Schubert polynomials $\widetilde{\mathfrak{S}}_w(x; y)$, $w \in S_\infty$, form a basis of $\mathbb{Z}[q][x, y]$: the quantization map $\mathcal{Q}$ of Lemma 2.1 is a $\mathbb{Z}[q][y]$-linear bijection of $\mathbb{Z}[q][x, y]$ onto itself, since $\mathcal{Q} - \mathrm{id}$ lowers the total x-degree by at least two and is therefore locally

nilpotent, so that $\sum_{r\geq 0}(-1)^r(\mathcal{Q}-\mathrm{id})^r$ is a well-defined inverse on every polynomial. It carries the basis $\{\mathfrak{S}_w(x;y)\}$ to $\{\widetilde{\mathfrak{S}}_w(x;y)\}$. We may therefore write

$$\widetilde{\mathfrak{S}}_u(x;y)\,\widetilde{\mathfrak{S}}_v(x;z)=\sum_{w,d} q^d c_{u,v}^{w,d}(y;z)\,\widetilde{\mathfrak{S}}_w(x;y),$$

the sum being over $w\in S_\infty$ and all degrees d, with only finitely many nonzero terms. The coefficients $c_{u,v}^{w,d}(y;z)$ are the *triple quantum Schubert structure constants*, which are polynomials in y and z with integer coefficients.

The finite-dimensional meaning of this expansion is the following. Given $u,v\in S_n$, choose N so large that every w occurring in the expansion lies in S_N and every occurring d has $d_i=0$ for $i\geq N$; this holds for all sufficiently large N, since there are finitely many terms. Then the expansion is an identity in $\mathbb{Z}[q_1,\dots,q_{N-1}][x_1,\dots,x_N,y,z]$, and reducing it modulo the ideal of Kim's presentation of $QH_T^*(\mathrm{Fl}_N)$ [19] (the ideal J_{qy}^N recalled in Appendix A) identifies it, by [21], with the expansion of the product of the two corresponding classes in the Schubert basis of $QH_T^*(\mathrm{Fl}_N)$, with $z=y$. Thus $c_{u,v}^{w,d}(y;y)$ is the equivariant quantum Littlewood–Richardson coefficient of Fl_N for every such N, and the degree vector d is the degree $\sum_i d_i\alpha_i^\vee$ of a rational curve in Fl_N. Note that $w\notin S_n$ and $d_i\neq 0$ for $i\geq n-1$ do occur: for instance $\widetilde{\mathfrak{S}}_{21}(x;y)^2=(y_2-y_1)\,\widetilde{\mathfrak{S}}_{21}(x;y)+\widetilde{\mathfrak{S}}_{231}(x;y)+q_1$, the last term coming from the quantum Monk formula in Fl_N for any $N\geq 3$.

The product respects the grading where for each i, $\deg x_i=\deg y_i=\deg z_i=1$ and $\deg q_i=2$, in which $\widetilde{\mathfrak{S}}_w$ is homogeneous of degree $\ell(w)$. In particular, if $c_{u,v}^{w,d}(y;z)\neq 0$, then

$$\deg c_{u,v}^{w,d}(y;z)=\ell(u)+\ell(v)-\ell(w)-2\sum_i d_i;$$

for instance the term q_1 in $\widetilde{\mathfrak{S}}_{21}^2$ above has a coefficient of degree $0=1+1-0-2$. Unlike the usual triple Schubert structure constants, they do not exhibit triangularity in Bruhat order, and indeed we may have $\ell(w)<\ell(u)+\ell(v)$. There is an ordering in which the quantum double Schubert polynomials are triangular, but we will not need this in full generality.

Throughout, a factorization $w=v_1\cdot v_2$ written with a dot is understood to be *reduced*, i.e. $\ell(w)=\ell(v_1)+\ell(v_2)$. The following identity will be used repeatedly to pass between the triple constants and the equivariant quantum structure constants $c_{u,v}^{w,d}(y;y)$. Setting $y=0$ in it recovers the expansion $\widetilde{\mathfrak{S}}_w(x;z)=\sum_{v_1\cdot v_2=w}\mathfrak{S}_{v_1}(0;z)\,\widetilde{\mathfrak{S}}_{v_2}(x)$ of Kirillov–Maeno [20] in terms of quantum Schubert polynomials.

Lemma 2.1 (Quantum Cauchy identity)**.** *For all $w\in S_\infty$,*

$$\widetilde{\mathfrak{S}}_w(x;z)=\sum_{v_1\cdot v_2=w}\mathfrak{S}_{v_1}(y;z)\;\widetilde{\mathfrak{S}}_{v_2}(x;y).$$

Consequently, for all u,v,w and d,

$$c_{u,v}^{w,d}(y;z)=\sum_{v_1\cdot v_2=v} c_{u,v_2}^{w,d}(y;y)\;\mathfrak{S}_{v_1}(y;z).$$

Proof. Let $\mathcal{Q}$ denote the quantization map of Fomin, Gelfand and Postnikov [13], used throughout the paper: the $\mathbb{Z}[q]$-linear map on polynomials in x which sends each standard elementary monomial $\prod_j e_{i_j}(x_1,\dots,x_j)$ to $\prod_j e_{i_j}^{(q)}(x_1,\dots,x_j)$, extended linearly over any further coefficient variables. Since the divided differences in the y-variables commute with $\mathcal{Q}$, the definition of $\widetilde{\mathfrak{S}}_w(x;y)$ by y-divided differences gives $\widetilde{\mathfrak{S}}_w(x;y)=\mathcal{Q}(\mathfrak{S}_w(x;y))$ [21, (10)]. Now apply $\mathcal{Q}$, which is $\mathbb{Z}[y,z]$-linear, to the classical Cauchy-type expansion $\mathfrak{S}_w(x;z)=\sum_{v_1\cdot v_2=w}\mathfrak{S}_{v_1}(y;z)\,\mathfrak{S}_{v_2}(x;y)$ [23, (6.3)]: the left side becomes $\widetilde{\mathfrak{S}}_w(x;z)$ and the right side becomes $\sum_{v_1\cdot v_2=w}\mathfrak{S}_{v_1}(y;z)\,\widetilde{\mathfrak{S}}_{v_2}(x;y)$. For the second assertion, multiply the first identity (with $w=v$) by $\widetilde{\mathfrak{S}}_u(x;y)$ and expand each $\widetilde{\mathfrak{S}}_u(x;y)\,\widetilde{\mathfrak{S}}_{v_2}(x;y)$ in the basis $\{\widetilde{\mathfrak{S}}_w(x;y)\}$ with coefficients $q^d c_{u,v_2}^{w,d}(y;y)$. □

Remark 2.2 ($\overset{k}{\rightsquigarrow}$ and the quantum k-Bruhat order). The *quantum k-Bruhat order* was introduced by Benedetti, Bergeron, Colmenarejo, Saliola, and Sottile [1] and studied further by Colmenarejo and Mayers [12]. That order $\preceq_k^{\mathbf{q}}$ on $S_\infty[\mathbf{q}]$ is generated by cover relations

$$u' \;\lessdot_k^{\mathbf{q}}\; \mathbf{q}^\gamma u' t_{ab}, \qquad a\leq k<b,$$

each of exactly two kinds: a *classical* cover, with $\gamma = 0$ and $\ell(u't_{ab}) = \ell(u') + 1$, or a *quantum* cover, with $\ell(u') + 1 = \ell(\mathbf{q}^\gamma u't_{ab}) = \ell(u't_{ab}) + 2(b-a)$, that is $\ell(u't_{ab}) = \ell(u') - 2(b-a) + 1$ and $\gamma = e_a + \cdots + e_{b-1}$. These are exactly the two alternatives of condition (4) of Definition 3.1. Consequently every sequence $t_{a_1b_1}, \ldots, t_{a_pb_p}$ witnessing $u \overset{k}{\rightsquigarrow} w$ is a saturated chain from u to $\mathbf{q}^\alpha w$ in the quantum k-Bruhat order, where $\mathbf{q}^\alpha$ is the monomial recorded by $q_k(u, w)$; in particular $u \overset{k}{\rightsquigarrow} w$ implies $u \preceq_k^{\mathbf{q}} \mathbf{q}^\alpha w$.

The relation $\overset{k}{\rightsquigarrow}$ is *not* the full order $\preceq_k^{\mathbf{q}}$, however. Definition 3.1 imposes the additional requirements that the lower indices $a_1, \ldots, a_p$ be pairwise distinct, so that $\overset{k}{\rightsquigarrow}$ admits only those saturated chains of a special form. Thus $\overset{k}{\rightsquigarrow}$ is strictly larger than the covering relation $\lessdot_k^{\mathbf{q}}$ but strictly smaller than $\preceq_k^{\mathbf{q}}$: a general chain in the quantum k-Bruhat order may reuse a lower index a_i, and such chains do not contribute to the Pieri expansion. This is the quantum counterpart of the classical situation, where the Pieri relation of [35, 33] is likewise the restriction of the k-Bruhat order to chains with distinct lower indices, and it is exactly this restriction that makes the Pieri coefficients well defined.

What we use below is only that each individual step of a $\overset{k}{\rightsquigarrow}$ chain is a cover, and hence is governed by the following length-free description of the covers [1, Proposition 3.1] (see also [12, Proposition 4.2]): with $i \le k < j$, the step $u' \lessdot_k^{\mathbf{q}} \mathbf{q}^\gamma u' t_{ij}$ is a classical cover exactly when $u'(i) < u'(j)$ and no l with $i < l < j$ satisfies $u'(i) < u'(l) < u'(j)$, and it is a quantum cover exactly when $u'(i) > u'(j)$ and every l with $i < l < j$ satisfies $u'(j) < u'(l) < u'(i)$. This characterization is an immediate consequence of the standard formula for the length of $u't_{ij}$ (Lemma 6.2 below), and it is the form in which the covers are used throughout Section 6.

2.2. **Positivity.** For $d = (0, \ldots, 0)$, this is the ordinary triple Schubert calculus structure constant, and it was proved in [15] that these have the positivity property that they are polynomials in the differences $y_i - z_j$ with nonnegative integer coefficients. We will refer to this as *triple Schubert positivity*. The following theorem shows that it persists in the quantum setting.

Theorem 2.3 (Triple quantum Schubert positivity)**.** *For all u, v, w and d, the structure constants $c_{u,v}^{w,d}(y; z)$ are polynomials in the differences $y_i - z_j$ with nonnegative integer coefficients.*

The proof, due to Tongren Xiao and Yihua Yue, is given in Appendix A.

2.3. **The parabolic case.** Unlike ordinary cohomology or equivariant cohomology, quantum cohomology and equivariant quantum cohomology do not demonstrate simple functoriality, in the sense that a projection of the complete flag variety G/B to a partial flag variety G/P does not induce a homomorphism of rings $QH_T^*(G/P) \to QH_T^*(G/B)$, and indeed the quantum cohomology of G/P is not isomorphic to the subring of the quantum cohomology of G/B generated by the Schubert classes indexed by W^P. This is reflected in the fact that there are parabolic quantum double Schubert polynomials defined in a different manner that represent Schubert classes in $QH_T^*(G/P)$ [21] and are not in general equal to the quantum double Schubert polynomials above. (The non-equivariant ring $QH^*(SL_n/P)$ itself is well understood: its presentation and its quantum Pieri and Giambelli formulas were proved by Ciocan-Fontanine [11] and, by elementary geometric arguments, by Buch [5].)

Definition 2.4 (Parabolic quantum double Schubert polynomials)**.** Following Lam and Shimozono [21], we define parabolic quantum double Schubert polynomials as follows. Fix an integer $n > 0$. A parabolic subgroup W_P of S_n, which has Coxeter generators $s_1, \ldots, s_{n-1}$, is generated by a subset $P \subseteq \{s_1, \ldots, s_{n-1}\}$. In terms of the subgroup's properties, it is most natural to identify P by the reflections that it *excludes*. Suppose P is generated by $S \setminus \{s_{i_1}, \ldots, s_{i_k}\}$ with $1 \le i_1 < \cdots < i_k \le n-1$, and set $i_0 = 0$ and $i_{k+1} = n$. We define the *block sizes* $n_1, \ldots, n_{k+1} > 0$, with

$$n_1 + \cdots + n_{k+1} = n,$$

by $n_p = i_p - i_{p-1}$ for $1 \le p \le k+1$, and put

$$N_j = n_1 + \cdots + n_j = i_j \qquad (0 \le j \le k+1),$$

so that $N_0 = 0$ and $N_{k+1} = n$. Thus $W_P = S_{n_1} \times \cdots \times S_{n_{k+1}}$ and SL_n/P is the variety of partial flags with subspaces of dimensions $N_1 < \cdots < N_k$. The extreme cases are $k = 0$, where $W_P = S_n$, $W^P = \{1\}$, SL_n/P is a point and the only polynomial defined below is $\widetilde{\mathfrak{S}}_1^P = 1$, and $P = \emptyset$ ($k = n-1$, all $n_p = 1$), which recovers the quantum double Schubert polynomials of the complete flag variety.

For such a setup, let D^P be the $n \times n$ matrix with $x_1, \ldots, x_n$ on the diagonal, -1 on the superdiagonal, the entry $(-1)^{n_{j+1}+1} q_j$ in position $(N_{j+1}, N_{j-1}+1)$ for $1 \leq j \leq k$, and all other entries 0, and let D^P_j be its upper left $N_j \times N_j$ submatrix. Notice that there are precisely k q variables, q_j corresponding to the excluded reflection s_{i_j}, and we set

$$\deg q_j = n_j + n_{j+1}.$$

Define

$$\Delta^P_j(t; x) = \det(D^P_j + t\,\mathrm{Id}) \qquad (1 \leq j \leq k),$$

so that for $P = \emptyset$ this is the polynomial $\Delta_j(t \mid x_1, \ldots, x_j)$ above (the matrices differ by a transposition). Then we define the parabolic quantum double Schubert polynomials $\widetilde{\mathfrak{S}}^P_w(x; y)$ for $w \in W^P$ by

$$\widetilde{\mathfrak{S}}^P_w(x; y) = (-1)^{\ell(w(w_0^P)^{-1})} \partial^{(y)}_{w(w_0^P)^{-1}} \prod_{j=1}^{k} \prod_{i=n-N_{j+1}+1}^{n-N_j} \Delta^P_j(-y_i; x),$$

where w_0^P is the longest element of W^P. For $P = \emptyset$ the product is $\prod_{j=1}^{n-1} \Delta_j(-y_{n-j} \mid x_1, \ldots, x_j) = \widetilde{\mathfrak{S}}_{w_0}(x; y)$.

Taken as defined, when we vary n these polynomials are not stable under inclusion and do not form a basis of $\mathbb{Z}[q][x_1, \ldots, x_n; y]$ as a module over $\mathbb{Z}[q][y]$. To obtain stability and a basis, instead we fix n and choose $N > n$ as large as we would like, maintaining $N_{k+1} = n$ and setting $n_{k+2} = N - n$. The added quantum parameter q_{k+1} has degree $n_{k+1} + N - n$, which tends to infinity; hence it cannot occur in any polynomial or product of fixed degree once N is sufficiently large. The remaining parameters are the original $q_1, \ldots, q_k$. From a cohomological perspective, this introduces "extraneous" polynomials that are indexed by elements not in W^P. The full indexing set is $W^{\widehat{P}} \subset S_\infty$, where

$$\widehat{P} = P \cup \{s_{n+1}, s_{n+2}, \ldots\}$$

The notation $W^{\widehat{P}}$ specifies the stable indexing set only. Any use of longest elements below is made in a finite symmetric group, with the enlarged parabolic generated by $\widehat{P} \cap \{s_1, \ldots, s_{N-1}\}$.

Example 2.5. As equivariant cohomology rings of finite dimensional projective varieties over $\mathbb{C}$ are finitely generated modules over the coefficient ring, it is fully expected that products of double Schubert polynomials may include terms indexed by permutations that do not lie within the indexing set of the basis of Schubert classes. For example, consider the following non-quantum double Schubert polynomial product, with $P = \{s_1, s_3\}$ and basis elements in S_4:

$$\begin{aligned}
\mathfrak{S}_{2413}(x; y)\, \mathfrak{S}_{3412}(x; y) = &-(\alpha_1 + \alpha_2)(\alpha_1 + \alpha_2 + \alpha_3)(\alpha_2 + \alpha_3)\ \mathfrak{S}_{3412}(x; y) \\
&+ (\alpha_1 + \alpha_2)(\alpha_1 + 2\alpha_2 + 2\alpha_3 + \alpha_4)\ \mathfrak{S}_{35124}(x; y) \\
&- (\alpha_1 + \alpha_2)\ \mathfrak{S}_{361245}(x; y) \\
&- (\alpha_1 + 2\alpha_2 + 2\alpha_3 + \alpha_4)\ \mathfrak{S}_{45123}(x; y) \\
&+ \mathfrak{S}_{461235}(x; y).
\end{aligned}$$

Not only does the product $\widetilde{\mathfrak{S}}^P_{2413}(x; y)\, \widetilde{\mathfrak{S}}^P_{3412}(x; y)$ include extra terms indexed by permutations in W^P, there are also terms with nonzero coefficients that are in $W^{\widehat{P}} \setminus W^P$:

$$\begin{aligned}
q_1(\alpha_1\alpha_2 + \alpha_1\alpha_3 + \alpha_2\alpha_3 + \alpha_2^2)\ \widetilde{\mathfrak{S}}^P_{132}(x; y) &+ q_1(\alpha_1 + \alpha_2)\ \widetilde{\mathfrak{S}}^P_{231}(x; y) \\
&+ q_1(\alpha_2 + \alpha_3)\ \widetilde{\mathfrak{S}}^P_{1423}(x; y) \\
&+ q_1(\alpha_1 + \alpha_2)\ \widetilde{\mathfrak{S}}^{\widehat{P}}_{13254}(x; y) \\
&+ q_1\ \widetilde{\mathfrak{S}}^{\widehat{P}}_{14253}(x; y) \\
&+ q_1\ \widetilde{\mathfrak{S}}^P_{15234}(x; y) \\
&+ q_1\ \widetilde{\mathfrak{S}}^P_{2413}(x; y)
\end{aligned}$$

Definition 2.6 (Coroot lattices, $\langle d\rangle_i$, Peterson–Woodward lifts)**.** Let $\alpha_1, \alpha_2, \ldots$ be the simple roots of S_∞ and $\alpha_i^\vee$ the simple coroots, and let

$$Q^\vee = \bigoplus_{i\geq 1} \mathbb{Z}\alpha_i^\vee$$

be the coroot lattice. We write its elements as integer vectors $d = (d_1, d_2, \ldots)$ with finitely many nonzero entries, $d = \sum_i d_i\alpha_i^\vee$; the degrees of Section 2 are the *effective* elements, those with all $d_i \geq 0$, and $q^d = \prod_i q_i^{d_i}$. (When working in S_n we take $i \leq n-1$ and $d \in \mathbb{Z}^{n-1}$.) For $d \in Q^\vee$ and $i \geq 1$ put

$$\langle d\rangle_i = \langle \alpha_i, d\rangle = 2d_i - d_{i-1} - d_{i+1},$$

with the convention $d_0 = 0$ (and $d_i = 0$ for $i \geq n$ when working in S_n), and for $i < j$

$$\langle d\rangle_{ij} = \sum_{l=i}^{j-1} \langle d\rangle_l = \langle \alpha_i + \cdots + \alpha_{j-1}, d\rangle = \langle e_i - e_j, d\rangle.$$

We write $\delta_i \in Q^\vee$ for the simple coroot $\alpha_i^\vee$, the vector with a single 1 in position i, and $|d| = \sum_i d_i$.

For a parabolic subgroup W_P generated by P, let

$$Q_P^\vee = \bigoplus_{s_i \in P} \mathbb{Z}\alpha_i^\vee = \{d \in Q^\vee \mid d_i = 0 \text{ whenever } s_i \notin P\}$$

be the coroot lattice of the Levi subgroup of P. The degrees of $QH_T^*(G/P)$ are indexed by the quotient $Q^\vee/Q_P^\vee$, which we identify with $\bigoplus_{s_i\notin P} \mathbb{Z}\alpha_i^\vee$ by sending the class of d to $d_P := (d_i)_{s_i\notin P}$, the vector of coordinates of d at the excluded reflections; d_P is called effective if these coordinates are nonnegative. In the notation of Definition 2.4, the coordinate of d_P at the excluded reflection s_{i_j} is the exponent of q_j, so that $q^{d_P} = \prod_{j=1}^k q_j^{d_{i_j}}$. (For $P = \emptyset$, $Q_P^\vee = 0$ and $d_P = d$.)

Let $d_P \in Q^\vee/Q_P^\vee$. A *Peterson–Woodward lift* of d_P is a $d \in Q^\vee$ such that $d + Q_P^\vee = d_P$ and $\langle d\rangle_{ij} \in \{0, -1\}$ for all $i < j$ such that $s_i, \ldots, s_{j-1} \in P$. In the language of roots, the second condition says that $\langle \alpha, d\rangle \in \{0, -1\}$ for every positive root α of the Levi subgroup of P, these being the roots $\alpha_i + \cdots + \alpha_{j-1}$ with $s_i, \ldots, s_{j-1} \in P$. Every d_P has exactly one Peterson–Woodward lift [37]; we denote it by d or λ_B when $d_P = \lambda_P$.

Method 2.7 (Checking the Peterson-Woodward lift property)**.** While backing out a Peterson-Woodward lift is difficult, going in the other direction is not. Suppose $d \in Q^\vee$ and P is parabolic. If there is any i with $s_i \in P$ such that $\langle d\rangle_i \notin \{0, -1\}$, then d is not a Peterson-Woodward lift. If this condition does hold, let $I_1, \ldots, I_m$ be the maximal intervals of consecutive indices i with $s_i \in P$. Since $\langle d\rangle_{ij} = \sum_{l=i}^{j-1} \langle d\rangle_l$, the condition $\langle d\rangle_{ij} \in \{0, -1\}$ for all $i < j$ with $s_i, \ldots, s_{j-1} \in P$ holds if and only if each interval I_r contains at most one index l with $\langle d\rangle_l = -1$. If and only if this is the case, d is the Peterson-Woodward lift of a unique $d_P \in Q^\vee/Q_P^\vee$, namely its image $d + Q_P^\vee$, which we denote by $\pi_P(d)$. In that case we let

$$z_P(d) = \{\, i \mid s_i \in P,\ \langle d\rangle_i = 0 \,\}.$$

When $P \subseteq \{s_1, \ldots, s_{N-1}\}$ is finite, write $w_{z_P(d)}$ for the longest element of the parabolic subgroup generated by $\{s_i \mid i \in z_P(d)\}$, and w_P for the longest element of $W_P \subseteq S_N$.

Example 2.8. Let $n = 3$ and $P = \{s_1\}$, so that $W_P = S_2 \times S_1$, $SL_3/P = \mathbb{P}^2$, $Q_P^\vee = \mathbb{Z}\alpha_1^\vee = \{(d_1, 0)\}$, and $Q^\vee/Q_P^\vee \cong \mathbb{Z}$ via $d \mapsto d_2$. The lifts of $d_P = 1$ are the vectors $(d_1, 1)$, with $\langle d\rangle_1 = 2d_1 - 1$; the only one with $\langle d\rangle_1 \in \{0, -1\}$ is $d = (0, 1)$, with $\langle d\rangle_1 = -1$ and $z_P(d) = \emptyset$, so that $w_P w_{z_P(d)} = s_1$. For $d_P = 2$ the lifts are $(d_1, 2)$ with $\langle d\rangle_1 = 2d_1 - 2$, and the Peterson–Woodward lift is $d = (1, 2)$, with $\langle d\rangle_1 = 0$, $z_P(d) = \{1\}$ and $w_P w_{z_P(d)} = 1$.

This algorithm is polynomial in the length of d, which in almost all currently feasible cases is quite small. In contrast, a naive search for d with d_P as the input has an exponential search space, though a more refined algorithm may exist.

The parabolic triple quantum Schubert structure constants $c_{u,v}^{w,d_P}(y; z)$ are defined by

$$\widetilde{\mathfrak{S}}_u^P(x; y)\, \widetilde{\mathfrak{S}}_v^P(x; z) = \sum_{w\in W^{\hat{P}},\, d_P} q^{d_P}\, c_{u,v}^{w,d_P}(y; z)\ \widetilde{\mathfrak{S}}_w^P(x; y).$$

We use the same notation for the finite-rank structure constants of $QH_T^*(SL_N/P)$ when the ambient rank is specified. We prove the following finite-rank result in Section 9, as a consequence of the Peterson–Woodward comparison formula of Huang and Li [17] and Lemma 2.1, and then apply it to the stabilized polynomial coefficients.

Proposition 2.9. *Fix N and a parabolic subgroup $W_P \subseteq S_N$ generated by $P \subseteq \{s_1, \ldots, s_{N-1}\}$, and let $u, v, w \in W^P$. For the structure constants of $QH_T^*(SL_N/P)$ and $QH_T^*(\mathrm{Fl}_N)$, and any $d_P \in Q^\vee/Q_P^\vee$ with Peterson–Woodward lift d, we have*

$$c_{u,v}^{w,d_P}(y;z) = c_{u,v}^{ww_P w_{z_P(d)},d}(y;z).$$

For the stabilized polynomial coefficients, start with the block sizes of Definition 2.4 and apply Proposition 2.9 after adjoining a final block of size $N-n$, with N sufficiently large to contain all permutations in the product expansion. At fixed total degree the added parameter has degree too large to occur, so its degree coordinate is zero. The Peterson–Woodward lift for the original blocks, extended by zero in positions $i \geq n$, is also the lift for the enlarged parabolic: its root pairings on the added block are zero. Thus the longest element of that block occurs in both w_P and $w_{z_P(d)}$ and cancels in their product. The twisted output permutation and the lift are consequently independent of further enlargement, and the finite-rank comparison gives the comparison for the stabilized coefficients. No longest element of $W_{\widehat{P}}$ is required.

Thus, though we lack a simple functoriality property, we can reduce the computation of parabolic triple quantum Schubert structure constants completely to the complete flag variety case with an efficient algorithm. We obtain a nontrivial corollary from this.

Corollary 2.10 (Triple quantum Schubert positivity for the parabolic case)**.** *Let $W_P \subseteq S_N$ be a parabolic subgroup and let $u, v, w \in W^P$. Then for every d_P the parabolic structure constant $c_{u,v}^{w,d_P}(y;z)$ exhibits triple Schubert positivity. The same holds for the stabilized polynomial coefficients. In particular this holds when u, v, w are k-Grassmannian and P is the corresponding maximal parabolic subgroup ($n_1 = k$, $n_2 = N-k$).*

Proof. By Proposition 2.9, $c_{u,v}^{w,d_P}(y;z)$ equals a structure constant $c_{u,v}^{w',d}(y;z)$ of the complete flag variety, which lies in $\mathbb{Z}_{\geq 0}[y_i - z_j]$ by Theorem 2.3. The stabilized assertion follows by the finite-rank interpretation just explained. □

Remark 2.11 (The Grassmannian case). When P is a maximal parabolic subgroup, $G/P = \mathrm{Gr}(k,n)$, and at $z = y$ the parabolic constants $c_{u,v}^{w,d_P}(y;y)$ are the structure constants of $QH_T^*(\mathrm{Gr}(k,n))$, whose Graham positivity is Mihalcea's theorem [25]. In this case a positive *combinatorial* rule is available. The equivariant "quantum equals classical" theorem of Buch and Mihalcea [9] (the equivariant form of [8]) identifies each $c_{u,v}^{w,d_P}(y;y)$ with an equivariant structure constant of a two-step flag variety $\mathrm{Fl}(k-d, k+d; n)$, and Buch's mutation algorithm for puzzles [6] expresses the latter as a Graham-positive sum over equivariant puzzles; together they give a Littlewood–Richardson rule for $QH_T^*(\mathrm{Gr}(k,n))$ (see also the earlier equivariant quantum Pieri rule and recursion of Mihalcea [24], and the non-equivariant puzzle rule for two-step flag varieties of [7]). Proposition 2.9 describes the same numbers as structure constants of the complete flag variety, with the degree d_P replaced by its Peterson–Woodward lift and w by $ww_Pw_{z_P(d)}$, which is how the package [34] computes them. For $z \neq y$ the triple Grassmannian constants are determined by the equivariant ones through the parabolic quantum Cauchy identity (proof of Corollary 9.2), $c_{u,v}^{w,d_P}(y;z) = \sum_{v_1 \cdot v_2 = v} c_{u,v_2}^{w,d_P}(y;y)\, \mathfrak{S}_{v_1}(y;z)$, but this expression is not manifestly positive in the differences $y_i - z_j$, and we do not know a positive combinatorial rule for them.

3. The quantum Pieri formula

Definition 3.1 ($\overset{k}{\rightsquigarrow}$, $\mathrm{fix}_k(u,w)$, $q_k(u,w)$, $E_{p,k}(y;z)$, $\mathbf{E}(y_A;z)$)**.** We define a relation $\overset{k}{\rightsquigarrow}$ on pairs of permutations, introduced in its non-quantum form by Sottile [35], which was used originally to indicate when the coefficient of a Schubert polynomial is nonzero in a suitable application of the Pieri formula. In the generalization to double Schubert polynomials in [33], the same relation applies to all nonzero coefficients in the expansion of the generalized Pieri formula. The relation plays the same role in our case.

Given $u, w \in S_\infty$ and a positive integer k, we declare that $u \overset{k}{\rightsquigarrow} w$ if there exists a p with $0 \leq p \leq k$ such that there are transpositions $t_{a_1 b_1}, \ldots, t_{a_p b_p}$ satisfying

(1) $a_i \leq k < b_i$ for all i

(2) $a_i \neq a_j$ whenever $i \neq j$
(3) $b_i \geq b_j$ whenever $i < j$
(4) We have either

$$\ell(ut_{a_1b_1}\cdots t_{a_ib_i}) = \ell(ut_{a_1b_1}\cdots t_{a_{i-1}b_{i-1}}) + 1$$

or

$$\ell(ut_{a_1b_1}\cdots t_{a_ib_i}) = \ell(ut_{a_1b_1}\cdots t_{a_{i-1}b_{i-1}}) - 2(b_i - a_i) + 1$$

for all i
(5) $w = ut_{a_1b_1}\cdots t_{a_pb_p}$

In contrast to the non-quantum case, $u \overset{k}{\rightsquigarrow} w$ does not imply that $\ell(u) \leq \ell(w)$, as the final condition can cause the length of the permutation to drop. For intuition, the condition $\ell(ut_{ab}) = \ell(u) - 2(b-a) + 1$ means that if we write the reflection t_{ab} as

$$s_{b-1}\cdots s_{a+1}s_as_{a+1}\cdots s_{b-1}$$

then each simple reflection in this word cancels a corresponding simple reflection in a reduced word for u, as opposite to the length-increase case where there is a word for which precisely one simple reflection is inserted, regardless of the length of the reflection as a permutation.

If $u \overset{k}{\rightsquigarrow} w$, define

$$\mathrm{fix}_k(u,w) = \{u(i) \mid i \leq k \text{ and } u(i) = w(i)\}$$

and

$$n_k(u,w) = k - |\mathrm{fix}_k(u,w)|.$$

Also, we define $q_k(u,w)$ recursively. If $u = w$, define $q_k(u,w) = 1$. If we have defined $q_k(u,w')$ with $w' = ut_{a_1b_1}\cdots t_{a_{p-1}b_{p-1}}$, suppose $\ell(w't_{a_pb_p}) = \ell(w') + 1$. Then define

$$q_k(u,w) = q_k(u,w')$$

If instead $\ell(w't_{a_pb_p}) = \ell(w') - 2(b_p - a_p) + 1$, define

$$q_k(u,w) = q_k(u,w')q_{a_pb_p}$$

Suppose y and z are sets of variables and $0 \leq p \leq k$ are integers. We define $E_{p,k}(y;z)$ to be the factorial elementary symmetric polynomial in y and z. For specificity, this is the double Schubert polynomial $\mathfrak{S}_{c[p,k]}(y;z)$ where

$$c[p,k] = s_{k-p+1}\cdots s_{k-1}s_k$$

If $p = 0$, the polynomial is identically 1, and we set $E_{p,k} = 0$ for $p < 0$. If $p = k$, $E_{k,k}(y;z)$ can be simply written as

$$E_{k,k}(y;z) = \prod_{i=1}^{k}(y_i - z_1)$$

An explicit formula can be written for general p without too much trouble as well, using e.g. [27]:

$$E_{p,k}(y;z) = \sum_{1\leq i_1<\cdots<i_p\leq k}\prod_{j=1}^{p}(y_{i_j} - z_{i_j-j+1})$$

For a set $A = \{i_1,\ldots,i_{|A|}\}$, we define

$$\mathbf{E}(y_A;z) = \prod_{i\in A}(y_i - z_1) = E_{|A|,|A|}(y_{i_1},\ldots,y_{i_{|A|}};z),$$

the specialization of the factorial elementary polynomial to which the coefficients of the Pieri formula reduce. It will be used to express those coefficients.

Definition 3.2 (Quantum factorial elementary polynomials $\widetilde{E}_{p,k}(x;z)$)**.** For $0 \leq p \leq k$ we define

$$\widetilde{E}_{p,k}(x;z) = \widetilde{\mathfrak{S}}_{c[p,k]}(x;z),$$

the quantum double Schubert polynomial of the Grassmannian permutation $c[p,k] = s_{k-p+1}\cdots s_k$, whose diagram is a single column of height p with descent at k. At $q = 0$ this is $E_{p,k}(x;z)$. We also set $\widetilde{E}_{0,k} = 1$ and $\widetilde{E}_{p,k} = 0$ for $p < 0$.

Lemma 3.3. *For $k \geq 1$,*

$$\widetilde{E}_{k,k}(x;z) = \Delta_k(-z_1 \mid x_1, \ldots, x_k) = \sum_{j=0}^{k} (-z_1)^{k-j} e_j^{(q)}(x_1, \ldots, x_k).$$

Proof. The permutation $c[k,k] = s_1 s_2 \cdots s_k$ has a unique reduced word, so its reduced factorizations are $v_1 = s_1 \cdots s_j$, $v_2 = s_{j+1} \cdots s_k$ for $0 \leq j \leq k$. By Lemma 2.1 with $y = 0$, $\widetilde{E}_{k,k}(x;z) = \sum_j \mathfrak{S}_{s_1 \cdots s_j}(0;z)\ \widetilde{\mathfrak{S}}_{s_{j+1} \cdots s_k}(x)$. Here $s_1 \cdots s_j$ is dominant with code (1^j), so $\mathfrak{S}_{s_1 \cdots s_j}(x;z) = \prod_{i \leq j}(x_i - z_1)$ and $\mathfrak{S}_{s_1 \cdots s_j}(0;z) = (-z_1)^j$; and $s_{j+1} \cdots s_k$ is the Grassmannian permutation with descent at k and a single column of height $k-j$, whose Schubert polynomial $e_{k-j}(x_1, \ldots, x_k)$ is a standard elementary monomial, so that its quantum Schubert polynomial is $e_{k-j}^{(q)}(x_1, \ldots, x_k)$ [13]. Summing gives the second expression, which equals $\Delta_k(-z_1 \mid x_1, \ldots, x_k)$ by the definition of the quantum elementary polynomials. □

Theorem 3.4 (The Pieri formula)**.** *Suppose $u \in S_\infty$ and $k \geq 1$. Then*

$$\widetilde{\mathfrak{S}}_u(x;y)\widetilde{E}_{k,k}(x;z) = \sum_{u \overset{k}{\rightsquigarrow} w} q_k(u,w)\mathbf{E}(y_{\mathrm{fix}_k(u,w)};z)\,\widetilde{\mathfrak{S}}_w(x;y).$$

The proof is given in Section 6. Note that the right-hand side is manifestly triple Schubert positive, in accordance with Theorem 2.3, and that at $q = 0$ it reduces to the case $p = k$ of the Pieri formula of [33, Theorem 7.1]. The general-degree version, Theorem 8.1, replaces $\mathbf{E}(y_{\mathrm{fix}_k(u,w)};z)$ by $E_{p-n_k(u,w),\,k-n_k(u,w)}(y_{\mathrm{fix}_k(u,w)};z)$.

Triple Schubert positivity does not by itself imply Graham positivity at $z = y$: a difference $y_i - y_j$ with $i < j$ is not a nonnegative combination of the negative simple roots $y_{l+1} - y_l$. The Pieri coefficients nevertheless are Graham-positive after specialization, for the same reason as in [33, Section 7]: they are factorial elementary polynomials evaluated at an increasing sequence of y-variables.

Corollary 3.5 (Graham positivity of the Pieri coefficients)**.** *Let $u \overset{k}{\rightsquigarrow} w$, $n = n_k(u,w)$ and $0 \leq p \leq k$. The specialization $E_{p-n,\,k-n}(y_{\mathrm{fix}_k(u,w)};z)\big|_{z=y}$ is a polynomial in the differences $y_{l+1} - y_l$ with nonnegative integer coefficients. Consequently, at $z = y$, every coefficient in Theorems 3.4 and 8.1 lies in $\mathbb{Z}_{\geq 0}[y_{l+1} - y_l][q_1, q_2, \ldots]$.*

Proof. Write $\mathrm{fix}_k(u,w) = \{a_1 < a_2 < \cdots < a_{k-n}\}$. Since the a_i are distinct positive integers, $a_i \geq i$ for all i. By the formula of Definition 3.1,

$$E_{p-n,\,k-n}(y_{\mathrm{fix}_k(u,w)};z) = \sum_{1 \leq i_1 < \cdots < i_{p-n} \leq k-n} \prod_{j=1}^{p-n} \left(y_{a_{i_j}} - z_{i_j - j + 1}\right),$$

and at $z = y$ each factor becomes $y_{a_{i_j}} - y_{i_j - j + 1}$ with $a_{i_j} \geq i_j \geq i_j - j + 1$, which is the sum $\sum_{l=i_j-j+1}^{a_{i_j}-1}(y_{l+1} - y_l)$ of negative simple roots (empty, hence 0, when $a_{i_j} = i_j - j + 1$). The product and the sum have nonnegative integer coefficients, and $q_k(u,w)$ is a monomial. □

4. Expansion formulas and the multiplication algorithms

4.1. **Dominant shapes.**

Definition 4.1 ($\mathfrak{c}$, (strict) dominant permutations, θ, μ, $\widetilde{\theta}$, $\widetilde{\mu}$, medium theta $\overline{\theta}$, $\overline{\mu}$)**.** We say that a permutation μ is *dominant* if $\mathfrak{c}_i(\mu) \geq \mathfrak{c}_{i+1}(\mu)$ for all i. A permutation μ is said to be *strict dominant* if μ is dominant and whenever $\mathfrak{c}_i(\mu) \neq 0$ we have $\mathfrak{c}_i(\mu) > \mathfrak{c}_{i+1}(\mu)$. For a permutation $v \in S_\infty$, we define a dominant permutation $\mu(v)$ recursively as follows. If v is already dominant, define $\mu(v) = v$. Otherwise, let i be the maximal index such that $\mathfrak{c}_i(v) < \mathfrak{c}_{i+1}(v)$, and define

$$\mu(v) = \mu(vs_i)$$

Define $\theta_i(v)$ to be $\mathfrak{c}_i(\mu(v))$.

Similarly, for a permutation $v \in S_\infty$, we define a strict dominant permutation $\widetilde{\mu}(v)$ recursively as follows. If v is already strict dominant, define $\widetilde{\mu}(v) = v$. Otherwise, let i be the maximal index such that $\mathfrak{c}_{i+1}(v) \neq 0$ and $\mathfrak{c}_i(v) \leq \mathfrak{c}_{i+1}(v)$, then define

$$\widetilde{\mu}(v) = \widetilde{\mu}(vs_i)$$

Define $\widetilde{\theta}_i(v)$ to be $\mathfrak{c}_i(\widetilde{\mu}(v))$.

Between these two lies the *medium theta* $\overline{\theta}(v)$, the shape actually iterated by the quantum multiplication algorithm (the routine `medium_theta` of [34]). Call an index $i \geq 1$ *defective* for v if either

(a) $\mathfrak{c}_i(v) < \mathfrak{c}_{i+1}(v)$, or
(b) $\mathfrak{c}_i(v) = \mathfrak{c}_{i+1}(v) \neq 0$, $i > 1$, and $\mathfrak{c}_{i-1}(v) \leq \mathfrak{c}_i(v) + 1$.

If v has no defective index, define $\overline{\mu}(v) = v$; otherwise let i be the *minimal* defective index and define

$$\overline{\mu}(v) = \overline{\mu}(vs_i).$$

Define $\overline{\theta}_i(v) = \mathfrak{c}_i(\overline{\mu}(v))$ and $|\overline{\theta}(v)| = \sum_i \overline{\theta}_i(v)$. In both cases (a) and (b) one has $\mathfrak{c}_i(v) \leq \mathfrak{c}_{i+1}(v)$, which forces $v(i) < v(i+1)$; hence $\ell(vs_i) = \ell(v) + 1$, $\mathfrak{c}(vs_i)$ is obtained from $\mathfrak{c}(v)$ by replacing $(\mathfrak{c}_i, \mathfrak{c}_{i+1})$ with $(\mathfrak{c}_{i+1} + 1, \mathfrak{c}_i)$ (in case (b): by incrementing $\mathfrak{c}_i$), and the recursion terminates, since a defective index has $\mathfrak{c}_{i+1}(v) \neq 0$, so that $v \in S_n$ implies $vs_i \in S_n$, and the length increases at each step. By construction $\overline{\theta}(v)$ is weakly decreasing, so $\overline{\mu}(v)$ is dominant; a pair of equal consecutive nonzero parts $\overline{\theta}_i(v) = \overline{\theta}_{i+1}(v)$ occurs only when $i = 1$ or $\overline{\theta}_{i-1}(v) \geq \overline{\theta}_i(v) + 2$, in particular no three consecutive nonzero parts are equal; and $\overline{\mu}(v) = v \cdot y$ is a reduced factorization with $\ell(\overline{\mu}(v)) = |\overline{\theta}(v)|$. Thus $\overline{\theta}(v)$ permits some, but not all, repeated parts, and lies between $\theta(v)$ (arbitrary repeats) and $\widetilde{\theta}(v)$ (no repeats). The number of nonzero parts of $\overline{\theta}(v)$ equals the number of nonzero parts of $\mathfrak{c}(v)$, since neither (a) nor (b) creates or destroys a zero. We say that a dominant permutation μ has *medium shape* if its code satisfies the condition just stated: whenever two consecutive nonzero parts $\mathfrak{c}_i(\mu) = \mathfrak{c}_{i+1}(\mu)$ are equal, either $i = 1$ or $\mathfrak{c}_{i-1}(\mu) \geq \mathfrak{c}_i(\mu) + 2$.

The three shapes are related by the right weak order, in which $u \leq_R w$ means $w = u \cdot y$ reduced, equivalently $N(u) \subseteq N(w)$ for the sets of *value inversions* $N(u) = \{(a, b) : a < b,\ u^{-1}(b) < u^{-1}(a)\}$ [2, Proposition 3.1.3]. Recall that a permutation is dominant if and only if it avoids the pattern 132; it is strict dominant if and only if it is decreasing and then increasing, i.e. if and only if it avoids both 132 and 231 (a strict dominant permutation has $\mu(i) > \mu(i+1)$ for all i up to its last descent, and conversely a permutation without peaks has no 132 and no 231).

Lemma 4.2. *Let $v \in S_\infty$ and let $i \geq 1$ be such that $v(i) < v(i+1)$ and $\mathfrak{c}_{i+1}(v) \neq 0$. Then every $w \geq_R v$ avoiding* 132 *and* 231 *satisfies $w \geq_R vs_i$. If moreover $\mathfrak{c}_i(v) < \mathfrak{c}_{i+1}(v)$, then every $w \geq_R v$ avoiding* 132 *satisfies $w \geq_R vs_i$.*

Proof. Put $a = v(i) < b = v(i+1)$; then $N(vs_i) = N(v) \cup \{(a, b)\}$, so we must show that b precedes a in w. Since $\mathfrak{c}_{i+1}(v) \neq 0$ there is a value $c < b$ at a position $j > i + 1$ of v, and $(c, b) \in N(v) \subseteq N(w)$ says that b precedes c in w. Suppose that a precedes b in w. If $c < a$, then $(c, a) \in N(v) \subseteq N(w)$, so w contains a, b, c in this order, a 231 pattern. If $a < c < b$, then w contains a, b, c in this order, a 132 pattern. In the first assertion both cases are excluded. For the second, note that $\mathfrak{c}_{i+1}(v) - \mathfrak{c}_i(v) = \#\{j > i + 1 : a < v(j) < b\}$ when $v(i) < v(i+1)$, so $\mathfrak{c}_i(v) < \mathfrak{c}_{i+1}(v)$ allows us to choose c with $a < c < b$, and only the 132 case arises. □

Proposition 4.3 (Weak order characterization)**.** *Let $v \in S_\infty$ and let $\mathcal{P} = \mathcal{P}(v)$ be the set of indices i with $\overline{\theta}_i(v) = \overline{\theta}_{i+1}(v) \neq 0$. Then:*

(i) *$\mu(v)$ is the minimum, for $\leq_R$, of the dominant permutations $w \geq_R v$. (This is the upper projection of v to the Tamari lattice, realized as the set of* 132*-avoiding permutations in the weak order* [3]*, i.e. to the quotient of the weak order by the sylvester congruence* [29, 30]*.)*
(ii) *$\widetilde{\mu}(v)$ is the minimum of the strict dominant permutations $w \geq_R v$.*
(iii) *The simple reflections s_i, $i \in \mathcal{P}$, commute pairwise, and*

$$\widetilde{\mu}(v) = \overline{\mu}(v) \cdot \prod_{i \in \mathcal{P}} s_i$$

is a reduced factorization. Equivalently, $\widetilde{\theta}(v)$ is obtained from $\overline{\theta}(v)$ by increasing by one the first entry of each retained equal pair, and $|\widetilde{\theta}(v)| = |\overline{\theta}(v)| + |\mathcal{P}(v)|$.

In (i) and (ii) the recursions of Definition 4.1 may take the defective indices in any order; $\overline{\mu}(v)$, by contrast, depends on the order (for $v =$ 2341 the permutations 4231 $= \overline{\mu}(v)$ and 3421 *are both minimal among the medium shape permutations above v), and it is normalized by the choice of the minimal defective index.*

Proof. (i) Each step $v \mapsto vs_i$ of the recursion for μ has $\mathfrak{c}_i(v) < \mathfrak{c}_{i+1}(v)$, so by Lemma 4.2 every dominant $w \geq_R v$ satisfies $w \geq_R vs_i$; by induction on $\ell(\mu(v)) - \ell(v)$, $w \geq_R \mu(v)$, and $\mu(v)$ is dominant and $\geq_R v$. (ii) is identical, each step of the recursion for $\widetilde{\mu}$ having $\mathfrak{c}_i(v) \leq \mathfrak{c}_{i+1}(v)$, i.e. $v(i) < v(i+1)$, and $\mathfrak{c}_{i+1}(v) \neq 0$, so

that the first assertion of Lemma 4.2 applies. (iii) Put $\mu = \overline{\mu}(v)$ and $w = \widetilde{\mu}(v)$. Each step of the recursion for $\overline{\mu}$ is of type (a) or (b) of Definition 4.1, and in both cases $v(i) < v(i+1)$ and $\mathfrak{c}_{i+1}(v) \neq 0$; so Lemma 4.2 and induction give $w \geq_R \mu$. The indices in $\mathcal{P}$ are pairwise at distance ≥ 2, since no three consecutive parts of $\overline{\theta}(v)$ are equal, so the s_i commute. For $i \in \mathcal{P}$ we have $\mu(i) < \mu(i+1)$ (as $\mathfrak{c}_i(\mu) = \mathfrak{c}_{i+1}(\mu)$) and $\mathfrak{c}_{i+1}(\mu) \neq 0$; applying Lemma 4.2 successively to μ, μs_{i_1}, $\mu s_{i_1} s_{i_2}, \ldots$ (the positions $i, i+1$ and the entry $\mathfrak{c}_{i+1}$ being unaffected by the previous $s_{i'}$, $|i - i'| \geq 2$) gives $w \geq_R \mu' := \mu \prod_{i \in \mathcal{P}} s_i$. Each s_i increases the length, replacing $(\mathfrak{c}_i, \mathfrak{c}_{i+1}) = (k, k)$ by $(k+1, k)$, so $\mu' = \mu \cdot \prod s_i$ is reduced and $\mathfrak{c}(\mu')$ is strictly decreasing on its nonzero part (the part preceding a retained pair is $\geq k+2 > k+1$): μ' is strict dominant and $\mu' \geq_R \mu \geq_R v$, whence $w \leq_R \mu'$ by (ii). Thus $w = \mu'$. For the final remark, $N(2341) = \{(1,2), (1,3), (1,4)\}$ is contained in $N(4231)$ and in $N(3421)$, both of which are medium shape with codes $(3,1,1)$ and $(2,2,1)$, and neither is below the other. □

4.2. **Expansion of a quantum double Schubert polynomial.**

Definition 4.4 ($\overset{i}{\Rightarrow}$, $\ell(u,v)$, $\sigma_i(u,v)$, $D_i(u,v)$, z_D)**.** Let $u, v \in S_\infty$ and $i \geq 1$ be an integer. We define $u \overset{i}{\Rightarrow} v$ if there exist distinct integers $b_1, b_2, \ldots, b_k$ such that $b_j \neq i$ for all j, $\ell(u t_{i,b_1} \cdots t_{i,b_j}) = \ell(u) + j$ for all j, and

$$v = u t_{i,b_1} \cdots t_{i,b_k}.$$

Thus $u \overset{i}{\Rightarrow} v$ means that v is reached from u by a chain of Bruhat covers $u \lessdot u t_{ib_1} \lessdot u t_{ib_1} t_{ib_2} \lessdot \cdots \lessdot v$, each of which exchanges the position i with a position not used before. The sequence $b_1, \ldots, b_k$ is determined by u and v: the product $u^{-1}v = t_{ib_1} \cdots t_{ib_k}$ is the cycle $i \mapsto b_k \mapsto b_{k-1} \mapsto \cdots \mapsto b_1 \mapsto i$, from which the b_j are read off. We write $\ell(u,v) = \ell(v) - \ell(u) = k$ and define

$$\sigma_i(u,v) = \#\{j \mid b_j < i\}, \qquad D_i(u,v) = \{u(i)\} \cup \{u(j) \mid u(j) \neq v(j)\} = \{u(i), u(b_1), \ldots, u(b_k)\},$$

a set of $k+1$ values. Finally, for a finite set $D = \{d_1 < \cdots < d_m\}$ of positive integers with $m = k - p + 1$ we write $E_{p,k}(x; z_D) = E_{p,k}(x; z_{d_1}, \ldots, z_{d_m})$ and $\widetilde{E}_{p,k}(x; z_D) = \widetilde{E}_{p,k}(x; z_{d_1}, \ldots, z_{d_m})$; since these polynomials are symmetric in $z_1, \ldots, z_{k-p+1}$ (Lemma 7.3), the order in which the elements of D are listed is immaterial. As in Definition 3.2, $E_{p,k} = \widetilde{E}_{p,k} = 0$ when $p < 0$.

The following formulas express a (quantum) double Schubert polynomial as a signed sum of products of (quantum) factorial elementary polynomials in nested initial alphabets; they are the v-side input of the multiplication algorithm, and they are proved in Section 7. All of them start from a dominant permutation μ lying above v^{-1} in the right weak order, that is $\mu = v^{-1} \cdot y$ with $\ell(\mu) = \ell(v) + \ell(y)$; the permutations $\mu(v^{-1})$, $\widetilde{\mu}(v^{-1})$ and $\overline{\mu}(v^{-1})$ are of this kind, since each step of their construction multiplies on the right by a simple reflection s_i with $\mathfrak{c}_i \leq \mathfrak{c}_{i+1}$, which forces $v(i) < v(i+1)$ and so increases the length. The implementation uses $\mu = \mu(v^{-1})$ in the classical case, and the parts of $\mathfrak{c}(\mu)$ are then the $\theta_i(v^{-1})$ of Definition 4.1. In the quantum case the dominant permutation cannot be arbitrary, because the quantum polynomial $\widetilde{\mathfrak{S}}_{\mu^{-1}}(x; z)$ which is being expanded factors into a product of quantum factorial elementary polynomials only when the column heights of the diagram of μ^{-1}, which are the parts of $\mathfrak{c}(\mu)$, are pairwise distinct (Lemma 7.4); for repeated column heights it is *not* a product (see Section 8). Theorem 4.6 accordingly uses the strict shape $\widetilde{\mu}(v^{-1})$. The implementation (the routine `schubmult_q_double_fast` of [34]) uses the smaller intermediate shape $\overline{\mu}(v^{-1})$, in which an equal pair of columns is allowed when there is room for it; the quantization of such a pair is a single "doubled" factor, which Theorem 4.7 isolates and which the algorithm multiplies out by the doubled Pieri formula of Theorem 8.3 and Corollary 8.5.

Theorem 4.5 (Formula for ordinary double Schubert polynomials)**.** *Let $v \in S_\infty$ and let μ be a dominant permutation with $\mu = v^{-1} \cdot y$ reduced, for instance $\mu = \mu(v^{-1})$. Let $\theta = \mathfrak{c}(\mu)$, with nonzero parts $\theta_1 \geq \cdots \geq \theta_k > 0$. Then*

$$\mathfrak{S}_v(x; z) = \sum_{(v_0, \ldots, v_k)} \prod_{i=1}^{k} (-1)^{\sigma_i(v_{i-1}, v_i)} E_{\theta_i - \ell(v_{i-1}, v_i),\, \theta_i}\big(x; z_{D_i(v_{i-1}, v_i)}\big),$$

the sum being over all sequences $1 = v_0 \overset{1}{\Rightarrow} v_1 \overset{2}{\Rightarrow} v_2 \overset{3}{\Rightarrow} \cdots \overset{k}{\Rightarrow} v_k = v\mu$.

Theorem 4.6 (Formula for quantum double Schubert polynomials)**.** *Let $v \in S_\infty$ and let μ be a strict dominant permutation with $\mu = v^{-1} \cdot y$ reduced, for instance $\mu = \widetilde{\mu}(v^{-1})$. Let $\widetilde{\theta} = \mathfrak{c}(\mu)$, with nonzero parts $\widetilde{\theta}_1 > \cdots > \widetilde{\theta}_k > 0$. Then*

$$\widetilde{\mathfrak{S}}_v(x;z) = \sum_{(v_0,\ldots,v_k)} \prod_{i=1}^{k} (-1)^{\sigma_i(v_{i-1},v_i)} \widetilde{E}_{\widetilde{\theta}_i - \ell(v_{i-1},v_i),\,\widetilde{\theta}_i}\big(x; z_{D_i(v_{i-1},v_i)}\big),$$

the sum being over all sequences $1 = v_0 \overset{1}{\Rightarrow} v_1 \overset{2}{\Rightarrow} \cdots \overset{k}{\Rightarrow} v_k = v\mu$.

For the medium shape we need a name for the quantization of a product of two factorial elementary polynomials in the same x-variables. Let $\mathcal{Q}$ be the quantization map of Fomin, Gelfand and Postnikov (Lemma 2.1), extended linearly over all coefficient variables, and for $0 \le p_1, p_2 \le k$ and two alphabets z, z' put

$$\widetilde{E}_{p_1,p_2;k}(x;z,z') := \mathcal{Q}\big(E_{p_1,k}(x;z)\, E_{p_2,k}(x;z')\big), \tag{4.1}$$

the *doubled quantum factorial elementary polynomial*; for $p_1 = p_2 = k$ it is the quantum double Schubert polynomial $\widetilde{\mathfrak{S}}_{d_2[k]}$ of the $k \times 2$ rectangle in the variables z_1, z_1' (Section 8). It is *not* the product $\widetilde{E}_{p_1,k}(x;z)\widetilde{E}_{p_2,k}(x;z')$ in general.

Theorem 4.7 (Formula for quantum double Schubert polynomials, medium shape)**.** *Let $v \in S_\infty$ and let μ be a dominant permutation of medium shape (Definition 4.1) with $\mu = v^{-1} \cdot y$ reduced, for instance $\mu = \overline{\mu}(v^{-1})$. Let $\overline{\theta} = \mathfrak{c}(\mu)$, with nonzero parts $\overline{\theta}_1 \ge \cdots \ge \overline{\theta}_m > 0$, let $\mathcal{P} = \{i : \overline{\theta}_i = \overline{\theta}_{i+1}\}$ be the set of* paired *indices and $\mathcal{S} = \{1,\ldots,m\} \setminus (\mathcal{P} \cup (\mathcal{P}+1))$ the set of* single *indices. Then*

$$\widetilde{\mathfrak{S}}_v(x;z) = \sum_{(v_0,\ldots,v_m)} \prod_{i=1}^{m} (-1)^{\sigma_i(v_{i-1},v_i)} \prod_{i\in\mathcal{S}} \widetilde{E}_{\overline{\theta}_i-\ell_i,\,\overline{\theta}_i}\big(x; z_{D_i}\big) \prod_{i\in\mathcal{P}} \widetilde{E}_{\overline{\theta}_i-\ell_i,\,\overline{\theta}_i-\ell_{i+1};\,\overline{\theta}_i}\big(x; z_{D_i}, z_{D_{i+1}}\big),$$

where $\ell_i = \ell(v_{i-1}, v_i)$, $D_i = D_i(v_{i-1}, v_i)$, and the sum is over all sequences $1 = v_0 \overset{1}{\Rightarrow} v_1 \overset{2}{\Rightarrow} \cdots \overset{m}{\Rightarrow} v_m = v\mu$. For a strict dominant μ this is Theorem 4.6.

By Proposition 4.3(iii), $|\widetilde{\theta}(v^{-1})| = |\overline{\theta}(v^{-1})| + |\mathcal{P}|$: the medium shape is smaller than the strict shape by exactly the number of retained pairs (for instance $\mathfrak{c}(v^{-1}) = (1,1)$ gives $\overline{\theta} = (1,1)$, $\widetilde{\theta} = (2,1)$, and $\mathfrak{c}(v^{-1}) = (2,2,2)$ gives $\overline{\theta} = (4,2,2)$, $\widetilde{\theta} = (4,3,2)$). Since $\sum_i \ell_i = |\mathfrak{c}(\mu)| - \ell(v)$ for every chain in these formulas, a smaller shape means shorter chains in the v-side sum and, in the multiplication algorithm, u-side Pieri steps of smaller degree; this is the reason the implementation prefers $\overline{\mu}(v^{-1})$ to $\widetilde{\mu}(v^{-1})$. Proposition 4.3(iii) also explains how Theorem 4.7 arises from Theorem 4.6: the dominant polynomial of the medium shape is obtained from that of the strict shape by one divided difference per retained pair, which turns the product of the two adjacent factors into the doubled factor (Corollary 7.7).

4.3. **The classical algorithm.** Coupled with the Pieri formulas, these formulas give an algorithm for computing products of (quantum) double Schubert polynomials, which is the one implemented in [34]. (Algorithms for computing quantum Schubert polynomials and Gromov–Witten invariants of partial flag manifolds were given by Buch [5].) We describe its classical form first. The u-side covers are generated by the classical Pieri rule, Theorem 8.1 at $q = 0$ (the enumeration is Method 4.12 below with the quantum covers discarded); the v-side is the following signed path expansion.

The starting point of the v-side primitive is the structure of a *dominant* double Schubert polynomial. If μ is dominant and the columns of its diagram have heights $\lambda_1, \ldots, \lambda_m$ (so that $(\lambda_1,\ldots,\lambda_m) = \mathfrak{c}(\mu^{-1})$), then $\mathfrak{S}_\mu(x;z)$ factors as a product of factorial elementary symmetric polynomials of maximal degree, one per column of the shape,

$$\mathfrak{S}_\mu(x;z) = \prod_{i=1}^{m} E_{\lambda_i,\,\lambda_i}(x; z_i)$$

(Lemma 7.4). For a general v one writes $\mathfrak{S}_v(x;z)$ by applying divided differences in the z-variables to the dominant polynomial $\mathfrak{S}_{\mu^{-1}}(x;z)$ with $\mu = \mu(v^{-1})$, whose column heights are the parts of $\theta(v^{-1})$; because that polynomial factors by column, the divided-difference action decouples and may be carried out one

column (one variable z_i) at a time. Applied to a single column this produces a *signed* descending expansion. The following primitive carries it out.

Definition 4.8 (k-strip descent beneath an edge, the down-Pieri problem)**.** Let $w, \mu \in S_n$ and $1 \le k < n$. Say w' lies *beneath the edge* μ if

$$\mathrm{inv}(w'\mu) = \mathrm{inv}(\mu) - \mathrm{inv}(w'),$$

that is, $\ell(w'\mu) = \ell(\mu) - \ell(w')$. A *$k$-strip descent* of w is a Bruhat descent $w \gtrdot wt_{ab}$ realized by a transposition t_{ab} that moves the value in the pivot position k either leftward (from a position $a < k$, with sign -1) or rightward (to a position $b > k$, with sign $+1$), touching only positions still holding their original value. Given a starting label w, the edge μ, a degree p, and the column k, the *down-Pieri problem* asks for the signed list of those length-t descending k-strip chains from w ($t \le p$) whose endpoint lies beneath μ.

Method 4.9 (**kdown**(w, μ, p, k))**.** The algorithm returns a signed list D of triples (w', t, s), one for each length-t descending k-strip chain from w ($t \le p$) whose endpoint w' lies beneath μ, with $s \in \{+1, -1\}$ its sign.

(1) Initialize $D \leftarrow [\,]$; if w is already beneath μ, append $(w, 0, +1)$. Set a frontier $F \leftarrow [(w, +1)]$.
(2) For $t = 1, \ldots, p$, form $F' \leftarrow [\,]$ and, for each $(w'', s) \in F$:
   (a) (leftward) for each position $a < k$ with $w''(a) = w(a)$ such that $w'' \gtrdot w''t_{a,k}$ is a Bruhat descent, set $w''' \leftarrow w''t_{a,k}$ and append $(w''', -s)$ to F'; if w''' is beneath μ, append $(w''', t, -s)$ to D;
   (b) (rightward) for each position $b > k$ with $w''(b) = w(b)$ such that $w'' \gtrdot w''t_{k,b}$ is a Bruhat descent, set $w''' \leftarrow w''t_{k,b}$ and append $(w''', +s)$ to F'; if w''' is beneath μ, append $(w''', t, +s)$ to D.

   Then set $F \leftarrow F'$.
(3) Return D.

Applied without the edge μ, the signed descending expansion produces many extraneous terms that ultimately cancel. The edge is exactly the bound that discards them: only endpoints beneath μ are retained in D. In the assembled v-side (Method 4.10 below), only these retained labels seed the next column, so each stage passes forward only labels compatible with the remaining shape.

Method 4.10 (**vpaths**(v), the v-side path dictionaries)**.** Let $\theta := \theta(v^{-1})$ with nonzero parts in columns $1, \ldots, L$, and let $\nu := v\,\mu(v^{-1})$. Processing columns from $i = L$ down to $i = 1$, maintain for each column a set of live labels (initialized at column L to the single label ν). At column i, with $k = i$ and edge $\mu_i := \mu$ of the shape truncated below column i, call **kdown**(w, μ_i, θ_i, k) for each live label w; record its signed output triples as the transitions out of w, and let their endpoints be the live labels seeding column $i - 1$. The collected transitions form the path dictionaries used by **schubmult**.

Method 4.11 (The algorithm **schubmult**)**.** Fix n and let $u, v \in S_n$. The algorithm **schubmult**(u, v) computes the expansion of $\mathfrak{S}_u(x; z)\ \mathfrak{S}_v(x; z)$ in the Schubert basis as follows. Set $\theta := \theta(v^{-1})$ and let

$$L := \#\{i \mid \theta_i \neq 0\}$$

be the number of nonzero parts. The algorithm proceeds in L layers. It maintains a *front*, a collection of nodes each consisting of an intermediate permutation on the u-side paired with a label on the v-side, initialized to the single node $(u,\ e)$. At layer i it replaces each node by the nodes obtained by combining one u-side cover, generated by the classical Pieri rule, i.e. by **qpieri**$(\,\cdot\,, \theta_i, \theta_i)$ of Method 4.12 below with the quantum covers discarded, with one v-side transition, generated by **vpaths**(v) (Method 4.10), accumulating coefficients in the coefficient ring. After all L layers the surviving coefficients at v-label $v\mu(v^{-1})$ give the structure constants.

4.4. **The quantum algorithm.** The same path organization gives the quantum multiplication algorithm, the routine `schubmult_q_double_fast` of [34]. The v-side is unchanged except for the shape: the path dictionaries of Method 4.10 are built over the columns of $\overline{\theta} := \overline{\theta}(v^{-1})$ (Definition 4.1), with $\nu := v\,\overline{\mu}(v^{-1})$ in place of $v\mu(v^{-1})$, and by Theorem 4.7 they enumerate the chains $1 = v_0 \overset{1}{\Longrightarrow} \cdots \overset{m}{\Longrightarrow} v_m = \nu$ together with the data ℓ_i, σ_i, D_i of each step. On the u-side, the classical covers are replaced by the covers of the quantum Bruhat graph (Definition 6.3, Remark 2.2), including the length-decreasing moves, whose contributions carry the corresponding q-monomials. We first describe the two u-side primitives, which enumerate the chains of Theorem 8.1 and the compatible pairs of Corollary 8.5.

Method 4.12 (**qpieri**(u,p,k))**.** Given $u \in S_\infty$, $k \geq 1$ and $0 \leq p \leq k$, the algorithm returns a list T of triples $(w,t,\mathbf{q})$ with $w \in S_\infty$, $0 \leq t \leq p$ and $\mathbf{q}$ a monomial in the q_i.

(1) Initialize $T \leftarrow [(u,0,1)]$ and a frontier $F \leftarrow [(u,1,\infty)]$ of triples $(w,\mathbf{q},b_{\mathrm{last}})$.
(2) For $t = 1,\dots,p$, form $F' \leftarrow [\,]$ and, for each $(w,\mathbf{q},b_{\mathrm{last}}) \in F$:
   (a) compute the set $A \leftarrow \{a \leq k :\ w(a) = u(a)\}$ of positions $\leq k$ not yet moved;
   (b) for each b with $k < b \leq b_{\mathrm{last}}$, in decreasing order, and each $a \in A$ such that t_{ab} is a cover of w (Definition 6.3: either $w(a) < w(b)$ and no $w(r)$, $a < r < b$, lies in $(w(a),w(b))$, or $w(a) > w(b)$ and every $w(r)$, $a < r < b$, lies in $(w(b),w(a))$), put $\mathbf{q}' = \mathbf{q}\,\kappa(w,t_{ab})$, i.e. $\mathbf{q}' = \mathbf{q}$ for a classical cover and $\mathbf{q}' = \mathbf{q}\,q_{ab} = \mathbf{q}\,q_a \cdots q_{b-1}$ for a quantum cover, and append $(wt_{ab},\mathbf{q}',b)$ to F' and $(wt_{ab},t,\mathbf{q}')$ to T.

   Then set $F \leftarrow F'$.
(3) Return T.

Only finitely many b qualify in (b): if w fixes every position $> N$ then no t_{ab} with $b > \max(N,k)+1$ is a cover of w (the values $w(r) = r$ at positions $\max(N,k) < r < b$ lie between $w(a)$ and $w(b) = b$), so b ranges over $k < b \leq \min(b_{\mathrm{last}}, \max(N,k)+1)$. The cover test in (b) is performed numerically: the implementation computes the signed length change

$$\ell(wt_{ab}) - \ell(w) = \pm 1 \pm 2\,\#\{r \mid a < r < b,\ w(r) \text{ between } w(a) \text{ and } w(b)\}$$

of Lemma 6.2 (the routine `count_bruhat`, with the sign $+$ if $w(a) < w(b)$ and $-$ otherwise) and accepts t_{ab} exactly when this number is 1 or $1 - 2(b-a)$, multiplying the current monomial by $q_a \cdots q_{b-1}$ in the second case. This is the routine `elem_sym_perms_q` of [34]; at $q = 0$, i.e. discarding the quantum covers, it enumerates the chains of the classical Pieri rule of [35, 33] (Theorem 8.1 at $q = 0$).

Lemma 4.13 (Correctness of **qpieri**)**.** *The list T returned by* **qpieri**(u,p,k) *contains exactly one triple $(w,t,\mathbf{q})$ for each $w \in S_\infty$ with $u \overset{k}{\rightsquigarrow} w$ and $n_k(u,w) \leq p$, and for it $t = n_k(u,w)$ and $\mathbf{q} = q_k(u,w)$. Consequently, for $0 \leq p' \leq p$,*

$$\widetilde{\mathfrak{S}}_u(x;y)\,\widetilde{E}_{p',k}(x;z) = \sum_{(w,t,\mathbf{q})\in T,\ t\leq p'} \mathbf{q}\,E_{p'-t,\,k-t}\big(y_{\mathrm{fix}_k(u,w)};z\big)\ \widetilde{\mathfrak{S}}_w(x;y),$$

where $\mathrm{fix}_k(u,w) = \{u(a) : a \leq k,\ u(a) = w(a)\}$ is read off from u and w.

Proof. A triple $(w,t,\mathbf{q})$ is appended to T for each sequence of transpositions $t_{a_1b_1},\dots,t_{a_tb_t}$ generated by the loop, and by construction these sequences are exactly those satisfying conditions (1)–(5) of Definition 3.1 with p replaced by $t \leq p$: $a_i \leq k < b_i$, the a_i are distinct (a position leaves A once it is moved and, being moved only once, never returns, since $w(a) \neq u(a)$ after the move), the b_i are weakly decreasing ($b \leq b_{\mathrm{last}}$), and each step is a cover (condition (4), by Definition 6.3). The weight $\mathbf{q}$ is the product of the κ's, which is $q_k(u,w)$ by the recursive definition in Definition 3.1. It remains to see that each w with $u \overset{k}{\rightsquigarrow} w$ arises from exactly one such sequence. By Lemma 6.8, a witnessing sequence is a pivot chain of $\pi = u^{-1}w$ in which the factors with a common b are consecutive and the distinct values of b decrease; the factors with a common b form the pivot factorization (6.1) of the cycle of π with top b, whose order is determined by that cycle, and the order of the cycles is determined by their tops. Hence the sequence is determined by w, and $t = n_k(u,w)$ is its length. The displayed formula is Theorem 8.1 with p' in place of p, the sum over $u \overset{k}{\rightsquigarrow} w$ with $n_k(u,w) \leq p'$ being the sum over the triples with $t \leq p'$. □

Method 4.14 (**doublepieri**(u,p_1,p_2,k))**.** Given $u \in S_\infty$, $k \geq 1$ and $0 \leq p_1,p_2 \leq k$, the algorithm returns, for each triple $(v,t_1,\mathbf{q}_1)$ of **qpieri**(u,p_1,k), a list $L(v,t_1,\mathbf{q}_1)$ of triples $(w,t_2,\mathbf{q}_2)$.

(1) Compute $T_1 \leftarrow$ **qpieri**(u,p_1,k).
(2) For each $(v,t_1,\mathbf{q}_1) \in T_1$: compute the cycle decomposition of $\pi_1 = u^{-1}v$, and record for each top B the set $\operatorname{supp} c_B \setminus \{B\}$ of non-top elements of the cycle c_B of π_1 with top B (empty if there is none); compute $T_2 \leftarrow$ **qpieri**(v,p_2,k); and for each $(w,t_2,\mathbf{q}_2) \in T_2$ compute the cycles of $\pi_2 = v^{-1}w$ and append $(w,t_2,\mathbf{q}_2)$ to $L(v,t_1,\mathbf{q}_1)$ unless some cycle d of π_2 has a non-top element in $\operatorname{supp} c_{\max d} \setminus \{\max d\}$.
(3) Return the lists L.

This is the routine `double_elem_sym_q` of [34].

Lemma 4.15 (Correctness of **doublepieri**). *The pairs* $\big((v,t_1,\mathbf{q}_1),(w,t_2,\mathbf{q}_2)\big)$ *with* $(w,t_2,\mathbf{q}_2)\in L(v,t_1,\mathbf{q}_1)$ *returned by* **doublepieri**(u,p_1,p_2,k) *are in bijection with the compatible pairs* $u \overset{k}{\rightsquigarrow} v \overset{k}{\rightsquigarrow} w$ *with* $n_k(u,v)\le p_1$ *and* $n_k(v,w)\le p_2$, *with* $t_1=n_k(u,v)$, $t_2=n_k(v,w)$, $\mathbf{q}_1\mathbf{q}_2=q_k(u,v)q_k(v,w)$. *Consequently, for* $0\le p_1'\le p_1$ *and* $0\le p_2'\le p_2$ *and two alphabets* z,z',

$$\widetilde{\mathfrak{S}}_u(x;y)\,\widetilde{E}_{p_1',p_2';k}(x;z,z') = \sum_{\substack{(v,t_1,\mathbf{q}_1)\in T_1\\ (w,t_2,\mathbf{q}_2)\in L(v,t_1,\mathbf{q}_1)\\ t_1\le p_1',\ t_2\le p_2'}} \mathbf{q}_1\mathbf{q}_2\,E_{p_1'-t_1,\,k-t_1}\big(y_{\mathrm{fix}_k(u,v)};z\big)\,E_{p_2'-t_2,\,k-t_2}\big(y_{\mathrm{fix}_k(v,w)};z'\big)\,\widetilde{\mathfrak{S}}_w(x;y).$$

Proof. By Lemma 4.13 the pairs examined in step 2 are in bijection with the pairs of chains $u \overset{k}{\rightsquigarrow} v \overset{k}{\rightsquigarrow} w$ with $n_k(u,v)\le p_1$, $n_k(v,w)\le p_2$, with the stated t's and $\mathbf{q}$'s. The rejection test is the negation of Definition 8.2: a cycle d of $v^{-1}w$ whose top $\max d$ is the top of a cycle c of $u^{-1}v$ must satisfy $c\cap d=\{\max d\}$, i.e. no non-top element of d may lie in c (it cannot equal $\max d=\max c$), and cycles of $v^{-1}w$ whose top is not a top of $u^{-1}v$ are unrestricted. The formula is Corollary 8.5 with (p_1',p_2') in place of (p_1,p_2). □

The implementation calls **qpieri** and **doublepieri** once per front node with p, respectively (p_1,p_2), equal to the largest degrees $k-\ell_i$, respectively $(k-\ell_i,k-\ell_{i+1})$, occurring among the v-side transitions of the layer, and then evaluates the coefficient of each transition by the displayed formulas with the actual degrees $p'\le p$ (the routine `elem_sym_func_q`, which reads $\mathrm{fix}_k(u,w)$ off the positions $a\le k$ with $u(a)=w(a)$ and the alphabet z_{D_i} off the transition); by the two lemmas this is the same as calling them separately for each transition.

Method 4.16 (The algorithm **schubmult**$_q$). Fix n and $u,v\in S_n$, and let $\overline{\theta}=\overline{\theta}(v^{-1})$ with nonzero parts $\overline{\theta}_1\ge\cdots\ge\overline{\theta}_m>0$, paired indices $\mathcal{P}$ and single indices $\mathcal{S}$ as in Theorem 4.7. The algorithm computes the expansion of $\widetilde{\mathfrak{S}}_u(x;y)\,\widetilde{\mathfrak{S}}_v(x;z)$ in the basis $\{\widetilde{\mathfrak{S}}_w(x;y)\}$. It maintains a front of nodes (w,λ), a u-side permutation paired with a v-side label, initialized to (u,e) with coefficient 1, and runs through the indices $i=1,\dots,m$, skipping $i\in\mathcal{P}+1$:

(1) If $i\in\mathcal{S}$, with $k=\overline{\theta}_i$: for each node (w,λ) and each v-side transition $\lambda\to\lambda'$ of column i, with data (ℓ_i,σ_i,D_i), and each $(w',t,\mathbf{q})\in$ **qpieri**$(w,k-\ell_i,k)$, create the node (w',λ') with coefficient

$$(-1)^{\sigma_i}\,\mathbf{q}\,E_{k-\ell_i-t,\,k-t}\big(y_{\mathrm{fix}_k(w,w')};z_{D_i}\big)$$

times the coefficient of (w,λ); by Lemma 4.13 this expands $\widetilde{\mathfrak{S}}_w(x;y)\,\widetilde{E}_{k-\ell_i,k}(x;z_{D_i})$.

(2) If $i\in\mathcal{P}$, with $k=\overline{\theta}_i=\overline{\theta}_{i+1}$: for each node (w,λ) and each pair of consecutive v-side transitions $\lambda\to\lambda'\to\lambda''$ of the columns i and $i+1$, with data (ℓ_i,σ_i,D_i) and $(\ell_{i+1},\sigma_{i+1},D_{i+1})$, and each pair $\big((w',t_1,\mathbf{q}_1),(w'',t_2,\mathbf{q}_2)\big)$ returned by **doublepieri**$(w,k-\ell_i,k-\ell_{i+1},k)$, create the node (w'',λ'') with coefficient

$$(-1)^{\sigma_i+\sigma_{i+1}}\,\mathbf{q}_1\mathbf{q}_2\,E_{k-\ell_i-t_1,\,k-t_1}\big(y_{\mathrm{fix}_k(w,w')};z_{D_i}\big)\,E_{k-\ell_{i+1}-t_2,\,k-t_2}\big(y_{\mathrm{fix}_k(w',w'')};z_{D_{i+1}}\big)$$

times the coefficient of (w,λ); by Lemma 4.15 this expands $\widetilde{\mathfrak{S}}_w(x;y)\,\widetilde{E}_{k-\ell_i,k-\ell_{i+1};k}(x;z_{D_i},z_{D_{i+1}})$ over the compatible pairs $w \overset{k}{\rightsquigarrow} w' \overset{k}{\rightsquigarrow} w''$.

Coefficients of coinciding nodes are added. After the last index the coefficient of (w,ν) is $\sum_d q^d c_{u,v}^{w,d}(y;z)$.

Proposition 4.17 (Correctness of **schubmult**$_q$). *Method 4.16 computes* $\sum_d q^d c_{u,v}^{w,d}(y;z)$ *for all* w.

Proof. By Theorem 4.7, $\widetilde{\mathfrak{S}}_v(x;z)$ is the sum over the v-side chains $1=v_0\overset{1}{\Longrightarrow}\cdots\overset{m}{\Longrightarrow}v_m=\nu$ of $\prod_i(-1)^{\sigma_i}$ times the product of the single factors $\widetilde{E}_{k-\ell_i,k}(x;z_{D_i})$, $i\in\mathcal{S}$, and the doubled factors $\widetilde{E}_{k-\ell_i,k-\ell_{i+1};k}(x;z_{D_i},z_{D_{i+1}})$, $i\in\mathcal{P}$; the path dictionaries enumerate these chains step by step, the label λ of a node recording the current v_i. Multiplying $\widetilde{\mathfrak{S}}_u(x;y)$ by these factors one at a time, in the order of the indices, by Lemma 4.13 for the single factors and Lemma 4.15 for the doubled ones, the coefficient of the node (w,λ) after the index i is the coefficient of $\widetilde{\mathfrak{S}}_w(x;y)$ in $\widetilde{\mathfrak{S}}_u(x;y)$ times the product of the factors of the columns $\le i$, summed over the partial chains ending at λ. After the last index, the nodes (w,ν) carry the coefficient of $\widetilde{\mathfrak{S}}_w(x;y)$ in $\widetilde{\mathfrak{S}}_u(x;y)\,\widetilde{\mathfrak{S}}_v(x;z)$. The two v-side transitions of a paired layer must be consumed together, since the doubled factor is not a product of two quantum factorial elementary polynomials. (The requirement that a retained

pair be preceded by a drop of at least two is what makes the lifted shape ν of Corollary 7.7 strict dominant, so that the two columns of the pair are the only ones whose quantization interacts.) □

The classical algorithm is the specialization $q = 0$ with $\overline{\theta}$ replaced by θ, in which the doubled factor is the product $E_{p_1,k}E_{p_2,k}$ and the paired step may be split into two ordinary ones. Thus the two algorithms share the same layer-by-layer structure, while their Pieri transitions reflect the classical or quantum multiplication rule, respectively.

Neither algorithm depends on the particular dominant permutation used for the v-side, beyond the hypotheses of the expansion theorems: the classical algorithm is correct for any dominant μ lying above v^{-1} in the right weak order, that is, with $\mu = v^{-1} \cdot y$ reduced, equivalently with μ^{-1} above v in the left weak order (Theorem 4.5); the quantum algorithm with single steps only is correct for any strict dominant such μ (Theorem 4.6); and the quantum algorithm with paired steps is correct for any such μ of medium shape (Theorem 4.7), strict dominant permutations being a special case. The canonical choices $\mu(v^{-1})$, $\widetilde{\mu}(v^{-1})$ and $\overline{\mu}(v^{-1})$ are simply the smallest admissible ones, and the length of μ governs the number of layers and the sizes of the path dictionaries, so they are the most efficient choices; the medium shape is preferred to the strict one for the same reason.

Example 4.18 (The algorithm **schubmult**$_q$ on $\widetilde{\mathfrak{S}}_{21}(x;y)\ \widetilde{\mathfrak{S}}_{312}(x;z)$)**.** Let $u = 21$ and $v = 312$. The v-side: $v^{-1} = 231$ has code $(1,1)$, which is of medium shape (the equal pair is at $i = 1$), so $\overline{\mu}(v^{-1}) = v^{-1}$, $\overline{\theta} = (1,1)$, $\mathcal{P} = \{1\}$, $\mathcal{S} = \emptyset$, and $\nu = v\,\overline{\mu}(v^{-1}) = 1$. The only chain $1 = v_0 \overset{1}{\Rightarrow} v_1 \overset{2}{\Rightarrow} v_2 = \nu$ is the trivial one, with $\ell_1 = \ell_2 = 0$, $\sigma_1 = \sigma_2 = 0$, $D_1 = \{v_0(1)\} = \{1\}$ and $D_2 = \{v_1(2)\} = \{2\}$; so Theorem 4.7 reads $\widetilde{\mathfrak{S}}_{312}(x;z) = \widetilde{E}_{1,1;1}(x;z_1,z_2) = \mathcal{Q}\big((x_1 - z_1)(x_1 - z_2)\big)$, and there is a single paired layer with $k = 1$ and $(p_1,p_2) = (1,1)$.

The u-side: **qpieri**$(21,1,1)$ examines the transpositions t_{1b}, $b \geq 2$, of $w = 21$: t_{12} has length change $-1 = 1 - 2(2-1)$ (a quantum cover, $w(1) = 2 > w(2) = 1$ with nothing between) and gives $(12, 1, q_1)$; t_{13} has $w(1) = 2 < w(3) = 3$ with the intermediate value $1 \notin (2,3)$, length change $+1$, and gives $(312,1,1)$; t_{14} has the intermediate value $3 \in (2,4)$ and is not a cover. Hence

$$T_1 = \{(21,0,1),\ (12,1,q_1),\ (312,1,1)\}.$$

doublepieri$(21,1,1,1)$ then computes, for each $(v', t_1, \mathbf{q}_1) \in T_1$, the list **qpieri**$(v',1,1)$ and discards the incompatible pairs:

- $v' = 21$ (trivial first step, no cycles): all of $T_2 = T_1$ is retained.
- $v' = 12$ (first step t_{12}, cycle $(1,2)$ with top 2, non-top element 1): $T_2 = \{(12,0,1),(21,1,1)\}$, the second step t_{13} not being a cover of 12. The pair $(12,1,q_1),(21,1,1)$ is *rejected*: the cycle $(1,2)$ of the second step has the top 2 of a first-step cycle and the non-top element 1 in common with it. Retained: $(12,0,1)$.
- $v' = 312$ (first step t_{13}, cycle $(1,3)$): $T_2 = \{(312,0,1),\ (132,1,q_1),\ (4123,1,1)\}$, from the quantum cover t_{12} $(3 > 1)$ and the classical cover t_{14} $(3 < 4$, intermediate values $1,2 \notin (3,4))$; t_{13} is not a cover of 312 $(3 > 2$ but the intermediate value $1 \notin (2,3))$. The second-step cycles $(1,2)$ and $(1,4)$ have tops $2, 4 \neq 3$, so both pairs are retained.

Each retained pair $\big((v', t_1, \mathbf{q}_1), (w, t_2, \mathbf{q}_2)\big)$ contributes $\mathbf{q}_1\mathbf{q}_2\, E_{1-t_1,1-t_1}(y_{\mathrm{fix}_1(u,v')}; z_1)\, E_{1-t_2,1-t_2}(y_{\mathrm{fix}_1(v',w)}; z_2)$ to the coefficient of $\widetilde{\mathfrak{S}}_w(x;y)$, where $E_{1,1}(y_a; z_i) = y_a - z_i$ and $E_{0,0} = 1$:

v'	w	$\mathbf{q}_1\mathbf{q}_2$	coefficient contributed
21	21	1	$(y_2 - z_1)(y_2 - z_2)$
21	12	q_1	$q_1(y_2 - z_1)$
21	312	1	$y_2 - z_1$
12	12	q_1	$q_1(y_1 - z_2)$
312	312	1	$y_3 - z_2$
312	132	q_1	q_1
312	4123	1	1

Adding the coefficients of coinciding w,

$$\widetilde{\mathfrak{S}}_{21}(x;y)\ \widetilde{\mathfrak{S}}_{312}(x;z) = (y_2 - z_1)(y_2 - z_2)\ \widetilde{\mathfrak{S}}_{21} + q_1(y_1 + y_2 - z_1 - z_2) + (y_2 + y_3 - z_1 - z_2)\ \widetilde{\mathfrak{S}}_{312} + q_1\ \widetilde{\mathfrak{S}}_{132} + \widetilde{\mathfrak{S}}_{4123},$$

all polynomials on the right being in $(x;y)$. The rejected pair would have contributed $q_1\,\widetilde{\mathfrak{S}}_{21}$, which is exactly the difference $q_1\,\widetilde{\mathfrak{S}}_u\,\widetilde{\mathfrak{S}}_{d_2[0]}$ between $\widetilde{\mathfrak{S}}_u\ \widetilde{E}_{1,1}(x;z_1)\widetilde{E}_{1,1}(x;z_2)$ and $\widetilde{\mathfrak{S}}_u\ \widetilde{\mathfrak{S}}_{312}(x;z)$ predicted by Lemma 8.6 below; see Example 8.4. Every coefficient is a nonnegative combination of products of differences $y_a - z_b$, as Theorem 2.3 requires, and has the degree $\ell(u)+\ell(v)-\ell(w)-2|d|$ of Section 2 (for instance $3-0-2=1$ for the term $q_1(y_1+y_2-z_1-z_2)$).

4.5. **Comparison with straightening methods.** The general-purpose alternative to a Pieri-driven algorithm is straightening: expand both factors in a basis of standard monomials of $QH^*(Fl_n)$, multiply, and reduce modulo the quantum ideal. The most recent form of this is the quantum rim-hook rule of Gu and Kalashnikov [16], which for the complete flag variety works in the basis of Fomin, Gelfand and Postnikov [13] of standard elementary monomials $\prod_m e_{j_m}(x_1,\dots,x_m)$, $0 \le j_m \le m$ (under flag duality, their monomials in the Chern classes of the tautological quotient bundles), and reduces a product vertex by vertex: $e_a(x_1,\dots,x_m)e_b(x_1,\dots,x_m)$ is expanded by the Pieri rule into two-column Schur polynomials, and each two-column shape is removed by a quantum rim-hook relation,

$$s_{(2^c,1^d)}[m] = e_{c+d}[m]\,e_c[m{+}1] - e_{c-1}[m]\,e_{c+d+1}[m{+}1] - q_m\,e_{c+d}[m]\,e_{c-2}[m{-}1] + q_m\,e_{c-1}[m]\,e_{c+d-1}[m{-}1]$$

(where $e_k[m] = e_k(x_1,\dots,x_m)$ and $e_k[0] = e_k[n] = \delta_{k0}$), which pushes a factor to a neighboring vertex, after which the process repeats. Since the quantization map is multiplicative on standard elementary monomials, Schubert classes are converted into and out of this basis by the classical transition matrix, of size $n! \times n!$.

We implemented this algorithm, memoizing the straightening, and verified it against `schubmult` on all 576 products in S_4 and 150 random products in S_5. Table 1 compares running times on the same random pairs $u, v \in S_n$ (non-equivariant, $y = z = 0$, the only case the rim-hook method covers), excluding the one-time construction of the transition matrix. For $n \le 6$ the two pure-Python implementations are comparable. At $n = 7$ the straightening is already five to seven times slower than the pure-Python kernel of `schubmult` and more than 500 times slower than its compiled kernel, and the cost of a single straightened product of two basis monomials grows by a factor of roughly 40 to 50 with each increment of n (18 ms, 0.69 s and 37 s on average at $n = 6, 7, 8$); at $n = 8$ a single Schubert product, which requires hundreds of such monomial products together with a 40320×40320 transition matrix, is out of reach, while `schubmult` takes 18 ms with the compiled kernel and 1.5 s in pure Python, and handles $n = 9$ in about 0.1 s. The reason is structural. The algorithm of Section 4 is not cancellation-free, since the expansion of Theorem 4.7 is signed, but it expands only one factor into elementary polynomials, and the Pieri formula then multiplies each product of elementary polynomials against the other factor directly in the Schubert basis, so that the cancellation is confined to that one signed expansion. Straightening expands both factors into a basis in which Schubert classes are dense, and every reduction step creates further terms that later cancel. The triple constants, with $y \ne z$, are not accessible to the rim-hook method at all.

n	products	`schubmult`, compiled	`schubmult`, pure Python	quantum rim-hook, pure Python
5	20	0.1 ms	2.5 ms	1.4 ms
6	20	0.7 ms	38 ms	46 ms
7	10	5 ms	0.53 s	2.7 s
8	5	18 ms	1.5 s	—
9	3	0.12 s	16 s	—

TABLE 1. Median wall-clock time per quantum product $\widetilde{\mathfrak{S}}_u\,\widetilde{\mathfrak{S}}_v$, $u, v \in S_n$ random, computed in the Schubert basis by `schubmult` (Pieri-based) and by the quantum rim-hook straightening of [16]; see Section 4.5.

5. Quantum-to-classical reduction

In this section we extend the quantum-to-classical reduction of Huang and Li [17] from the equivariant quantum structure constants $c^{w,d}_{u,v}(y;y)$ to the triple constants. The quantum-to-classical principle originates with Buch, Kresch and Tamvakis [8], who reduce the quantum product of Schubert classes of a Grassmannian to a classical product on a two-step flag variety; its equivariant (and K-theoretic) version is due to Buch and Mihalcea [9], and in combination with Buch's puzzle rule for two-step flag varieties [6] it yields a positive

combinatorial rule for $QH_T^*(\mathrm{Gr}(k,n))$ (Remark 2.11). For the complete flag variety, Leung and Li [22] showed that Gromov–Witten invariants of different degrees coincide and lower the degree d of a structure constant by one simple coroot at a time, and Huang and Li [17] extended this to equivariant quantum cohomology, which is the form we use. We write $\mathrm{sgn}_i(w) = 1$ if $w(i) > w(i+1)$ and $\mathrm{sgn}_i(w) = 0$ otherwise, and we use the notation $\delta_i = \alpha_i^\vee$, $|d| = \sum_i d_i$ and $\langle d\rangle_i = 2d_i - d_{i-1} - d_{i+1}$ (with $d_0 = 0$) of Definition 2.6. We say that $u \in S_\infty$ is *k-Grassmannian* if k is its only descent.

Theorem 5.1 ([17, Theorem 2.7])**.** *We have the following.*

(1) $c_{u,v}^{w,d}(y;y) = 0$ *unless* $\mathrm{sgn}_i(w) + \langle d\rangle_i \le \mathrm{sgn}_i(u) + \mathrm{sgn}_i(v)$ *for all* i.
(2) *If* $\mathrm{sgn}_k(w) + \langle d\rangle_k = \mathrm{sgn}_k(u) + \mathrm{sgn}_k(v) = 2$ *for some* k, *then*

$$c_{u,v}^{w,d}(y;y) = c_{us_k,vs_k}^{w,d-\delta_k}(y;y) = \begin{cases} c_{us_k,v}^{ws_k,d-\delta_k}(y;y) & \text{if } \mathrm{sgn}_k(w) = 0 \\ c_{us_k,v}^{ws_k,d}(y;y) & \text{if } \mathrm{sgn}_k(w) = 1 \end{cases}$$

This is symmetric in u and v.

Corollary 5.2 ([17, Corollary 2.8])**.** *Let $u, v, w \in S_\infty$ and $d \in Q^\vee$. Fix an index i.*

(1) *If* $\langle d\rangle_i = \mathrm{sgn}_i(u) = 0$ *and* $\mathrm{sgn}_i(w) = \mathrm{sgn}_i(v) = 1$, *then*

$$c_{u,v}^{w,d}(y;y) = c_{u,vs_i}^{ws_i,d}(y;y).$$

(2) *If* $\langle d\rangle_i = \mathrm{sgn}_i(u) = 1$ *and* $\mathrm{sgn}_i(w) = \mathrm{sgn}_i(v) = 0$, *then*

$$c_{u,v}^{w,d}(y;y) = c_{us_i,v}^{ws_i,d-\delta_i}(y;y).$$

This is symmetric in u and v.

Theorem 5.3 (Quantum reduction holds for triple quantum Schubert coefficients)**.** *We have the following.*

(1) $c_{u,v}^{w,d}(y;z) = 0$ *unless* $\mathrm{sgn}_i(w) + \langle d\rangle_i \le \mathrm{sgn}_i(u) + \mathrm{sgn}_i(v)$ *for all* i.
(2) *If* $\mathrm{sgn}_k(w) + \langle d\rangle_k = \mathrm{sgn}_k(u) + \mathrm{sgn}_k(v) = 2$ *for some* k, *then*

$$c_{u,v}^{w,d}(y;z) = c_{us_k,vs_k}^{w,d-\delta_k}(y;z) = \begin{cases} c_{us_k,v}^{ws_k,d-\delta_k}(y;z) & \text{if } \mathrm{sgn}_k(w) = 0 \\ c_{us_k,v}^{ws_k,d}(y;z) & \text{if } \mathrm{sgn}_k(w) = 1 \end{cases}$$

This is symmetric in u and v.

Proof. For (1), we note that since $\mathrm{sgn}_i(v_2) \le \mathrm{sgn}_i(v)$ for any factorization $v_1 \cdot v_2 = v$, we have that $\mathrm{sgn}_i(w) + \langle d\rangle_i \le \mathrm{sgn}_i(u) + \mathrm{sgn}_i(v_2)$ is a necessary condition for $c_{u,v_2}^{w,d}(y;y) \neq 0$. The result then follows from Lemma 2.1.

For (2), we note that if $\mathrm{sgn}_k(w) + \langle d\rangle_k = \mathrm{sgn}_k(u) + \mathrm{sgn}_k(v) = 2$, then since $\mathrm{sgn}_k(v_2) = \mathrm{sgn}_k(v)$ for any factorization $v_1 \cdot v_2 = v$ such that $c_{u,v_2}^{w,d}(y;y) \neq 0$, Theorem 5.1(2) holds for each term in the sum in Lemma 2.1, and the result follows. □

Leung and Li [22, Theorem 1.2] observed that when one of the two factors is Grassmannian, Theorem 5.1 can be iterated until no quantum parameter remains. Since their argument uses nothing but the two parts of that theorem, it applies verbatim to the triple constants.

Proposition 5.4 (Grassmannian factor: reduction to classical triple constants)**.** *Let v be k-Grassmannian, let $u, w \in S_\infty$ and let $d \neq 0$ be effective. Then either $c_{u,v}^{w,d}(y;z) = 0$, or there is an index m with $\langle d\rangle_m > 0$, $\mathrm{sgn}_m(u) = 1$ and $\mathrm{sgn}_m(w) = 0$ such that*

$$c_{u,v}^{w,d}(y;z) = c_{us_m,v}^{ws_m,\,d-\delta_m}(y;z).$$

Consequently $c_{u,v}^{w,d}(y;z)$ is either 0 or equal to a classical triple constant $c_{u',v}^{w',0}(y;z)$ with $u' = us_{m_1}\cdots s_{m_{|d|}}$ and $w' = ws_{m_1}\cdots s_{m_{|d|}}$, for a sequence of indices determined by (u,w,d).

Proof. Let $I = \{m : \langle d\rangle_m > 0\}$. If I were empty, d would be a nonpositive combination of the fundamental coweights, hence of the simple coroots, contradicting $d \neq 0$ effective; so $I \neq \emptyset$. Suppose first that $|I| \ge 2$ and choose $m \in I$ with $m \neq k$, so that $\mathrm{sgn}_m(v) = 0$. By Theorem 5.3(1), $c_{u,v}^{w,d}(y;z) = 0$ unless $\mathrm{sgn}_m(w) + \langle d\rangle_m \le \mathrm{sgn}_m(u) \le 1$, which forces $\mathrm{sgn}_m(u) = 1$, $\mathrm{sgn}_m(w) = 0$ and $\langle d\rangle_m = 1$. Then the quadruple (u, vs_m, ws_m, d) satisfies $\mathrm{sgn}_m(ws_m) + \langle d\rangle_m = 2 = \mathrm{sgn}_m(u) + \mathrm{sgn}_m(vs_m)$, and Theorem 5.3(2) applied to it gives, on the one hand, $c_{u,vs_m}^{ws_m,d} = c_{us_m,v}^{ws_m,d-\delta_m}$, and on the other hand, since $\mathrm{sgn}_m(ws_m) = 1$, $c_{u,vs_m}^{ws_m,d} = c_{u,v}^{w,d}$ by the second equality

with the roles of u and v exchanged. Suppose next that $I = \{m\}$. Then $\langle d\rangle_m \geq 2$ by [22, Lemma 2.10], and Theorem 5.3(1) forces $m = k$, $\mathrm{sgn}_k(u) = 1$, $\mathrm{sgn}_k(w) = 0$ and $\langle d\rangle_k = 2$, whereupon Theorem 5.3(2) gives $c_{u,v}^{w,d} = c_{us_k,v}^{ws_k,d-\delta_k}$. In either case $|d|$ drops by one, and the last assertion follows by induction on $|d|$. □

Remark 5.5. Proposition 5.4 with $v = c[p,k]$ expresses every coefficient of a Pieri product as a classical triple Pieri coefficient, which [33, Theorem 7.1] describes positively. At $z = y$ this is exactly the route by which Huang and Li obtain their equivariant quantum Pieri rule [17, Theorem 3.10], with Robinson's rule [31] in the role of the classical formula; the same route with the triple formula in that role yields a triple quantum Pieri rule of their form, in which the terms are indexed by a degree d, the twists (u', w') of (u, w) that d determines, and classical chains from u' to w'. Theorem 3.4 is a statement of a different form: its terms are indexed by chains in the quantum Bruhat graph from u itself, and the degree and the weight are read off from the chain. We have checked numerically, for $v = d_2[2]$ and all $u \in S_4$, that the recursion of the proposition carries each of the 125 coefficients of positive degree in q of $\widetilde{\mathfrak{S}}_u(x;y)\,\widetilde{\mathfrak{S}}_v(x;z)$ to the predicted classical triple constant.

6. Proof of the Pieri formula

Throughout this section permutations act on positions: $(vt_{ab})(a) = v(b)$, $(vt_{ab})(b) = v(a)$, and $(vt_{ab})(r) = v(r)$ otherwise. A cycle is written $(a_p, \ldots, a_1, b)$ with the meaning $a_p \mapsto a_{p-1} \mapsto \cdots \mapsto a_1 \mapsto b \mapsto a_p$; when $b = \max\{a_1, \ldots, a_p, b\}$ we call b the *top* of the cycle and refer to

$$(a_p, \ldots, a_1, b) = t_{a_1 b}\, t_{a_2 b} \cdots t_{a_p b} \tag{6.1}$$

as its *pivot factorization*. (Indeed, the right-hand side sends $a_p \mapsto b \mapsto a_{p-1}$, $b \mapsto a_p$ and $a_1 \mapsto b$.)

6.1. The single-variable Monk formula.

Lemma 6.1. *Let $u \in S_\infty$ and let $i, j \geq 1$. Then*

$$(x_i - z_j)\,\widetilde{\mathfrak{S}}_u(x;y) = (y_{u(i)} - z_j)\,\widetilde{\mathfrak{S}}_u(x;y) + \sum_{\substack{p \neq i \\ \ell(ut_{ip}) = \ell(u)+1}} \mathrm{sign}(p-i)\,\widetilde{\mathfrak{S}}_{ut_{ip}}(x;y)$$

$$+ \sum_{\substack{p \neq i \\ \ell(ut_{ip}) = \ell(u)-2|p-i|+1}} \mathrm{sign}(p-i) q_{ip}\,\widetilde{\mathfrak{S}}_{ut_{ip}}(x;y)$$

where q_{ip} stands for $q_{\min(i,p)\max(i,p)}$.

Proof. The quantum double Schubert polynomials represent the Schubert classes in the torus-equivariant quantum cohomology of the flag manifold [21], so they satisfy the equivariant quantum Chevalley formula of Mihalcea [26, Theorem 1] (see [10] for a proof via curve neighborhoods; at $y = 0$ it is the quantum Monk formula of [13], for which Buch [4] gave a direct geometric proof). In type A, with $\widetilde{\mathfrak{S}}_{s_m}(x;y) = \sum_{a \leq m}(x_a - y_a)$, that formula reads

$$\widetilde{\mathfrak{S}}_{s_m}(x;y)\,\widetilde{\mathfrak{S}}_u(x;y) = \sum_{a \leq m} (y_{u(a)} - y_a)\,\widetilde{\mathfrak{S}}_u(x;y) + \sum_{\substack{a \leq m < b \\ \ell(ut_{ab}) = \ell(u)+1}} \widetilde{\mathfrak{S}}_{ut_{ab}}(x;y) + \sum_{\substack{a \leq m < b \\ \ell(ut_{ab}) = \ell(u)-2(b-a)+1}} q_{ab}\,\widetilde{\mathfrak{S}}_{ut_{ab}}(x;y),$$

the diagonal coefficient being the localization of $\widetilde{\mathfrak{S}}_{s_m}(x;y)$ at u, and the two sums ranging over the classical and quantum covers of u in the quantum Bruhat graph (for $\alpha = e_a - e_b$ one has $\langle \omega_m, \alpha^\vee\rangle = 1$ exactly when $a \leq m < b$, and $\langle 2\rho, \alpha^\vee\rangle = 2(b-a)$). By stability we may work in S_n for n large. Now

$$x_i - z_j = \widetilde{\mathfrak{S}}_{s_i}(x;y) - \widetilde{\mathfrak{S}}_{s_{i-1}}(x;y) + y_i - z_j,$$

with the term $\widetilde{\mathfrak{S}}_{s_{i-1}}$ omitted when $i = 1$. Subtracting the two Chevalley formulas, the diagonal coefficients combine to $(y_{u(i)} - y_i) + (y_i - z_j) = y_{u(i)} - z_j$, and the reflections t_{ab} with $a < i < b$ occur in both sums with the same coefficient and cancel. What remains are the reflections t_{ib} with $b > i$ (from $\widetilde{\mathfrak{S}}_{s_i}$, with sign $+1$) and t_{ai} with $a < i$ (from $\widetilde{\mathfrak{S}}_{s_{i-1}}$, with sign -1), each carrying the factor 1 or q_{ab} according to whether the cover is classical or quantum. This is the asserted formula. □

6.2. **Covers and tight cycles.** The following classical count is the only fact about lengths that we shall use.

Lemma 6.2. *Let $v \in S_\infty$ and $a < b$. If $v(a) < v(b)$ then*

$$\ell(vt_{ab}) = \ell(v) + 1 + 2\,\#\{r \mid a < r < b,\ v(a) < v(r) < v(b)\},$$

and if $v(a) > v(b)$ then

$$\ell(vt_{ab}) = \ell(v) - 1 - 2\,\#\{r \mid a < r < b,\ v(b) < v(r) < v(a)\}.$$

In particular $\ell(vt_{ab}) \geq \ell(v) + 1$ if $v(a) < v(b)$, and $\ell(vt_{ab}) \geq \ell(v) - (2(b-a) - 1)$ always.

Proof. Passing from v to vt_{ab} changes the inversion status of the pair of positions (a, b), and of the pairs (a, r), (r, b) with $a < r < b$ exactly when $v(r)$ lies strictly between $v(a)$ and $v(b)$ (in which case both pairs change); all other pairs are unaffected. □

Definition 6.3 (Covers, $\lhd$, κ)**.** For $v \in S_\infty$ and $a < b$ we write $v \lhd vt_{ab}$, and say that t_{ab} is a *cover* of v, if either $\ell(vt_{ab}) = \ell(v) + 1$ (a *classical* cover) or $\ell(vt_{ab}) = \ell(v) - 2(b-a) + 1$ (a *quantum* cover). We set $\kappa(v, t_{ab}) = 1$ in the first case and $\kappa(v, t_{ab}) = q_{ab}$ in the second. By Lemma 6.2, t_{ab} is a classical cover of v if and only if $v(a) < v(b)$ and no r with $a < r < b$ satisfies $v(a) < v(r) < v(b)$, and a quantum cover if and only if $v(a) > v(b)$ and every r with $a < r < b$ satisfies $v(b) < v(r) < v(a)$. These are the covers of the quantum Bruhat graph: the classical covers are the terms of Monk's formula and the quantum covers the terms carrying q_{ab} in the quantum Chevalley formula [14, 4]; condition (4) of Definition 3.1 says precisely that each step of a witnessing sequence is a cover (Remark 2.2).

It is convenient to separate the two values being exchanged from the values lying between them. For distinct numbers $\alpha \neq \beta$ and a finite set J of numbers not containing α, β, say that the pair (α, β) is *J-admissible* if either $\alpha < \beta$ and $J \cap (\alpha, \beta) = \emptyset$, or $\alpha > \beta$ and $J \subseteq (\beta, \alpha)$. Thus t_{ab} is a cover of v if and only if $(v(a), v(b))$ is J-admissible for $J = \{v(r) \mid a < r < b\}$; the cover is classical when $\alpha < \beta$ and quantum when $\alpha > \beta$.

Lemma 6.4. *Let α, β, γ be distinct numbers and I, J, J' finite sets of numbers avoiding them.*

(i) *(α, β) is $(J \cup J')$-admissible if and only if it is J-admissible and J'-admissible. In particular $J' \subseteq J$ and J-admissibility imply J'-admissibility.*
(ii) *(Composition.) If $(\alpha_1, \alpha_0), (\alpha_2, \alpha_1), \ldots, (\alpha_n, \alpha_{n-1})$ are I-admissible then so is (α_n, α_0).*
(iii) *(Triangle.) If (γ, β) is I-admissible and (α, γ) is $(J \cup \{\beta\} \cup I)$-admissible, then (α, β) is I-admissible and (α, γ) is J-admissible.*

Proof. (i) is immediate from the definition. For (ii), if $I = \emptyset$ there is nothing to prove. Otherwise I cuts the complement of I into intervals ("blocks"), among them $V_- = (-\infty, \min I)$ and $V_+ = (\max I, \infty)$. An I-admissible pair (α', α) with $\alpha' < \alpha$ has α, α' in the same block, and one with $\alpha' > \alpha$ has $\alpha \in V_-$ and $\alpha' \in V_+$. Hence along the chain the values descend within a block, except for at most one jump from V_- to V_+, after which they remain in V_+. Either $\alpha_n < \alpha_0$ lie in a common block, or $\alpha_0 \in V_-$ and $\alpha_n \in V_+$; in both cases (α_n, α_0) is I-admissible. For (iii), the second assertion is (i). For the first there are four cases. If $\gamma < \beta$ and $\alpha < \gamma$: then $\alpha < \beta$ and $I \cap (\alpha, \beta) \subseteq (I \cap (\alpha, \gamma)) \cup (I \cap [\gamma, \beta)) = \emptyset$. If $\gamma < \beta$ and $\alpha > \gamma$: then $\beta \in (\gamma, \alpha)$ gives $\beta < \alpha$, and $I \subseteq (\gamma, \alpha)$ together with $I \cap (\gamma, \beta) = \emptyset$ gives $I \subseteq (\beta, \alpha)$. If $\gamma > \beta$ and $\alpha < \gamma$: then $\beta \notin (\alpha, \gamma)$ forces $\beta < \alpha$, and $I \subseteq (\beta, \gamma)$ with $I \cap (\alpha, \gamma) = \emptyset$ gives $I \subseteq (\beta, \alpha)$. The case $\gamma > \beta$, $\alpha > \gamma$ cannot occur, since it would require $\beta \in (\gamma, \alpha)$. □

Definition 6.5 (Descents of a cycle, $\mathrm{desc}_u(c)$)**.** Let c be a cycle and suppose $u \in S_\infty$. Write c as

$$(a_1, \ldots, a_n)$$

where $a_n > a_i$ for all $i < n$ (this is a unique way to express the cycle since the a_j are distinct). Then for any i with $1 \leq i < n$ we say that a_i is a *descent* of c with respect to u if $u(a_i) > u(a_{i+1})$. We denote the set of all descents of c with respect to u by $\mathrm{desc}_u(c)$.

In the notation $c = (a_p, \ldots, a_1, b)$ of (6.1), with $a_0 := b$, we have $a_j \in \mathrm{desc}_u(c)$ if and only if $u(a_j) > u(a_{j-1})$; the top b is never a descent.

Definition 6.6 ($L_u(\sigma)$, $\mathbf{q}_u(\sigma)$, m-shaped, u-tight)**.** Let $u, \sigma \in S_\infty$ and let $c_1, \dots, c_n$ be the nontrivial cycles of σ, with $c_i = (a_{i,p_i}, \dots, a_{i,1}, b_i)$ and $b_i = \max c_i$. Put

$$L_u(\sigma) = \sum_{i=1}^{n} \Big(p_i - 2 \sum_{a \in \mathrm{desc}_u(c_i)} (b_i - a) \Big), \qquad \mathbf{q}_u(\sigma) = \prod_{i=1}^{n} \prod_{a \in \mathrm{desc}_u(c_i)} q_{ab_i}.$$

Both depend only on the restriction of u to $\mathrm{supp}(\sigma)$, and $\deg \mathbf{q}_u(\sigma) = \sum_i p_i - L_u(\sigma)$. We say that σ is *m-shaped* if every nontrivial cycle of σ contains exactly one element greater than m, and that σ is *u-tight* if

$$\ell(u\sigma) - \ell(u) = L_u(\sigma).$$

A *pivot chain* of σ is a sequence $t_1, \dots, t_N$ ($N = \sum_i p_i$) of transpositions obtained by interleaving the pivot factorizations (6.1) of $c_1, \dots, c_n$ in any manner that preserves the order of the factors within each cycle; since the cycles are disjoint, $t_1 \cdots t_N = \sigma$.

Lemma 6.7 (Tightness)**.** *Let $u, \sigma \in S_\infty$, let $t_1, \dots, t_N$ be a pivot chain of σ, and put $u_0 = u$, $u_j = u_{j-1} t_j$.*

(i) *If $t_j = t_{a_{i,l} b_i}$ then $u_{j-1}(a_{i,l}) = u(a_{i,l})$ and $u_{j-1}(b_i) = u(a_{i,l-1})$ (with $a_{i,0} = b_i$). Thus the j-th step exchanges the values $u(a_{i,l})$ and $u(a_{i,l-1})$, and $a_{i,l} \in \mathrm{desc}_u(c_i)$ exactly when the larger of the two sits at the smaller position $a_{i,l}$ before the step.*

(ii) *$\ell(u\sigma) - \ell(u) \geq L_u(\sigma)$.*

(iii) *σ is u-tight if and only if $u_{j-1} \lhd u_j$ for every j. In that case the j-th step is a classical cover if $a_{i,l} \notin \mathrm{desc}_u(c_i)$ and a quantum cover if $a_{i,l} \in \mathrm{desc}_u(c_i)$, and $\prod_j \kappa(u_{j-1}, t_j) = \mathbf{q}_u(\sigma)$.*

(iv) *If $\sigma = \rho\tau$ where ρ and τ are the products of two complementary sets of cycles of σ, then σ is u-tight if and only if ρ is u-tight and τ is $u\rho$-tight. Moreover $L_u(\tau) = L_{u\rho}(\tau)$ and $\mathbf{q}_u(\tau) = \mathbf{q}_{u\rho}(\tau)$.*

(v) *If σ is u-tight then so is $t_1 \cdots t_j$ for every j, and u_j is obtained from u by a sequence of covers.*

Proof. (i) The position $a_{i,l}$ is moved only by the factor $t_{a_{i,l} b_i}$, so before that factor is applied it carries $u(a_{i,l})$; the position b_i is moved only by the factors of c_i, and after $t_{a_{i,l-1} b_i}$ it carries $u(a_{i,l-1})$ (and $u(b_i)$ before any factor of c_i). For (ii) and (iii), by (i) and Lemma 6.2 the j-th step increases the length by at least 1 if $a_{i,l} \notin \mathrm{desc}_u(c_i)$ and by at least $-(2(b_i - a_{i,l}) - 1)$ if $a_{i,l} \in \mathrm{desc}_u(c_i)$; in both cases equality holds exactly when t_j is a cover of u_{j-1}, classical in the first case and quantum in the second (a step with $u_{j-1}(a_{i,l}) < u_{j-1}(b_i)$ increases the length and so cannot be a quantum cover, and symmetrically). Summing over j gives (ii), and shows that equality in (ii) holds exactly when every step is a cover; the product of the κ's is then the product of $q_{a_{i,l} b_i}$ over the descents, which is $\mathbf{q}_u(\sigma)$. For (iv), $\ell(u\rho\tau) - \ell(u) = (\ell(u\rho) - \ell(u)) + (\ell(u\rho\tau) - \ell(u\rho)) \geq L_u(\rho) + L_{u\rho}(\tau) = L_u(\rho) + L_u(\tau) = L_u(\sigma)$, where $L_{u\rho}(\tau) = L_u(\tau)$ because $u\rho$ and u agree on $\mathrm{supp}(\tau)$; equality holds if and only if both inequalities are equalities. (v) The partial product $t_1 \cdots t_j$ is again a product of disjoint cycles of the form $(a_{i,l}, \dots, a_{i,1}, b_i)$ (a cycle of σ truncated after its first l factors), and $t_1, \dots, t_j$ is a pivot chain of it, all of whose steps are covers by (iii). □

With this language the cycle description of $\overset{k}{\rightsquigarrow}$ becomes a statement in which the chain disappears entirely.

Lemma 6.8. *Suppose $u, w \in S_\infty$ and let $k \geq 0$ be an integer, and put $\pi = u^{-1} w$. Then $u \overset{k}{\rightsquigarrow} w$ if and only if π is k-shaped and u-tight. In that case the pivot chain of π in which the cycles are taken in order of decreasing tops is a sequence $t_{a_1 b_1}, \dots, t_{a_p b_p}$ as in Definition 3.1, every pivot chain of π is a chain of covers, and*

$$q_k(u, w) = \mathbf{q}_u(\pi).$$

In particular $q_k(u, w)$ does not depend on the witnessing sequence, and $\ell(u, w) + \deg q_k(u, w)$ equals the sum of the lengths of the cycles of π, each diminished by one.

Proof. Suppose first that $t_{a_1 b_1}, \dots, t_{a_p b_p}$ satisfy conditions (1)–(5) of Definition 3.1. The b_i are weakly decreasing, so the factors sharing a common value b of b_i are consecutive, and by (6.1) their product $t_{a_i b} \cdots t_{a_{i+r} b}$ is the cycle $(a_{i+r}, \dots, a_i, b)$, whose top is b because $a_l \leq k < b$. Cycles belonging to distinct values of b are disjoint, since the a_i are pairwise distinct and no a_i equals any b_j. Hence π is the product of these cycles, each of which contains exactly one element greater than k, namely its top: π is k-shaped. The given sequence is a pivot chain of π, and condition (4) says that each of its steps is a cover; so π is u-tight by Lemma 6.7(iii), and the recursive definition of $q_k(u, w)$ assigns to each step exactly the factor κ, whence $q_k(u, w) = \mathbf{q}_u(\pi)$.

Conversely, suppose π is k-shaped and u-tight. Order the nontrivial cycles of π by decreasing tops and take the corresponding pivot chain. The top of each cycle is its unique element exceeding k, so condition (1) holds;

(2) holds because the cycles are disjoint and the a's within a cycle are distinct; (3) holds by the chosen order; (4) holds by Lemma 6.7(iii); and (5) is $t_1 \cdots t_N = \pi$. The final assertion is $\deg \mathbf{q}_u(\pi) = \sum_i p_i - L_u(\pi)$. □

Corollary 6.9 (Cycle formula for the quantum weight)**.** *Suppose $u \overset{k}{\rightsquigarrow} w$, and write the nontrivial cycles in the decomposition of $u^{-1}w$ as $c_i = (a_{i,p_i}, \ldots, a_{i,1}, b_i)$, as in Lemma 6.8. Then*

$$q_k(u,w) = \prod_i \prod_{a \in \mathrm{desc}_u(c_i)} q_{ab_i}.$$

Equivalently, the exponent of q_r in $q_k(u,w)$ is

$$\#\{(i,a) : a \in \mathrm{desc}_u(c_i),\ a \le r < b_i\}.$$

Moreover, $q_k(u,w)$ is multiplicative over the cycles of $u^{-1}w$, and the factor attached to a cycle depends only on the values of u on that cycle.

Proof. This is $q_k(u,w) = \mathbf{q}_u(\pi)$ together with $q_{ab} = q_a \cdots q_{b-1}$. □

We next show that a tight cycle is very constrained: the values of u along it, read cyclically, decrease except at exactly one place. For a cycle $c = (a_p, \ldots, a_1, b)$ with top b and a permutation u we write

$$\beta_0 = u(b), \qquad \beta_j = u(a_j) \quad (1 \le j \le p),$$

so that $a_j \in \mathrm{desc}_u(c)$ if and only if $\beta_j > \beta_{j-1}$, and we put $u_j = u\, t_{a_1 b} \cdots t_{a_j b}$. By Lemma 6.7(i), the j-th step of the pivot chain exchanges β_j (at a_j) with β_{j-1} (at b); moreover $u_m(a_l) = \beta_{l-1}$ for $l \le m$ and $u_m(a_l) = \beta_l$ for $l > m$. If c is u-tight, Definition 6.3 gives, for every j and every position r with $a_j < r < b$:

$$\text{if } \beta_j < \beta_{j-1} \text{ (classical step): } u_{j-1}(r) \notin (\beta_j, \beta_{j-1}); \tag{6.2}$$

$$\text{if } \beta_j > \beta_{j-1} \text{ (quantum step): } u_{j-1}(r) \in (\beta_{j-1}, \beta_j). \tag{6.3}$$

Lemma 6.10. *Let $c = (a_p, \ldots, a_1, b)$ be a cycle with top b which is u-tight. Then $\mathrm{desc}_u(c)$ has at most one element. In particular this applies to every cycle of $u^{-1}w$ whenever $u \overset{k}{\rightsquigarrow} w$.*

Proof. Suppose $i < j$ are two descents (quantum steps) with no descent strictly between them, so that $\beta_i > \beta_{i-1}$, $\beta_j > \beta_{j-1}$ and $\beta_i > \beta_{i+1} > \cdots > \beta_{j-1}$.

If $a_j > a_i$, then (6.3) for the step i and $r = a_j$ gives $\beta_j = u_{i-1}(a_j) \in (\beta_{i-1}, \beta_i)$, so $\beta_{j-1} < \beta_j < \beta_i$. Hence $j \ge i+2$, and there is m with $i < m < j$ and $\beta_m < \beta_j < \beta_{m-1}$. The step m is classical. If $a_j > a_m$, then (6.2) for the step m and $r = a_j$ is violated by $u_{m-1}(a_j) = \beta_j$. So $a_m > a_j$, and then (6.3) for the step j and $r = a_m$ requires $u_{j-1}(a_m) = \beta_{m-1} \in (\beta_{j-1}, \beta_j)$, contradicting $\beta_{m-1} > \beta_j$.

If $a_j < a_i$, then (6.3) for the step j and $r = a_i$ gives $u_{j-1}(a_i) = \beta_{i-1} \in (\beta_{j-1}, \beta_j)$, so $\beta_{j-1} < \beta_{i-1} < \beta_i$. Hence again $j \ge i+2$ and there is m with $i < m < j$ and $\beta_m < \beta_{i-1} < \beta_{m-1}$. If $a_i > a_m$, (6.2) for the step m and $r = a_i$ is violated by $u_{m-1}(a_i) = \beta_{i-1}$. So $a_m > a_i$, and (6.3) for the step i and $r = a_m$ requires $u_{i-1}(a_m) = \beta_m \in (\beta_{i-1}, \beta_i)$, contradicting $\beta_m < \beta_{i-1}$. □

Definition 6.11 (Redefinition of $\mathrm{desc}_u(c)$)**.** If c is a tight cycle, we redefine $\mathrm{desc}_u(c)$ to be the unique element of the previously defined set $\mathrm{desc}_u(c)$ if it exists, or otherwise 0.

Lemma 6.12. *Let $c = (a_p, \ldots, a_1, b)$ be a cycle with top b which is u-tight, and suppose $\mathrm{desc}_u(c) = a_q$ with $q \ge 1$. Then*

$$\beta_q > \beta_{q+1} > \cdots > \beta_p > \beta_0 > \beta_1 > \cdots > \beta_{q-1}.$$

In particular $u(a_j) > u(b)$ for all $j \ge q$.

Proof. All steps other than the q-th are classical, which gives the two decreasing runs; it remains to prove $\beta_l > \beta_0$ for $l \ge q$.

First let $l = q$. For $l' < q$ with $a_{l'} < a_q$, (6.2) for the step l' and $r = a_q$ gives $\beta_q = u_{l'-1}(a_q) \notin (\beta_{l'}, \beta_{l'-1})$. If some $l' < q$ has $a_{l'} > a_q$, let l^* be the least such; then (6.3) for the step q and $r = a_{l^*}$ gives $u_{q-1}(a_{l^*}) = \beta_{l^*-1} \in (\beta_{q-1}, \beta_q)$, so $\beta_q > \beta_{l^*-1}$, and descending induction on $l' < l^*$ using $\beta_q \notin (\beta_{l'}, \beta_{l'-1})$ yields $\beta_q > \beta_{l'-1}$ for all $l' \le l^*$, in particular $\beta_q > \beta_0$. If no such l' exists, the same induction starting from $\beta_q > \beta_{q-1}$ gives $\beta_q > \beta_0$.

Now let $l > q$ and assume $\beta_{l-1} > \beta_0$ (and $\beta_l < \beta_{l-1}$). Suppose, for a contradiction, that $\beta_l < \beta_0$. We claim that $a_{l'} < a_l$ and $\beta_l < \beta_{l'}$ for $1 \le l' \le q-1$. Indeed, assume $\beta_l < \beta_{l'-1}$ (true for $l' = 1$). If $a_{l'} > a_l$, then (6.2)

for the step l and $r = a_{l'}$ requires $u_{l-1}(a_{l'}) = \beta_{l'-1} \notin (\beta_l, \beta_{l-1})$, which is false as $\beta_l < \beta_{l'-1} \le \beta_0 < \beta_{l-1}$. So $a_{l'} < a_l$, and (6.2) for the step l' and $r = a_l$ gives $\beta_l \notin (\beta_{l'}, \beta_{l'-1})$, whence $\beta_l < \beta_{l'}$. The claim yields $\beta_l < \beta_{q-1}$. Finally consider the quantum step q: if $a_q < a_l$, (6.3) with $r = a_l$ requires $\beta_l \in (\beta_{q-1}, \beta_q)$, contradicting $\beta_l < \beta_{q-1}$; if $a_q > a_l$, (6.2) for the step l with $r = a_q$ requires $u_{l-1}(a_q) = \beta_{q-1} \notin (\beta_l, \beta_{l-1})$, contradicting $\beta_l < \beta_{q-1} < \beta_0 < \beta_{l-1}$. Hence $\beta_l > \beta_0$, and the induction is complete. □

Corollary 6.13 (Cyclic monotonicity)**.** *Let $c = (a_p, \dots, a_1, b)$ be a u-tight cycle with top b. Then the cyclic sequence $(\beta_0, \beta_1, \dots, \beta_p)$ has exactly one cyclic ascent: either $\beta_0 > \beta_1 > \cdots > \beta_p$ (and $\operatorname{desc}_u(c) = 0$), or $\operatorname{desc}_u(c) = a_q$ with $q \ge 1$ and $\beta_q > \cdots > \beta_p > \beta_0 > \cdots > \beta_{q-1}$. Consequently*

$$L_u(c) = \begin{cases} p & \text{if the ascent is at the wrap } (\beta_p, \beta_0), \\ p - 2(b - a_q) & \text{if the ascent is } (\beta_{q-1}, \beta_q), \end{cases} \qquad \mathbf{q}_u(c) = \begin{cases} 1, \\ q_{a_q b}. \end{cases}$$

We shall repeatedly use the following elementary observation about such sequences. Call a cyclic sequence of distinct numbers *unimodal* if it has exactly one cyclic ascent, i.e. if it is a cyclic rotation of a decreasing sequence. Every cyclically consecutive block of a unimodal sequence is unimodal, and so is any subsequence. If Γ is unimodal and $\eta \notin \Gamma$, there is exactly one *gap* of Γ (a pair of cyclically consecutive entries) into which η can be inserted so that the result is again unimodal, namely: the descent gap (γ, γ') with $\gamma > \eta > \gamma'$ if $\min\Gamma < \eta < \max\Gamma$, and the ascent gap $(\min\Gamma, \max\Gamma)$ if $\eta < \min\Gamma$ or $\eta > \max\Gamma$. We call it the *insertion gap* of η in Γ.

6.3. **The inductive step: Monk neighbors at position k.** We organize the inductive step of Theorem 3.4 around the position k. The following lemma records two facts about the interaction of $\overset{k}{\rightsquigarrow}$ and $\overset{k-1}{\rightsquigarrow}$ at position k.

Lemma 6.14. *Suppose $u, w \in S_\infty$ and let $k > 1$ be an integer.*

(i) *If $u(k) = w(k)$, then $u \overset{k}{\rightsquigarrow} w$ if and only if $u \overset{k-1}{\rightsquigarrow} w$, and in that case $q_k(u, w) = q_{k-1}(u, w)$ and $\operatorname{fix}_k(u, w) = \operatorname{fix}_{k-1}(u, w) \cup \{u(k)\}$.*

(ii) *If $u(k) \ne w(k)$ and $u \overset{k}{\rightsquigarrow} w$, then $u \overset{k-1}{\not\rightsquigarrow} w$.*

Proof. Let $\pi = u^{-1}w$. If $u(k) = w(k)$ then k is a fixed point of π, so no nontrivial cycle of π contains k, and such a cycle has exactly one element greater than k if and only if it has exactly one element greater than $k-1$. Thus π is k-shaped if and only if it is $(k-1)$-shaped, while tightness and $\mathbf{q}_u(\pi)$ do not involve k at all; (i) follows from Lemma 6.8. For (ii), the cycle of π containing k is nontrivial and, π being k-shaped, contains an element greater than k; so it contains two elements greater than $k-1$ and π is not $(k-1)$-shaped. □

For the rest of this section we fix $u, w \in S_\infty$ and $k > 1$, and write

$$\pi = u^{-1}w, \qquad C = \text{the cycle of } \pi \text{ containing } k \ (C = \{k\} \text{ if } \pi(k) = k), \qquad B = \max C.$$

For $m \ge 0$ we set $c_m(u, w) = q_m(u, w)\mathbf{E}(y_{\operatorname{fix}_m(u,w)}; z)$ if $u \overset{m}{\rightsquigarrow} w$ and $c_m(u, w) = 0$ otherwise. The *Monk summands* at w are indexed by the reflections $t \in \{t_{ak} \mid a < k\} \cup \{t_{kb} \mid b > k\}$: writing $v = wt$ and $\sigma = u^{-1}v = \pi t$, the summand of t is

$$\operatorname{sign}(t)\, \kappa(v, t)\, c_{k-1}(u, v), \qquad \operatorname{sign}(t_{ak}) = -1,\ \operatorname{sign}(t_{kb}) = +1,$$

where the summand is understood to be 0 unless $v \lhd w$. (These are precisely the terms of (6.8) and (6.9) below.) The summand of t is therefore nonzero if and only if $v \lhd w$ and $u \overset{k-1}{\rightsquigarrow} v$, that is, σ is $(k-1)$-shaped and u-tight.

Lemma 6.15 (Classification of the Monk neighbors)**.** *Suppose the Monk summand of t is nonzero, and let $P = \operatorname{fix}_{k-1}(u, w)$. Then every nontrivial cycle of π not meeting t has exactly one element greater than k, and exactly one of the following holds.*

(U) *$t = t_{kB}$ with $B > k$; C contains exactly two elements greater than $k-1$, namely k and B; and $\operatorname{fix}_{k-1}(u, v) = P$. Writing $C = (a_p, \dots, a_1, B)$ with $a_j = k$, the cycles of σ are those of π other than C together with*

$$C' = (a_{j-1}, \dots, a_1, B) \quad \text{and} \quad C'' = (a_p, \dots, a_{j+1}, k).$$

(S) *$t = t_{ak}$ with $a \in C$ and $B > k$; C contains exactly two elements greater than $k-1$, namely k and B; in the notation of (U), $a = a_m$ for some $m > j$;* $\mathrm{fix}_{k-1}(u,v) = P$*; and the cycles of σ are those of π other than C together with*

$$C'_m = (a_p, \dots, a_{m+1}, a_m, a_{j-1}, \dots, a_1, B) \quad \text{and} \quad C''_m = (a_{m-1}, \dots, a_{j+1}, k).$$

(P) *$t = t_{ak}$ with $a \in C$ and $B = k$; writing $C = (a_p, \dots, a_1, k)$ we have $a = a_p = \pi(k) = u^{-1}(w(k))$;* $\mathrm{fix}_{k-1}(u,v) = P \sqcup \{w(k)\}$*; and the cycles of σ are those of π other than C together with $C_- = (a_{p-1}, \dots, a_1, k)$ (trivial if $p = 1$).*

(M) *$t = t_{ak}$ with $a \notin C$; then $B = k$, the cycle D of π containing a (possibly $D = \{a\}$) satisfies $D \subseteq [1, k-1]$, and* $\mathrm{fix}_{k-1}(u,v) = P \setminus \{u(a)\}$*, which equals P unless $D = \{a\}$. The cycles of σ are those of π other than C and D together with the single cycle E with* $\mathrm{supp}(E) = \mathrm{supp}(C) \cup \mathrm{supp}(D)$ *obtained by setting $E(a) = \pi(k)$, $E(k) = \pi(a)$ and $E = \pi$ elsewhere.*

Proof. A cycle of π disjoint from $\mathrm{supp}(t)$ is a cycle of σ, hence has exactly one element greater than $k-1$, and since it does not contain k this element is greater than k. Recall that right multiplication of a permutation by a transposition t_{xy} merges the cycles of x and y if they are distinct, and otherwise splits their common cycle $(x_1, \dots, x_n)$, $x = x_s$, $y = x_t$, $s < t$, into $(x_t, x_{s+1}, \dots, x_{t-1})$ and $(x_s, x_{t+1}, \dots, x_n, x_1, \dots, x_{s-1})$.

Let $t = t_{kb}$. If $b \notin C$, the cycle of σ through k contains k and $b > k$, which is forbidden. So $b \in C$ and t splits C into a cycle through k and a cycle through b; each must contain exactly one element greater than $k-1$, so $C \cap (k-1, \infty) = \{k, b\}$ and $b = B$. The explicit form of the two cycles is the splitting formula applied to $(x_1, \dots, x_{p+1}) = (a_p, \dots, a_1, B)$ with $x_s = k$, $x_t = B$. Since v agrees with w at all positions less than k, $\mathrm{fix}_{k-1}(u,v) = P$. This is (U).

Let $t = t_{ak}$ with $a \in C$, so t splits C. If $B > k$, the piece not containing k must contain exactly one element greater than $k-1$, which forces $C \cap (k-1, \infty) = \{k, B\}$ and that piece to contain B. With $(x_1, \dots, x_{p+1}) = (a_p, \dots, a_1, B)$, $k = a_j$ and $a = a_m$: if $m < j$ the piece through a_m is $(a_m, a_{j-1}, \dots, a_{m+1})$, which has no element greater than $k-1$ unless it is trivial ($m = j-1$), in which case the other piece $(k, a_{m-1}, \dots, a_1, B, a_p, \dots, a_{j+1})$ contains both k and B; both are forbidden. So $m > j$, and the splitting formula gives C'_m and C''_m. Here $v(a_m) = w(k) = u(a_{j-1}) \neq u(a_m)$ and $w(a_m) \neq u(a_m)$, so a_m is a fixed position neither for v nor for w, and $\mathrm{fix}_{k-1}(u,v) = P$. This is (S). If $B = k$, the piece not containing k has no element greater than $k-1$ and must be trivial, i.e. $\sigma(a) = a$, i.e. $\pi(k) = a$; in the notation $C = (a_p, \dots, a_1, k)$ this is $a = a_p$, and $u(a_p) = u(\pi(k)) = w(k)$. Then σ has the cycle C_-, and $v(a_p) = w(k) = u(a_p)$ makes a_p a new fixed position with value $w(k) \notin P$. This is (P).

Finally let $t = t_{ak}$ with $a \notin C$, so t merges C with the cycle D of a into a cycle E containing k; E may contain no further element greater than $k-1$, so $C \subseteq [1, k]$, i.e. $B = k$, and $D \subseteq [1, k-1]$. We have $v(a) = w(k) = u(\pi(k)) \neq u(a)$ because $\pi(k) \in C \not\ni a$, so a is not a fixed position of v; it is one of w exactly when $D = \{a\}$. All other positions less than k agree. This is (M). □

The case $B > k$: the up-neighbor and the split down-neighbors.

Lemma 6.16 (Up/down pairing)**.** *Let $\psi \in S_\infty$, let $k < B$, let $z_1, \dots, z_s$ ($s \geq 0$) be distinct positions less than k, let $X = (z_s, \dots, z_1, k, B)$ and $w = \psi X$. Consider the following sequences of transpositions, each with product X:*

$$\begin{aligned} F_{\mathrm{std}} &: \ t_{kB},\, t_{z_1 B},\, \dots,\, t_{z_s B} && \text{(the pivot factorization of } X\text{)}, \\ F_{\mathrm{up}} &: \ t_{z_1 k},\, \dots,\, t_{z_s k},\, t_{kB}, && \\ F_m &: \ t_{z_1 k},\, \dots,\, t_{z_{m-1} k},\, t_{z_m B},\, \dots,\, t_{z_s B},\, t_{z_m k} && (1 \leq m \leq s). \end{aligned}$$

Call such a sequence a cover path *(from ψ) if each of its steps is a cover of the permutation reached before it, and define its* weight *to be the product of the κ's of its steps.*

(a) *If F_{std} is a cover path, then F_{up} is a cover path of the same weight, and no F_m is a cover path.*
(b) *If F_{up} is a cover path but F_{std} is not, then exactly one F_m is a cover path, and its weight equals that of F_{up}.*
(c) *If some F_m is a cover path, then so is F_{up}.*

Proof. (The products are X: F_{up} is $(z_s, \dots, z_1, k)\, t_{kB}$, and F_m is $(z_{m-1}, \dots, z_1, k)\, (z_s, \dots, z_m, B)\, t_{z_m k}$; both send $k \mapsto B \mapsto z_s \mapsto \cdots \mapsto z_1 \mapsto k$.) If $s = 0$ then $F_{\mathrm{std}} = F_{\mathrm{up}}$ and there is nothing to prove, so let $s \geq 1$.

Put

$$\beta = \psi(B), \qquad \zeta_0 = \psi(k), \qquad \zeta_i = \psi(z_i)\ (1 \le i \le s), \qquad I = \{\psi(r) \mid k < r < B\}.$$

The positions strictly between k and B are never moved, so I is the set of values found there at every stage of every path. Let $\phi_m = \psi t_{z_1 k} \cdots t_{z_{m-1} k}$ $(1 \le m \le s+1)$; thus $\phi_m(k) = \zeta_{m-1}$, $\phi_m(z_i) = \zeta_{i-1}$ for $i < m$, $\phi_m(z_i) = \zeta_i$ for $i \ge m$, $\phi_m(B) = \beta$, and ϕ_m agrees with ψ elsewhere. Let $J_i = \{\phi_i(r) \mid z_i < r < k\}$. We introduce the predicates

$$\begin{aligned} U_i &: \ (\zeta_i, \zeta_{i-1}) \text{ is } J_i\text{-admissible} && (1 \le i \le s),\\ T_i &: \ (\zeta_i, \beta) \text{ is } I\text{-admissible} && (0 \le i \le s),\\ A_i &: \ (\zeta_i, \zeta_{i-1}) \text{ is } (J_i \cup \{\beta\} \cup I)\text{-admissible} && (1 \le i \le s). \end{aligned}$$

U_i says that the i-th step of F_{up} (from ϕ_i) is a cover, and T_s that its last step is; so F_{up} is a cover path if and only if $U_1, \dots, U_s, T_s$ hold. For $1 \le m \le s+1$ let $\mathcal{S}_m$ be the statement that $t_{kB}, t_{z_m B}, \dots, t_{z_s B}$ is a cover path from ϕ_m; thus F_{std} is a cover path if and only if $\mathcal{S}_1$, and $\mathcal{S}_{s+1} = T_s$. The first step of $\mathcal{S}_m$ exchanges ζ_{m-1} at k with β at B across I; the step $t_{z_i B}$ $(i \ge m)$ exchanges ζ_i at z_i with ζ_{i-1} at B across the positions in (z_i, k), which carry the same values as in ϕ_i (the positions z_l, $l < i$, carry ζ_{l-1} and the others their ψ-values), the position k, which carries β, and the positions in (k, B). Hence

$$\mathcal{S}_m \iff T_{m-1} \wedge A_m \wedge \cdots \wedge A_s.$$

Let $\mathcal{D}_m$ $(1 \le m \le s)$ be the statement that $t_{z_m B}, \dots, t_{z_s B}, t_{z_m k}$ is a cover path from ϕ_m; then F_m is a cover path if and only if $U_1, \dots, U_{m-1}$ and $\mathcal{D}_m$ hold. The steps of $\mathcal{D}_m$ are: $D^{(m)}$, exchanging ζ_m at z_m with β at B across $J_m \cup \{\zeta_{m-1}\} \cup I$ (the position k now carries ζ_{m-1}); for $i > m$, $D^{(i)}$, exchanging ζ_i with ζ_{i-1} across $J_i^* \cup I$, where J_i^* consists of the values at the positions of $(z_i, k]$, so that $J_i^* = J_i \cup \{\beta\}$ if $z_m > z_i$ and $J_i^* = J_i \cup \{\zeta_{m-1}\}$ if $z_m < z_i$ (the position z_m carries β and k carries ζ_{m-1}); and D^{last}, exchanging β at z_m with ζ_{m-1} at k across $J^{**} = N_m \cup \{\zeta_{l-1} \mid l > m,\ z_l > z_m\}$, where N_m denotes the set of values carried by the positions of (z_m, k) other than $z_{m+1}, \dots, z_s$ (these values are the same in ϕ_m and at the end of $\mathcal{D}_m$), so that $J_m = N_m \cup \{\zeta_l \mid l > m,\ z_l > z_m\}$. For $i > m$ the set across which $D^{(i)}$ is taken contains J_i, and the sets of $D^{(m)}, \dots, D^{(s)}$ all contain ζ_{m-1}, the value at k.

Step 1: $\mathcal{S}_m \Rightarrow \mathcal{S}_{m+1}$, and $\mathcal{S}_m$ implies $U_m, \dots, U_s, T_s$. By Lemma 6.4(iii) applied with $(\alpha, \gamma, \beta) = (\zeta_m, \zeta_{m-1}, \beta)$ and $J = J_m$, the conjunction $T_{m-1} \wedge A_m$ implies $T_m \wedge U_m$. The first claim follows, and the second by iterating it down to $\mathcal{S}_{s+1} = T_s$, using $A_i \Rightarrow U_i$ (Lemma 6.4(i)). In particular $\mathcal{S}_1$ implies that F_{up} is a cover path.

Step 2: $\mathcal{D}_m$ implies $U_m, \dots, U_s, T_s$. For $i > m$, $D^{(i)}$ gives U_i by Lemma 6.4(i). By the same lemma, $D^{(m)}$ and $D^{(i)}$ $(i > m)$ give that (ζ_m, β) and (ζ_i, ζ_{i-1}) are I-admissible, so (ζ_s, β) is I-admissible by Lemma 6.4(ii): this is T_s. Likewise the pairs $(\zeta_m, \beta), (\zeta_{m+1}, \zeta_m), \dots, (\zeta_s, \zeta_{s-1})$ are $\{\zeta_{m-1}\}$-admissible, which as in the proof of Lemma 6.4(ii) means:

$$\begin{gathered} \text{the values } \beta, \zeta_m, \zeta_{m+1}, \dots, \zeta_s \text{ decrease on each side of } \zeta_{m-1},\\ \text{with at most one jump from below } \zeta_{m-1} \text{ to above it.} \end{gathered} \tag{6.4}$$

For U_m, admissibility of (ζ_m, β) with respect to $\{\zeta_{m-1}\}$ leaves three possibilities: (1) $\zeta_{m-1} > \beta > \zeta_m$; (2) $\zeta_m > \zeta_{m-1} > \beta$; (3) $\beta > \zeta_m > \zeta_{m-1}$. We use $D^{(m)}$ restricted to J_m and D^{last} restricted to N_m, and treat the remaining elements ζ_l $(l > m,\ z_l > z_m)$ of J_m by (6.4). In case (1), $J_m \cap (\zeta_m, \beta) = \emptyset$ and $N_m \cap (\beta, \zeta_{m-1}) = \emptyset$, while by (6.4) each ζ_l $(l > m)$ is either less than ζ_m or greater than ζ_{m-1}; hence $J_m \cap (\zeta_m, \zeta_{m-1}) = \emptyset$. In case (2), $J_m \subseteq (\beta, \zeta_m)$ and $N_m \cap (\beta, \zeta_{m-1}) = \emptyset$ give $N_m \subseteq (\zeta_{m-1}, \zeta_m)$, while by (6.4) the jump occurs at (β, ζ_m) and so $\zeta_{m-1} < \zeta_l < \zeta_m$ for $l > m$; hence $J_m \subseteq (\zeta_{m-1}, \zeta_m)$. In case (3), $J_m \cap (\zeta_m, \beta) = \emptyset$ and $N_m \subseteq (\zeta_{m-1}, \beta)$ give $N_m \subseteq (\zeta_{m-1}, \zeta_m)$, while by (6.4) all of $\beta, \zeta_m, \dots, \zeta_s$ lie above ζ_{m-1} and decrease, so again $\zeta_{m-1} < \zeta_l < \zeta_m$ for $l > m$; hence $J_m \subseteq (\zeta_{m-1}, \zeta_m)$. In each case this is U_m. Together with the prefix $U_1, \dots, U_{m-1}$ this proves (c).

For the remainder assume that F_{up} is a cover path. Then the cycle $(z_s, \dots, z_1, k)$ is ψ-tight, so by Corollary 6.13 the cyclic sequence $\Gamma = (\zeta_0, \zeta_1, \dots, \zeta_s)$ is unimodal; let $\delta \in \{0, \dots, s\}$ be the index of its ascent ($\zeta_\delta > \zeta_{\delta-1}$ if $\delta \ge 1$; $\delta = 0$ means the ascent is the wrap (ζ_s, ζ_0)), and let $g \in \{0, \dots, s\}$ be the insertion gap of β in Γ, where gap $g \ge 1$ is the gap (ζ_{g-1}, ζ_g) and gap 0 is the wrap. Explicitly: if $\min \Gamma < \beta < \max \Gamma$, then g is determined by $\zeta_{g-1} > \beta > \zeta_g$ (indices modulo $s+1$); otherwise $g = \delta$.

Step 3: $\mathcal{S}_m$ implies $m > g$. $\mathcal{S}_m$ says that the cycle $(z_s, \dots, z_m, k, B)$ is ϕ_m-tight, so by Corollary 6.13 the cyclic sequence $(\beta, \zeta_{m-1}, \dots, \zeta_s)$ is unimodal; that is, the wrap gap (ζ_s, ζ_{m-1}) of the block $\Gamma' = (\zeta_{m-1}, \dots, \zeta_s)$

is the insertion gap of β in Γ'. Suppose $1 \le m \le g$. Then ζ_{g-1}, ζ_g both lie in Γ' and are consecutive there. If $\min\Gamma < \beta < \max\Gamma$ then $\zeta_{g-1} > \beta > \zeta_g$, so the insertion gap of β in Γ' is (ζ_{g-1}, ζ_g), which is not the wrap gap of Γ' since $g - 1 \ne s$. If β lies outside $[\min\Gamma, \max\Gamma]$ then $g = \delta$, so $\zeta_g = \max\Gamma$ and $\zeta_{g-1} = \min\Gamma$ lie in Γ', whence β lies outside the range of Γ' and its insertion gap in Γ' is the ascent gap (ζ_{g-1}, ζ_g) of Γ', again not the wrap gap. In both cases $\mathcal{S}_m$ fails.

Step 4: $\mathcal{S}_{g+1}$ holds; hence $\mathcal{S}_m$ holds if and only if $m > g$. We must verify T_g and A_i for $i > g$; by Lemma 6.4(i) we treat the sets J_i, $\{\beta\}$ and I separately. Admissibility of (ζ_i, ζ_{i-1}) with respect to J_i is U_i. With respect to $\{\beta\}$: if $i > g$ and $i \ne \delta$, the gap (ζ_{i-1}, ζ_i) is a descent gap and $\beta \notin (\zeta_i, \zeta_{i-1})$, for otherwise it would be the insertion gap of β; if $i = \delta > g$, then $\beta \in (\min\Gamma, \max\Gamma) = (\zeta_{\delta-1}, \zeta_\delta)$, for otherwise $g = \delta$. With respect to I, and for T_g: the sequence $s_0 = \beta, s_1 = \zeta_g, \dots, s_n = \zeta_s$ is a consecutive block of the unimodal sequence obtained by inserting β into Γ, hence unimodal, and T_s says that (s_n, s_0) is I-admissible. If $s_0 > s_1 > \cdots > s_n$, then T_s is classical, $I \cap (s_n, s_0) = \emptyset$, and every pair (s_i, s_{i-1}) is I-admissible. Otherwise the linear ascent is at some $j \ge 1$, so $s_j > \cdots > s_n > s_0 > \cdots > s_{j-1}$; then T_s is quantum, $I \subseteq (s_0, s_n)$, and the pairs (s_i, s_{i-1}) with $i < j$ lie below s_0, the pair (s_j, s_{j-1}) has $I \subseteq (s_0, s_n) \subseteq (s_{j-1}, s_j)$, and the pairs with $i > j$ lie above s_n; all are I-admissible. The pairs (s_i, s_{i-1}) are exactly (ζ_g, β) and (ζ_i, ζ_{i-1}), $i > g$. This proves $\mathcal{S}_{g+1}$, and the second assertion follows from Steps 1 and 3.

Step 5: $\mathcal{D}_m$ implies $m = g$. Restricting $D^{(m)}$ to $\{\zeta_{m-1}\}$ leaves the three cases (1)–(3) of Step 2. In case (1) the gap (ζ_{m-1}, ζ_m) is a descent gap containing β, so $g = m$. In case (2) the gap is the ascent gap, $m = \delta$, and $\beta < \zeta_{m-1} = \min\Gamma$, so $g = \delta = m$. In case (3), $m = \delta$ and $\beta > \zeta_m = \max\Gamma$, so again $g = m$.

Step 6: if $g \ge 1$ then $\mathcal{D}_g$ holds. Write $a = z_g$. We are in one of the cases (1) $\zeta_{g-1} > \beta > \zeta_g$ ($g \ne \delta$), (2) $g = \delta$ and $\beta < \zeta_{g-1} = \min\Gamma$, (3) $g = \delta$ and $\beta > \zeta_g = \max\Gamma$. For $D^{(g)}$: admissibility of (ζ_g, β) with respect to I is T_g (Step 4); with respect to $\{\zeta_{g-1}\}$ it holds in each of the three cases by inspection; with respect to J_g it follows from U_g: in case (1), $J_g \cap (\zeta_g, \zeta_{g-1}) = \emptyset$ gives $J_g \cap (\zeta_g, \beta) = \emptyset$; in case (2), $J_g \subseteq (\zeta_{g-1}, \zeta_g) \subseteq (\beta, \zeta_g)$; in case (3), $J_g \subseteq (\zeta_{g-1}, \zeta_g)$ is disjoint from (ζ_g, β). For $D^{(i)}$, $i > g$: if $z_g > z_i$ the required statement is A_i (Step 4); if $z_g < z_i$ then $z_g \in (z_i, k)$, so $\zeta_{g-1} = \phi_i(z_g) \in J_i$ already, $J_i^* = J_i$, and the statement follows from U_i and the I-admissibility established in Step 4. For D^{last}, the pair (β, ζ_{g-1}) must be admissible with respect to $N_g \subseteq J_g$ and to the values ζ_{l-1}, $l > g$, $z_l > z_g$: in case (1) it is classical, and $(\beta, \zeta_{g-1}) \subseteq (\zeta_g, \zeta_{g-1})$ contains no element of J_g (by U_g) and no ζ_l (as ζ_{g-1}, ζ_g are consecutive in the decreasing order of Γ); in case (2) it is classical, and (β, ζ_{g-1}) lies below $\min\Gamma$ and below $J_g \subseteq (\zeta_{g-1}, \zeta_g)$; in case (3) it is quantum, and $J_g \subseteq (\zeta_{g-1}, \zeta_g) \subseteq (\zeta_{g-1}, \beta)$ while every ζ_{l-1} with $l > g$ lies in $(\zeta_{g-1}, \zeta_g] \subseteq (\zeta_{g-1}, \beta)$.

Step 7: weights. Suppose F_{up} is a cover path and $g \ge 1$. The two paths F_{up} and F_g share the prefix $t_{z_1 k}, \dots, t_{z_{g-1} k}$; after it, the quantum steps of F_{up} are $t_{z_\delta k}$ (if $\delta \ge g$) and t_{kB} (if $\zeta_s > \beta$), while those of F_g are $t_{z_g B}$ (if $\zeta_g > \beta$), $t_{z_\delta B}$ (if $\delta > g$) and $t_{z_g k}$ (if $\beta > \zeta_{g-1}$). In case (1) with $\delta > g$ we have $\zeta_s > \zeta_0 \ge \zeta_{g-1} > \beta$, and both weights are $q_{z_\delta k} q_{kB} = q_{z_\delta B}$; in case (1) with $\delta < g$ we have $\zeta_s < \zeta_g < \beta$ and both weights are 1; in case (2) both are $q_{z_g k} q_{kB} = q_{z_g B}$; in case (3) both are $q_{z_g k}$. If instead $g = 0$, so that $\mathcal{S}_1$ holds, the quantum steps of F_{std} are t_{kB} (if $\zeta_0 > \beta$) and $t_{z_\delta B}$ (if $\delta \ge 1$); if $\zeta_s > \beta > \zeta_0$ (so $\delta \ge 1$) both F_{std} and F_{up} have weight $q_{z_\delta B}$; if $\beta < \min\Gamma$ (so $\delta = 0$) both have weight q_{kB}; and if $\beta > \max\Gamma$ both have weight 1.

Conclusion. (a): $\mathcal{S}_1$ gives F_{up} (Step 1) with the same weight (Step 7), and $g = 0$ (Step 3), so no $\mathcal{D}_m$ holds (Step 5). (b): if F_{up} is a cover path and $\mathcal{S}_1$ fails, then $g \ge 1$ (Step 4), $\mathcal{D}_g$ holds (Step 6) and no other $\mathcal{D}_m$ does (Step 5); as $U_1, \dots, U_{g-1}$ hold, F_g is the unique F_m which is a cover path, and its weight is that of F_{up} (Step 7). (c) is Step 2. □

Lemma 6.17. *Suppose $u, w \in S_\infty$ and let $k > 1$ be an integer. If $u \overset{k}{\rightsquigarrow} w$ but $u \overset{k-1}{\not\rightsquigarrow} w$, then there is a unique $b > k$ such that $wt_{kb} \lhd w$ and $u \overset{k-1}{\rightsquigarrow} wt_{kb}$, namely $b = B$. Furthermore, in that case there does not exist an $a < k$ such that $wt_{ak} \lhd w$ and $u \overset{k-1}{\rightsquigarrow} wt_{ak}$, and*

$$\kappa(wt_{kB}, t_{kB})\, q_{k-1}(u, wt_{kB}) = q_k(u, w), \qquad \mathrm{fix}_{k-1}(u, wt_{kB}) = \mathrm{fix}_k(u, w).$$

Proof. By Lemma 6.8, π is k-shaped and u-tight but not $(k-1)$-shaped, so $u(k) \ne w(k)$ and $B > k$ (Lemma 6.14). Write $C = (a_p, \dots, a_1, B)$ with $a_j = k$, let ρ be the product of the other nontrivial cycles of π, let $C' = (a_{j-1}, \dots, a_1, B)$, and put $\psi = u\rho C'$ and $X = (a_p, \dots, a_{j+1}, k, B)$, so that $w = \psi X$ and X has the form of Lemma 6.16 with $s = p - j$ and $z_i = a_{j+i}$. Take the pivot chain of π consisting of the pivot factorization of ρ, then that of C; by Lemma 6.7(iii),(v) all its steps are covers, ρ is u-tight, C' is $u\rho$-tight (it is C truncated after $j - 1$ factors), and F_{std} is a cover path from ψ. By Lemma 6.16(a), F_{up} is a cover path of the same weight. Its first s steps form the pivot factorization of $C'' = (a_p, \dots, a_{j+1}, k)$, so $\rho C' C'' = u^{-1} w t_{kB}$ is u-tight

by Lemma 6.7(iii),(iv); it is $(k-1)$-shaped, its cycles being those of ρ (each with exactly one element greater than k), C' (top B, other elements less than k) and C'' (top k, other elements less than k). Hence $u \overset{k-1}{\rightsquigarrow} wt_{kB}$, and the last step of F_{up} says $wt_{kB} \lhd w$. Uniqueness of b is Lemma 6.15(U). The weights are $\mathbf{q}_u(\rho)\mathbf{q}_u(C')$ times the weights of F_{std}, respectively F_{up}, which agree; and $\mathrm{fix}_{k-1}(u, wt_{kB}) = \mathrm{fix}_{k-1}(u,w) = \mathrm{fix}_k(u,w)$ since $u(k) \neq w(k)$.

Suppose finally that $a < k$ satisfies $wt_{ak} \lhd w$ and $u \overset{k-1}{\rightsquigarrow} wt_{ak}$. By Lemma 6.15 we are in case (S): $a = a_m$ with $m > j$, and $\sigma = \rho C'_m C''_m$ is u-tight. Taking the pivot chain of σ in the order: factors of ρ, the first $j-1$ factors of C'_m (which are those of C'), the factors of C''_m, the remaining factors $t_{a_m B}, \dots, t_{a_p B}$ of C'_m, and appending the cover $t_{a_m k}$, we obtain that F_{m-j} is a cover path from ψ, contradicting Lemma 6.16(a). □

The case $B = k$: the peel and the merges.

Lemma 6.18 (Peel)**.** *Let $\omega \in S_\infty$ and let $C = (a_p, \dots, a_1, k)$ $(p \geq 1)$ be a cycle with top k; put $w = \omega C$, $C_- = (a_{p-1}, \dots, a_1, k)$ and $v = wt_{a_p k} = \omega C_-$. Then C is ω-tight if and only if C_- is ω-tight and $v \lhd w$, and in that case $\kappa(v, t_{a_p k})\, \mathbf{q}_\omega(C_-) = \mathbf{q}_\omega(C)$.*

Proof. The pivot factorization of C is that of C_- followed by $t_{a_p k}$, and $\omega C_- = v$. Apply Lemma 6.7(iii). □

Lemma 6.19 (Merge)**.** *Let $\omega \in S_\infty$, $k > 1$, $\eta = \omega(k)$, and let D be a cycle with* $\mathrm{supp}(D) \subseteq [1, k-1]$*, allowing D to be trivial, $D = \{a\}$ with $a \leq k-1$. Put $w = \omega D$, so $w(k) = \eta$. For $a \in$* $\mathrm{supp}(D)$ *let $\tau_a = Dt_{ak}$, the cycle with support* $\mathrm{supp}(D) \cup \{k\}$ *sending $a \mapsto k \mapsto D(a)$, and $v_a = wt_{ak} = \omega\tau_a$. Then:*

(a) *there is at most one $a \in$* $\mathrm{supp}(D)$ *such that τ_a is ω-tight and $v_a \lhd w$;*
(b) *such an a exists if and only if D is ω-tight and* $\max \mathrm{supp}(D) = k-1$ *(for trivial $D = \{a\}$: if and only if $a = k-1$);*
(c) *for this a, $\kappa(v_a, t_{ak})\, \mathbf{q}_\omega(\tau_a) = q_{k-1}\, \mathbf{q}_\omega(D)$.*

Proof. Trivial $D = \{a\}$. Then $w = \omega$, $\tau_a = t_{ak}$, and τ_a is ω-tight if and only if $w \lhd wt_{ak}$. If also $wt_{ak} \lhd w$, then both $\ell(wt_{ak}) - \ell(w)$ and its negative lie in $\{1, 1-2(k-a)\}$, which forces $k - a = 1$. Conversely for $a = k-1$ the transposition $t_{k-1,k}$ changes the length by ± 1, so both relations hold, one classical and one quantum with $\kappa = q_{k-1}$; thus $\kappa(v_a, t_{ak})\mathbf{q}_\omega(\tau_a) = q_{k-1} = q_{k-1}\mathbf{q}_\omega(D)$.

Nontrivial $D = (d_r, \dots, d_1, d)$, $r \geq 1$, $d = \max \mathrm{supp}(D)$. Put $d_0 = d$, $\gamma_0 = \omega(d)$, $\gamma_i = \omega(d_i)$ and $\epsilon = k - d \geq 1$. For $a = d_m$ $(0 \leq m \leq r)$ the cycle $\tau_m := \tau_{d_m}$ is

$$\tau_m = (d_{m-1}, \dots, d_1, d, d_r, \dots, d_m, k),$$

whose pivot factorization

$$t_{d_m k}\, t_{d_{m+1} k} \cdots t_{d_r k}\, t_{dk}\, t_{d_1 k} \cdots t_{d_{m-1} k}$$

exchanges, in order, the values $\eta, \gamma_m, \gamma_{m+1}, \dots, \gamma_r, \gamma_0, \gamma_1, \dots, \gamma_{m-1}$: the sequence of values of τ_m in the sense of Corollary 6.13 is the cyclic sequence $\Gamma = (\gamma_0, \dots, \gamma_r)$ of D with η inserted into the gap (γ_{m-1}, γ_m) (the wrap gap (γ_r, γ_0) when $m = 0$). Note $w(d_m) = \gamma_{m-1}$ for $m \geq 1$ and $w(d) = \gamma_r$; we write $w(a)$ for the value at $a = d_m$ in either case.

Necessity. Suppose τ_m is ω-tight and $v_m := v_{d_m} \lhd w$. By Corollary 6.13 the cyclic sequence of τ_m is unimodal; deleting η shows that Γ is unimodal, and then m must be the insertion gap of η in Γ. This already proves (a). Let δ be the ascent index of Γ ($\gamma_\delta > \gamma_{\delta-1}$, or $\delta = 0$ for the wrap), so that by Corollary 6.13 and Definition 6.6

$$L_\omega(D) = r - 2(d - d_\delta), \qquad \mathbf{q}_\omega(D) = q_{d_\delta d}$$

(read as r and 1 when $\delta = 0$). Now compare lengths. Tightness gives $\ell(v_m) - \ell(\omega) = L_\omega(\tau_m)$, Lemma 6.7(ii) gives $\ell(w) - \ell(\omega) \geq L_\omega(D)$ with equality if and only if D is ω-tight, and $v_m \lhd w$ gives $\ell(v_m) - \ell(w) = -1$ if $w(a) > \eta$ and $\ell(v_m) - \ell(w) = 2(k-a) - 1$ if $w(a) < \eta$. Hence

$$L_\omega(\tau_m) - L_\omega(D) \;\geq\; \begin{cases} -1 & \text{if } w(a) > \eta, \\ 2(k-a) - 1 & \text{if } w(a) < \eta, \end{cases} \tag{6.5}$$

with equality if and only if D is ω-tight. We compute the left side from Corollary 6.13 in the three possible configurations of the insertion gap.

(I) $\min \Gamma < \eta < \max \Gamma$, so $\gamma_{m-1} > \eta > \gamma_m$ and $m \neq \delta$. The ascent of the sequence of τ_m is the (interior) step onto γ_δ, whose position is d_δ, so $L_\omega(\tau_m) = (r+1) - 2(k - d_\delta)$ and $L_\omega(\tau_m) - L_\omega(D) = 1 - 2\epsilon$. Here $w(a) = \gamma_{m-1} > \eta$, so (6.5) reads $1 - 2\epsilon \geq -1$, forcing $\epsilon = 1$ and equality.

(II) $\eta < \min\Gamma$, so $m = \delta$. The ascent of the sequence of τ_m is its first step (η, γ_δ), at position d_δ; again $L_\omega(\tau_m) - L_\omega(D) = 1 - 2\epsilon$, and $w(a) = \gamma_{\delta-1} > \eta$ forces $\epsilon = 1$ and equality.

(III) $\eta > \max\Gamma$, so $m = \delta$. The sequence of τ_m is decreasing, its ascent being the wrap, so $L_\omega(\tau_m) = r+1$ and $L_\omega(\tau_m) - L_\omega(D) = 1+2(d-d_\delta)$. Here $w(a) = \gamma_{\delta-1} < \eta$, and (6.5) reads $1+2(d-d_\delta) \geq 2(k-d_\delta)-1$, i.e. $\epsilon \leq 1$; so $\epsilon = 1$ and equality.

In every case $d = k-1$ and D is ω-tight, proving the "only if" part of (b).

Sufficiency. Suppose D is ω-tight and $d = k-1$; then Γ is unimodal (Corollary 6.13). Let m be the insertion gap of η and $a = d_m$. The computations above show that (6.5) holds with equality, and since $\ell(w) - \ell(\omega) = L_\omega(D)$, the tightness of τ_m is equivalent to $\ell(v_m) - \ell(w)$ taking the value -1 (if $w(a) > \eta$) or $2(k-a)-1$ (if $w(a) < \eta$), i.e. to $v_m \lhd w$. It therefore suffices to show $wt_{ak} \lhd w$, that is, that $(w(a), \eta)$ is J-admissible for $J = \{w(r) \mid a < r < k\}$. The positions in (a, k) are: $d = k-1$ (if $a \neq d$), carrying γ_r; the $d_l > a$, carrying γ_{l-1}; and positions outside $\mathrm{supp}(D)$, carrying their ω-values. If $a = d = k-1$ the set J is empty and we are done; so assume $m \geq 1$.

(I) $\gamma_{m-1} > \eta > \gamma_m$: the pair $(w(a), \eta) = (\gamma_{m-1}, \eta)$ is classical and we need $J \cap (\eta, \gamma_{m-1}) = \emptyset$. No γ_l lies in (γ_m, γ_{m-1}), these being consecutive in the decreasing order of Γ. The m-th step of the pivot chain of D is classical, so by (6.2) every position $r \in (d_m, d)$ outside $\mathrm{supp}(D)$ has $\omega(r) \notin (\gamma_m, \gamma_{m-1})$.

(II) $m = \delta$, $\eta < \gamma_{\delta-1} = \min\Gamma$: the pair $(\gamma_{\delta-1}, \eta)$ is classical and we need $J \cap (\eta, \gamma_{\delta-1}) = \emptyset$; all $\gamma_l \geq \gamma_{\delta-1}$, and by (6.3) for the quantum step δ of D the positions $r \in (d_\delta, d)$ outside $\mathrm{supp}(D)$ have $\omega(r) \in (\gamma_{\delta-1}, \gamma_\delta)$.

(III) $m = \delta$, $\eta > \gamma_\delta = \max\Gamma$: the pair $(\gamma_{\delta-1}, \eta)$ is quantum and we need $J \subseteq (\gamma_{\delta-1}, \eta)$; the values γ_l occurring in J are different from $\gamma_{\delta-1}$ (which sits at a) and lie in $(\gamma_{\delta-1}, \gamma_\delta]$, and the remaining positions have $\omega(r) \in (\gamma_{\delta-1}, \gamma_\delta)$ by (6.3).

This proves the "if" part of (b).

Weights. In cases (I) and (II), $\kappa(v_a, t_{ak}) = 1$ and $\mathbf{q}_\omega(\tau_m) = q_{d_\delta k} = q_{d_\delta, k-1} q_{k-1} = \mathbf{q}_\omega(D) q_{k-1}$ (read $q_{dk} = q_{k-1}$ and $\mathbf{q}_\omega(D) = 1$ when $\delta = 0$). In case (III), $\kappa(v_a, t_{ak}) = q_{ak} = q_{d_\delta k}$ and $\mathbf{q}_\omega(\tau_m) = 1$, giving the same product. This is (c). □

The cancellations. We now assemble the Monk summands. Throughout, ρ denotes the product of the nontrivial cycles of π other than C and other than the cycle D of π containing $k-1$ (we set $D = \{k-1\}$ if $k-1$ is fixed by π, and we do not single out D when $k-1 \in C$). By Lemma 6.15, Lemma 6.8 and Lemma 6.7(iv), each of the quantities $c_{k-1}(u, w)$, $c_{k-2}(u, w)$, $c_k(u, w)$ and each Monk summand vanishes unless ρ is u-tight and each cycle of ρ contains exactly one element greater than k; indeed each requires ρ to be a tight product of cycles with exactly one element greater than k or than $k-1$ or than $k-2$, and these conditions agree for cycles containing neither k nor $k-1$. (For $c_{k-2}(u, w)$, note that $k-1$ lies in C or in D, hence not in a cycle of ρ.) *We assume this from now on*; otherwise all terms considered below are zero.

Lemma 6.20 (Reflection–drop cancellation)**.** *Suppose $u, w \in S_\infty$, let $k > 1$, and suppose $u(k) = w(k)$. Then*

$$\sum_{a<k,\ wt_{ak} \lhd w} \mathrm{sign}(a-k)\, \kappa_{ak}\, c_{k-1}(u, wt_{ak}) \ + \sum_{b>k,\ wt_{kb} \lhd w} \mathrm{sign}(b-k)\, \kappa_{kb}\, c_{k-1}(u, wt_{kb}) \ + \ q_{k-1}\, c_{k-2}(u, w) = 0,$$

where $\kappa_{rs} = \kappa(wt_{rs}, t_{rs})$. In fact the two reflection sums contribute at most one nonzero summand in total: it occurs precisely when $u \overset{k-2}{\rightsquigarrow} w$ (equivalently, when the quantum term is nonzero), it is indexed by some $a < k$, and it equals $-q_{k-1} c_{k-2}(u, w)$.

Proof. Here $C = \{k\}$, so by Lemma 6.15 only summands of type (M) can be nonzero: $t = t_{ak}$ with the π-cycle D_a of a contained in $[1, k-1]$. If D_a is nontrivial then it is not a cycle of ρ (those have an element greater than k), so $D_a = D$ and $\max D = k-1$; otherwise a is a fixed point of π. Let a be such a candidate and put $\omega_a = wD_a^{-1}$, so that $w = \omega_a D_a$, $\omega_a(k) = w(k)$, and $\omega_a = u\rho$ when $D_a = D$. Then $u^{-1} wt_{ak} = (\pi D_a^{-1})\, \tau_a$ with $\tau_a = D_a t_{ak}$ as in Lemma 6.19, where πD_a^{-1} is the product of the cycles of π other than D_a; by Lemma 6.7(iv), $u^{-1} wt_{ak}$ is u-tight if and only if πD_a^{-1} is u-tight and τ_a is ω_a-tight. Now apply Lemma 6.19 to (ω_a, D_a). If a is a fixed point, the summand can be nonzero only for $a = k-1$, which requires $k-1$ to be fixed, i.e. $D = \{k-1\}$; in that case $\pi D_a^{-1} = \rho$ and the summand is nonzero. If $D_a = D$ is nontrivial, at most one $a \in D$ works, and one does if and only if D is $u\rho$-tight. Altogether at most one a gives a nonzero summand, and one exists if and only if either $D = \{k-1\}$, or D is nontrivial with $\max D = k-1$ and

$u\rho$-tight. On the other hand $u \overset{k-2}{\rightsquigarrow} w$ if and only if $\pi = \rho D$ is $(k-2)$-shaped and u-tight, i.e. (given our standing assumption on ρ) if and only if D is trivial or has $k-1$ as its unique element greater than $k-2$, and D is $u\rho$-tight: the same condition. When it holds, with $v = wt_{ak}$ the unique contributing neighbor and $\omega = \omega_a = u\rho$, Lemma 6.19(c) and Lemma 6.8 give

$$\kappa_{ak}\, q_{k-1}(u,v) = \mathbf{q}_u(\rho)\,\kappa_{ak}\mathbf{q}_\omega(\tau_a) = \mathbf{q}_u(\rho)\,q_{k-1}\mathbf{q}_\omega(D) = q_{k-1}\,q_{k-2}(u,w),$$

and Lemma 6.15(M) gives $\mathrm{fix}_{k-1}(u,v) = \mathrm{fix}_{k-1}(u,w) \setminus \{u(a)\}$, which equals $\mathrm{fix}_{k-2}(u,w)$: if $D = \{k-1\}$ then $a = k-1$ and we remove the fixed value $u(k-1)$; if D is nontrivial then neither a nor $k-1$ is a fixed position of w. Hence the summand is $-\kappa_{ak}c_{k-1}(u,v) = -q_{k-1}c_{k-2}(u,w)$, cancelling the quantum term. If the condition fails, both the reflection sums and the quantum term vanish. □

Lemma 6.21 (Moved reflection cancellation)**.** *Let $u, w \in S_\infty$, let $k > 1$, and suppose $u(k) \neq w(k)$, so that C is nontrivial.*

(a) *If $B > k$, then $c_{k-1}(u,w) = c_{k-2}(u,w) = 0$, and the sum of the Monk summands equals $c_k(u,w)$. More precisely: if $u \overset{k}{\rightsquigarrow} w$ the only nonzero Monk summand is that of t_{kB}, and it equals $c_k(u,w)$; if $u \overset{k}{\not\rightsquigarrow} w$, the nonzero Monk summands are either none or exactly two, one of type (U) and one of type (S), with equal absolute value and opposite signs.*

(b) *If $B = k$, then $c_k(u,w) = 0$, no summand of type (U) occurs, and*

$$(y_{w(k)} - z_1)\, c_{k-1}(u,w) \;-\; \sum_{a<k,\ wt_{ak}\lhd w} \kappa_{ak}\, c_{k-1}(u, wt_{ak}) \;+\; q_{k-1}c_{k-2}(u,w) = 0.$$

In fact the diagonal term cancels against the summand of type (P), and the quantum term against the (at most one) summand of type (M).

Proof. (a) C contains k and B, two elements greater than $k-1$, so π is neither $(k-1)$- nor $(k-2)$-shaped and $c_{k-1}(u,w) = c_{k-2}(u,w) = 0$. By Lemma 6.15, a nonzero Monk summand is of type (U) or (S), and requires $C \cap (k-1,\infty) = \{k, B\}$; together with the standing assumption on ρ this means that π is k-shaped, which we may assume (otherwise all Monk summands vanish and $c_k(u,w) = 0$). Write $C = (a_p, \ldots, a_1, B)$ with $a_j = k$, $C' = (a_{j-1}, \ldots, a_1, B)$, $\psi = u\rho C'$ and $X = (a_p, \ldots, a_{j+1}, k, B)$, so that $w = \psi X$ as in Lemma 6.16 with $s = p - j$, $z_i = a_{j+i}$. If C' is not $u\rho$-tight then nothing is tight (Lemma 6.7(iv),(v): C' is a truncation of C and of each C'_m, and a cycle of $u^{-1}wt_{kB}$), all Monk summands vanish, and $c_k(u,w) = 0$. So assume C' is $u\rho$-tight. Then, exactly as in the proof of Lemma 6.17: $u \overset{k}{\rightsquigarrow} w$ if and only if F_{std} is a cover path from ψ; the summand of type (U) is nonzero if and only if F_{up} is a cover path, and then equals $\mathbf{q}_u(\rho)\mathbf{q}_u(C')\,\mathrm{wt}(F_{\mathrm{up}})\,\mathbf{E}(y_P; z)$ with $P = \mathrm{fix}_{k-1}(u,w)$; and the summand of type (S) indexed by a_m $(m > j)$ is nonzero if and only if F_{m-j} is a cover path, and then equals $-\mathbf{q}_u(\rho)\mathbf{q}_u(C')\,\mathrm{wt}(F_{m-j})\,\mathbf{E}(y_P; z)$ (Lemma 6.15(S) for the fixed set). If $u \overset{k}{\rightsquigarrow} w$, Lemma 6.16(a) and Lemma 6.17 show that the only nonzero summand is that of t_{kB} and that it equals $q_k(u,w)\mathbf{E}(y_{\mathrm{fix}_k(u,w)}; z) = c_k(u,w)$. If $u \overset{k}{\not\rightsquigarrow} w$, Lemma 6.16(b),(c) show that either no summand is nonzero, or exactly two are, namely (U) and one (S), with equal weights; they cancel, and $c_k(u,w) = 0$.

(b) $C \subseteq [1,k]$ has no element greater than k, so π is not k-shaped and $c_k(u,w) = 0$; type (U) is excluded by Lemma 6.15. Write $C = (a_p, \ldots, a_1, k)$. For the diagonal and peel terms, let $\omega_0 = u\rho D$ if $k-1 \notin C$ and $\omega_0 = u\rho$ if $k-1 \in C$ (where D is the cycle of $k-1$, trivial or not), so that $w = \omega_0 C$ and ω_0 agrees with u on $\mathrm{supp}(C)$. The diagonal term is nonzero if and only if π is $(k-1)$-shaped and tight, i.e. (given the standing assumption) D is trivial or has an element greater than k, D is $u\rho$-tight, and C is ω_0-tight (the conditions on D being vacuous when $k-1 \in C$). The summand of type (P) is nonzero if and only if $\rho D C_-$ is $(k-1)$-shaped and tight and $v = wt_{a_pk} \lhd w$, i.e. the same conditions on D together with: C_- is ω_0-tight and $v \lhd w$. By Lemma 6.18 these two sets of conditions are equivalent, and when they hold, $\kappa_{a_pk}q_{k-1}(u,v) = \mathbf{q}_u(\rho D)\kappa_{a_pk}\mathbf{q}_{\omega_0}(C_-) = \mathbf{q}_u(\rho D)\mathbf{q}_{\omega_0}(C) = q_{k-1}(u,w)$, while $\mathrm{fix}_{k-1}(u,v) = \mathrm{fix}_{k-1}(u,w) \sqcup \{w(k)\}$ by Lemma 6.15(P). Since $(y_{w(k)} - z_1)\mathbf{E}(y_{\mathrm{fix}_{k-1}(u,w)}; z) = \mathbf{E}(y_{\mathrm{fix}_{k-1}(u,w)\cup\{w(k)\}}; z)$, the diagonal term and the (negatively signed) summand of type (P) cancel.

For the merge and quantum terms, a summand of type (M) requires $a \notin C$ with the cycle D_a of a contained in $[1, k-1]$; as in the proof of Lemma 6.20, either a is fixed by π or $D_a = D$ with $\max D = k-1$ (in particular $k-1 \notin C$). If $k-1 \in C$, then $k-1$ is not fixed and no such D exists; since a fixed-point candidate must be $a = k-1$ by the trivial case of Lemma 6.19 (applied as below), there are no nonzero

summands of type (M), and $c_{k-2}(u,w)=0$ since C contains the two elements $k-1,k$ greater than $k-2$. So suppose $k-1\notin C$. For a candidate a put $\omega_a = wD_a^{-1}$, so that $w=\omega_a D_a$ and $\omega_a(k)=w(k)$; then $u^{-1}wt_{ak}=\rho_a E$, where ρ_a is the product of the cycles of π other than C and D_a and $E=C\tau_a$ is the cycle of Lemma 6.15(M), $\tau_a = D_a t_{ak}$. The pivot factorization of E is that of C followed by that of τ_a, so by Lemma 6.7(iii),(iv), $u^{-1}wt_{ak}$ is u-tight if and only if ρ_a is u-tight, C is $u\rho_a$-tight and τ_a is ω_a-tight (note $u\rho_a C=\omega_a$), and then $\mathbf{q}_u(E)=\mathbf{q}_u(C)\mathbf{q}_{\omega_a}(\tau_a)$. By Lemma 6.19 applied to (ω_a,D_a): a fixed-point candidate contributes only if $a=k-1$, i.e. $D=\{k-1\}$; and if $D_a=D$ is nontrivial, at most one $a\in D$ contributes, and one does if and only if D is ω_a-tight, where now $\omega_a = u\rho C =: \omega$. Altogether at most one summand of type (M) is nonzero, and one is if and only if C is $u\rho$-tight and either $D=\{k-1\}$, or D is nontrivial with $\max D = k-1$ and ω-tight. This is exactly the condition for $u \overset{k-2}{\rightsquigarrow} w$: $\pi=\rho CD$ is $(k-2)$-shaped (as $k-1\notin C$, C has the unique element k greater than $k-2$, and D the unique element $k-1$) and it is u-tight if and only if C is $u\rho$-tight and D is ω-tight. When it holds, Lemma 6.19(c) gives

$$\kappa_{ak} q_{k-1}(u,wt_{ak}) = \mathbf{q}_u(\rho)\mathbf{q}_u(C)\,\kappa_{ak}\mathbf{q}_\omega(\tau_a) = \mathbf{q}_u(\rho)\mathbf{q}_u(C)\,q_{k-1}\mathbf{q}_\omega(D) = q_{k-1}q_{k-2}(u,w),$$

and $\mathrm{fix}_{k-1}(u,wt_{ak}) = \mathrm{fix}_{k-1}(u,w)\setminus\{u(a)\} = \mathrm{fix}_{k-2}(u,w)$ as in the proof of Lemma 6.20. Thus the summand of type (M) equals $-q_{k-1}c_{k-2}(u,w)$ and cancels the quantum term; when the condition fails, both vanish. □

We now record the recurrence satisfied by the quantum factorial elementary polynomials, which drives the induction.

Lemma 6.22. *For all integers $k\geq 2$ we have*

$$\widetilde{E}_{k,k}(x;z) = (x_k - z_1)\widetilde{E}_{k-1,k-1}(x;z) + q_{k-1}\widetilde{E}_{k-2,k-2}(x;z),$$

with the conventions $\widetilde{E}_{0,0}=1$ and $\widetilde{E}_{k,k}=0$ if $k<0$. Moreover, the ordinary polynomials $\mathbf{E}(y_A;z) = \prod_{i\in A}(y_i - z_1)$ satisfy, for any set A and any $c\notin A$, the single-variable identity

$$\mathbf{E}(y_{A\cup\{c\}};z) = (y_c - z_1)\,\mathbf{E}(y_A;z).$$

Proof. By Lemma 3.3, $\widetilde{E}_{k,k}(x;z) = \Delta_k(-z_1 \mid x_1,\dots,x_k)$, and expanding the tridiagonal determinant $\Delta_k(t \mid x_1,\dots,x_k)$ along its last row gives $\Delta_k = (x_k+t)\Delta_{k-1} + q_{k-1}\Delta_{k-2}$. The second identity is clear. □

The first recurrence is the quantization of the classical relation $E_{k,k} = (x_k - z_1)E_{k-1,k-1}$; the extra term $q_{k-1}\widetilde{E}_{k-2,k-2}$ is the only feature absent from the non-quantum setting. Discarding all quantum terms recovers the classical Pieri formula [33, Theorem 7.1], whose term-by-term cancellation is carried out in [33, Proof of Proposition 4.4] by means of the reflection-matching Lemmas 7.6–7.9 of that paper; the lemmas above are the quantum counterparts, the essential new points being the length-decreasing covers permitted by Lemma 6.8, the attendant bookkeeping of the powers of q, and the extra term of the recurrence, which is absorbed by the merge summands of Lemma 6.19.

Proof of Theorem 3.4. For a permutation w, write $[\widetilde{\mathfrak{S}}_w]F$ for the coefficient of $\widetilde{\mathfrak{S}}_w(x;y)$ in an expansion F of a quantum double Schubert polynomial in the basis $\{\widetilde{\mathfrak{S}}_w(x;y)\}$. With this notation the theorem is the assertion that, for all u and all $k\geq 1$,

$$[\widetilde{\mathfrak{S}}_w]\big(\,\widetilde{\mathfrak{S}}_u(x;y)\widetilde{E}_{k,k}(x;z)\big) = \begin{cases} q_k(u,w)\,\mathbf{E}(y_{\mathrm{fix}_k(u,w)};z) & \text{if } u \overset{k}{\rightsquigarrow} w,\\ 0 & \text{otherwise.}\end{cases}$$

For brevity, when $u \overset{m}{\rightsquigarrow} w$ we set

$$c_m(u,w) = q_m(u,w)\,\mathbf{E}(y_{\mathrm{fix}_m(u,w)};z),$$

and $c_m(u,w)=0$ if $u \overset{m}{\not\rightsquigarrow} w$; note that $u \overset{0}{\rightsquigarrow} w$ only for $w=u$, with $c_0(u,u)=1$, in accordance with $\widetilde{E}_{0,0}=1$. The claim is that $[\widetilde{\mathfrak{S}}_w](\widetilde{\mathfrak{S}}_u\,\widetilde{E}_{k,k}) = c_k(u,w)$.

We argue by strong induction on k.

Base case $k=1$. We have $\widetilde{E}_{1,1}(x;z) = x_1 - z_1$, and Lemma 6.1 with $i=j=1$ gives

$$(x_1-z_1)\,\widetilde{\mathfrak{S}}_u(x;y) = (y_{u(1)}-z_1)\,\widetilde{\mathfrak{S}}_u(x;y) + \sum_{\substack{b>1\\ \ell(ut_{1b})=\ell(u)+1}} \widetilde{\mathfrak{S}}_{ut_{1b}}(x;y) + \sum_{\substack{b>1\\ \ell(ut_{1b})=\ell(u)-2(b-1)+1}} q_{1b}\,\widetilde{\mathfrak{S}}_{ut_{1b}}(x;y).$$

The permutations w with $u \overset{1}{\rightsquigarrow} w$ are exactly $w = u$ and $w = ut_{1b}$ for $b > 1$ with $u \lhd ut_{1b}$, since Definition 3.1 permits at most one reflection when $k = 1$. For $w = u$ we have $q_1 = 1$, $\mathrm{fix}_1(u,u) = \{u(1)\}$, and $\mathbf{E}(y_{\{u(1)\}}; z) = y_{u(1)} - z_1$, matching the diagonal term. For $w = ut_{1b}$ we have $\mathrm{fix}_1 = \emptyset$ and $\mathbf{E}(y_\emptyset; z) = 1$; if the step is length-increasing then $q_1(u,w) = 1$, matching the second sum, and if it is length-decreasing then $q_1(u,w) = q_{1b}$, matching the third sum. All other coefficients vanish on both sides. This proves the base case.

Inductive step. Fix $k > 1$ and assume the claim at all levels $m < k$. By Lemma 6.22,

$$\widetilde{\mathfrak{S}}_u \, \widetilde{E}_{k,k} = (x_k - z_1)\big(\widetilde{\mathfrak{S}}_u \, \widetilde{E}_{k-1,k-1}\big) + q_{k-1} \, \widetilde{\mathfrak{S}}_u \, \widetilde{E}_{k-2,k-2}. \tag{6.6}$$

By the inductive hypothesis, $\widetilde{\mathfrak{S}}_u \, \widetilde{E}_{k-1,k-1} = \sum_v c_{k-1}(u,v) \, \widetilde{\mathfrak{S}}_v$. Applying Lemma 6.1 with $i = k$, $j = 1$ to each $\widetilde{\mathfrak{S}}_v$ and extracting the coefficient of $\widetilde{\mathfrak{S}}_w$ from (6.6), we obtain

$$[\widetilde{\mathfrak{S}}_w]\big(\widetilde{\mathfrak{S}}_u \, \widetilde{E}_{k,k}\big) = (y_{w(k)} - z_1)\, c_{k-1}(u,w) \tag{6.7}$$

$$+ \sum_{\substack{b>k,\ wt_{kb} \lhd w \\ \ell(w) = \ell(wt_{kb})+1}} c_{k-1}(u, wt_{kb}) + \sum_{\substack{b>k,\ wt_{kb} \lhd w \\ \ell(w) = \ell(wt_{kb})-2(b-k)+1}} q_{kb}\, c_{k-1}(u, wt_{kb}) \tag{6.8}$$

$$- \sum_{\substack{a<k,\ wt_{ak} \lhd w \\ \ell(w) = \ell(wt_{ak})+1}} c_{k-1}(u, wt_{ak}) - \sum_{\substack{a<k,\ wt_{ak} \lhd w \\ \ell(w) = \ell(wt_{ak})-2(k-a)+1}} q_{ak}\, c_{k-1}(u, wt_{ak}) \tag{6.9}$$

$$+ q_{k-1}\, c_{k-2}(u,w), \tag{6.10}$$

where we used that $\mathrm{sign}(b-k) = 1$ for $b > k$ and $\mathrm{sign}(a-k) = -1$ for $a < k$, that the term $\widetilde{\mathfrak{S}}_w$ arises from $\widetilde{\mathfrak{S}}_v$ with $v = wt$ exactly when $v \lhd w$, and that a summand in (6.8)–(6.9) is present precisely when $u \overset{k-1}{\rightsquigarrow} wt_{kb}$, respectively $u \overset{k-1}{\rightsquigarrow} wt_{ak}$ (otherwise $c_{k-1} = 0$). In the terminology introduced before Lemma 6.15, (6.8) and (6.9) are the Monk summands at w, with κ_{kb}, respectively κ_{ak}, equal to 1 in the first sum and to q_{kb}, respectively q_{ak}, in the second. We must show that (6.7)–(6.10) equals $c_k(u,w)$. We distinguish three cases according to whether $u(k) = w(k)$ and, when $u(k) \neq w(k)$, whether the cycle C of $u^{-1}w$ containing k has its top B greater than k or equal to k.

Case 1: $u(k) = w(k)$. By Lemma 6.14(i) we have $u \overset{k}{\rightsquigarrow} w$ if and only if $u \overset{k-1}{\rightsquigarrow} w$; suppose first that both hold. Then, by the same lemma,

$$\mathrm{fix}_k(u,w) = \mathrm{fix}_{k-1}(u,w) \cup \{u(k)\}, \qquad q_k(u,w) = q_{k-1}(u,w),$$

so the diagonal contribution (6.7) gives

$$(y_{u(k)} - z_1)\, c_{k-1}(u,w) = q_{k-1}(u,w)\,(y_{u(k)} - z_1)\,\mathbf{E}(y_{\mathrm{fix}_{k-1}(u,w)}; z) = q_k(u,w)\,\mathbf{E}(y_{\mathrm{fix}_k(u,w)}; z) = c_k(u,w),$$

by Lemma 6.22. The remaining contributions are the reflection sums (6.8), (6.9) and the quantum term $q_{k-1}c_{k-2}(u,w)$ of (6.10), which sum to 0 by Lemma 6.20. Hence the total is $c_k(u,w)$, as required. If instead $u(k) = w(k)$ but $u \overset{k}{\not\rightsquigarrow} w$ (equivalently $u \overset{k-1}{\not\rightsquigarrow} w$), then $c_k(u,w) = 0$ and the diagonal term vanishes, while the reflection sums and quantum term again cancel by Lemma 6.20; the total is $0 = c_k(u,w)$.

Case 2: $u(k) \neq w(k)$ and $B > k$. By Lemma 6.21(a), the diagonal term (6.7) and the quantum term (6.10) vanish, and the Monk summands (6.8)–(6.9) sum to $c_k(u,w)$. (When $u \overset{k}{\rightsquigarrow} w$, Lemma 6.17 identifies the single surviving summand: it is indexed by t_{kB}, with $v = wt_{kB}$, $u \overset{k-1}{\rightsquigarrow} v$, $\mathrm{fix}_k(u,w) = \mathrm{fix}_{k-1}(u,v)$ and $\kappa_{kB}\, q_{k-1}(u,v) = q_k(u,w)$, the factor κ_{kB} being 1 or q_{kB} according as t_{kB} is a classical or a quantum cover of v; this is the step at which the quantum drop terms of Lemma 6.1 and the quantum weight of Definition 3.1 interlock.)

Case 3: $u(k) \neq w(k)$ and $B = k$. Here $u \overset{k}{\not\rightsquigarrow} w$, since C contains no element greater than k, so $c_k(u,w) = 0$; and (6.7)–(6.10) vanishes by Lemma 6.21(b).

By Lemma 6.14 these cases are exhaustive. In all cases $[\widetilde{\mathfrak{S}}_w](\widetilde{\mathfrak{S}}_u \, \widetilde{E}_{k,k}) = c_k(u,w)$, completing the induction and the proof. □

7. Proofs of the expansion formulas

In this section we prove Theorems 4.5, 4.6 and 4.7. Throughout, divided difference operators act on the z-variables, and in order to free the subscript for the skew operators we write

$$\partial^a = \partial_a^{(z)}, \qquad \partial^w = \partial_w^{(z)} = \partial^{a_1} \cdots \partial^{a_m} \quad (w = s_{a_1} \cdots s_{a_m} \text{ reduced}),$$

the rightmost operator being applied first. The simple reflection s_a acts on polynomials by exchanging z_a and z_{a+1}, and a permutation w acts by $w(z_i) = z_{w(i)}$, so that $w = s_{a_1} \circ \cdots \circ s_{a_m}$ as operators. All operators of this section are linear over $\mathbb{Z}[q][x]$, which we treat as the ring of scalars. Recall that the quantization map $\mathcal{Q}$ of Fomin, Gelfand and Postnikov (Lemma 2.1) is linear over $\mathbb{Z}[q][z]$, sends every standard elementary monomial $\prod_j e_{i_j}(x_1, \ldots, x_j)$ to $\prod_j e_{i_j}^{(q)}(x_1, \ldots, x_j)$, and satisfies $\mathcal{Q}(\mathfrak{S}_w(x; z)) = \widetilde{\mathfrak{S}}_w(x; z)$.

7.1. **Skew divided difference operators.** We use the skew divided difference operators of [32, 33]. They are the operators ∂_u^w, $u, w \in S_\infty$, determined by the Leibniz formula

$$\partial^w(PQ) = \sum_{u \in S_\infty} \partial^u(P)\, \partial_u^w(Q), \tag{7.1}$$

which differs by a normalization from the convention of Macdonald [23]. We shall use the following facts.

Lemma 7.1. *Let $u, w \in S_\infty$.*

(i) *(Subword formula.) If $(a_1, \ldots, a_m)$ is a reduced word for w, then ∂_u^w is the sum, over all reduced subwords $(b_1, \ldots, b_m)$ for u (that is, $b_j \in \{1, s_{a_j}\}$ for each j, and the nonunit entries form a reduced word for u), of the compositions $\partial_{b_1}^{a_1} \cdots \partial_{b_m}^{a_m}$, where $\partial_1^a = \partial^a$ and $\partial_{s_a}^a = s_a$.*

(ii) *$\partial_u^w = 0$ unless $u \le w$ in Bruhat order; $\partial_1^w = \partial^w$ and $\partial_w^w = w$; and ∂_u^w is homogeneous of degree $\ell(u) - \ell(w)$ in the z-variables. In particular, for a scalar c, $\partial_u^w(c) = c$ if $u = w$ and $\partial_u^w(c) = 0$ otherwise.*

(iii) *(Ascending Leibniz formula.) For polynomials $f_1, \ldots, f_k$,*

$$\partial_u^w(f_1 f_2 \cdots f_k) = \sum_{u = u_0 \le u_1 \le \cdots \le u_k = w} \prod_{j=1}^{k} \partial_{u_{j-1}}^{u_j}(f_j).$$

Proof. (i) is proved in [33] (it goes back to Macdonald [23]). (ii) follows from (i): a reduced subword for u of a reduced word for w exists only if $u \le w$; the only subword for $u = 1$ is the empty one and the only one for $u = w$ is the full word, whose composition $s_{a_1} \circ \cdots \circ s_{a_m}$ is the action of w; and each composition in (i) consists of $\ell(u)$ reflections, which preserve degree, and $\ell(w) - \ell(u)$ divided differences, which lower it by one. (iii) is a rearrangement of the generalized Leibniz rule of [32, Corollary 3.5.4]: in the notation of that work $\partial_u^w = \sum_v c_{u,v}^w \partial_v$, the coefficients $c_{u_0,u_1,\ldots,u_k}^w$ of the k-fold rule factor as $\sum_{v_1,\ldots,v_{k-1}} c_{u_0,u_1}^{v_1} c_{v_1,u_2}^{v_2} \cdots c_{v_{k-1},u_k}^w$ by associativity, and $\sum_{u_j} c_{v_{j-1},u_j}^{v_j} \partial_{u_j}(f_j) = \partial_{v_{j-1}}^{v_j}(f_j)$. By (ii) only chains $u_0 \le u_1 \le \cdots \le u_k$ contribute. □

Lemma 7.2 (Linear forms)**.** *Let $u, w \in S_\infty$ and $i \ge 1$. Then*

$$\partial_u^w(z_i) = \begin{cases} z_{u(i)} & \text{if } w = u, \\ 1 & \text{if } w = u t_{ib} \text{ with } b > i \text{ and } \ell(w) = \ell(u) + 1, \\ -1 & \text{if } w = u t_{bi} \text{ with } b < i \text{ and } \ell(w) = \ell(u) + 1, \\ 0 & \text{otherwise.} \end{cases}$$

Proof. The case $w = u$ is Lemma 7.1(ii), and by the same lemma $\partial_u^w(z_i) = 0$ unless $u < w$ and $\ell(w) = \ell(u)+1$. So let $w = u t_{ab}$ with $a < b$ and $\ell(w) = \ell(u) + 1$, so that $u(a) < u(b)$. Fix a reduced word $(a_1, \ldots, a_m)$ for w and put $\sigma_p = s_{a_{p+1}} \cdots s_{a_m}$. The reduced subwords for u omit exactly one letter, and omitting the p-th letter gives the element $w\, \sigma_p^{-1} s_{a_p} \sigma_p = w\, t_{\sigma_p^{-1}(a_p),\, \sigma_p^{-1}(a_p+1)}$. The reflections $\sigma_p^{-1} s_{a_p} \sigma_p$, $1 \le p \le m$, are pairwise distinct (they are the m reflections t with $\ell(wt) < \ell(w)$, see [18, Section 1.6–1.7]), so exactly one p yields u, namely the one with $\{\sigma_p^{-1}(a_p), \sigma_p^{-1}(a_p + 1)\} = \{a, b\}$, and by Lemma 7.1(i)

$$\partial_u^w(z_i) = s_{a_1} \cdots s_{a_{p-1}}\, \partial^{a_p}\, \sigma_p(z_i) = s_{a_1} \cdots s_{a_{p-1}}\, \partial^{a_p}\big(z_{\sigma_p(i)}\big),$$

which is 1 if $\sigma_p(i) = a_p$, -1 if $\sigma_p(i) = a_p + 1$, and 0 otherwise. It remains to see that $\sigma_p^{-1}(a_p) = a$ and $\sigma_p^{-1}(a_p + 1) = b$. The word $(a_1, \ldots, a_p)$ is reduced and ends with a_p, so $\tau = s_{a_1} \cdots s_{a_p}$ satisfies

$\tau(a_p) > \tau(a_p+1)$; since $w = \tau\sigma_p$ this says $w(\sigma_p^{-1}(a_p)) > w(\sigma_p^{-1}(a_p+1))$. As $w(a) = u(b) > u(a) = w(b)$, the claim follows. □

7.2. **Factorial elementary polynomials and dominant permutations.**

Lemma 7.3. *For $0 \le p \le k$,*

$$E_{p,k}(x;z) = \sum_{m=0}^{p} (-1)^{p-m} h_{p-m}(z_1,\dots,z_{k-p+1})\, e_m(x_1,\dots,x_k),$$

$$\widetilde{E}_{p,k}(x;z) = \sum_{m=0}^{p} (-1)^{p-m} h_{p-m}(z_1,\dots,z_{k-p+1})\, e^{(q)}_m(x_1,\dots,x_k).$$

In particular $E_{p,k}(x;z)$ and $\widetilde{E}_{p,k}(x;z)$ are symmetric in $z_1,\dots,z_{k-p+1}$, and $\widetilde{E}_{p,k}(x;z) = \mathcal{Q}\big(E_{p,k}(x;z)\big)$.

Proof. Lemma 2.1 with $y = 0$ gives $\widetilde{\mathfrak{S}}_w(x;z) = \sum_{v_1\cdot v_2 = w} \mathfrak{S}_{v_1}(0;z)\,\widetilde{\mathfrak{S}}_{v_2}(x)$. The permutation $c[p,k] = s_{k-p+1}\cdots s_k$ has a unique reduced word, so its reduced factorizations are $v_1 = s_{k-p+1}\cdots s_j$, $v_2 = s_{j+1}\cdots s_k$ with $k-p \le j \le k$. Here v_2 is the Grassmannian permutation with descent at k whose diagram is a single column of height $m = k-j$, so $\mathfrak{S}_{v_2}(x) = e_m(x_1,\dots,x_k)$ and $\widetilde{\mathfrak{S}}_{v_2}(x) = e^{(q)}_m(x_1,\dots,x_k)$ [13]; and $\mathfrak{S}_{v_1}(0;z) = (-1)^{\ell(v_1)}\,\mathfrak{S}_{v_1^{-1}}(z)$, where $v_1^{-1} = s_j\cdots s_{k-p+1}$ is the Grassmannian permutation with descent at $k-p+1$ whose diagram is a single row of length $\ell(v_1) = p-m$, so that $\mathfrak{S}_{v_1^{-1}}(z) = h_{p-m}(z_1,\dots,z_{k-p+1})$. Summing over j gives the second formula, and the first is its specialization at $q=0$. The symmetry is clear from the formulas, and the last assertion holds because $\mathcal{Q}$ fixes $\mathbb{Z}[z]$ and sends the standard elementary monomial $e_m(x_1,\dots,x_k)$ to $e^{(q)}_m(x_1,\dots,x_k)$. □

Lemma 7.4 (Dominant permutations)**.** *Let μ be a dominant permutation with $\mathfrak{c}(\mu) = \theta$, $\theta_1 \ge \cdots \ge \theta_k > 0$. Then μ^{-1} is dominant, the i-th column of its diagram has height θ_i, and*

$$\mathfrak{S}_{\mu^{-1}}(x;z) = \prod_{i=1}^{k} E_{\theta_i,\theta_i}(x;z_i) = \prod_{i=1}^{k}\prod_{j=1}^{\theta_i}(x_j - z_i).$$

If moreover $\theta_1 > \cdots > \theta_k$, then $\widetilde{\mathfrak{S}}_{\mu^{-1}}(x;z) = \prod_{i=1}^{k} \widetilde{E}_{\theta_i,\theta_i}(x;z_i)$.

Proof. A permutation is dominant if and only if its diagram is the Young diagram of its code, placed in the upper left corner, and for such a permutation ρ one has $\mathfrak{S}_\rho(x;z) = \prod_{(a,b)\in D(\rho)}(x_a - z_b)$ [23]. The diagram of μ^{-1} is the transpose of that of μ, so $D(\mu^{-1}) = \{(j,i) \mid 1 \le i \le k,\ 1 \le j \le \theta_i\}$, which is again a Young diagram in the upper left corner; this gives the first assertions and the factorization. For the quantum statement, expand each factor by Lemma 3.3 at $q=0$: $E_{\theta_i,\theta_i}(x;z_i) = \sum_m (-z_i)^{\theta_i - m} e_m(x_1,\dots,x_{\theta_i})$. Multiplying out, each term $\prod_i e_{m_i}(x_1,\dots,x_{\theta_i})$ is a standard elementary monomial because the alphabets $x_1,\dots,x_{\theta_i}$ have pairwise distinct lengths, and $\mathcal{Q}$ sends it to $\prod_i e^{(q)}_{m_i}(x_1,\dots,x_{\theta_i})$. Hence $\widetilde{\mathfrak{S}}_{\mu^{-1}}(x;z) = \mathcal{Q}(\mathfrak{S}_{\mu^{-1}}(x;z)) = \prod_i \sum_m (-z_i)^{\theta_i - m} e^{(q)}_m(x_1,\dots,x_{\theta_i}) = \prod_i \widetilde{E}_{\theta_i,\theta_i}(x;z_i)$ by Lemma 3.3. □

7.3. **The skew divided differences of $E_{k,k}(x;z_i)$.** For $i \ge 1$ and an integer $b \ne i$ put $\epsilon_i(b) = 1$ if $b < i$ and $\epsilon_i(b) = -1$ if $b > i$. If $u \overset{i}{\Rightarrow} w$ with chain $b_1,\dots,b_k$ as in Definition 4.4, then $\prod_j \epsilon_i(b_j) = (-1)^{\ell(u,w)-\sigma_i(u,w)}$. The heart of the matter is the following lemma, which governs the cancellation of the chains through position i that revisit a position.

Lemma 7.5 (Rotation)**.** *Let $\psi \in S_\infty$, $i \ge 1$, and let $\gamma_1,\dots,\gamma_m$ ($m \ge 2$) be distinct integers different from i, with indices read modulo m. For $1 \le r \le m$ let*

$$R_r = (\gamma_r, \gamma_{r+1}, \dots, \gamma_{r+m-1}, \gamma_r),$$

a sequence of length $m+1$, and call R_r valid *if the sequence of permutations ψ, $\psi t_{i\gamma_r}$, $\psi t_{i\gamma_r} t_{i\gamma_{r+1}}$, … obtained by applying its entries in turn is a chain of Bruhat covers. Then:*

- (i) *$t_{i\gamma_r} t_{i\gamma_{r+1}} \cdots t_{i\gamma_{r+m-1}} t_{i\gamma_r} = C$ for every r, where C is the cycle $\gamma_j \mapsto \gamma_{j-1}$ ($1 \le j \le m$), which fixes i. In particular all valid R_r lead from ψ to the same permutation ψC.*
- (ii) *Listed in cyclic order of r, the valid R_r alternate between those with $\gamma_r < i$ and those with $\gamma_r > i$. In particular the number of valid R_r with $\gamma_r < i$ equals the number of valid R_r with $\gamma_r > i$.*

Proof. (i) Since $t_{ib}t_{ic}t_{ib} = t_{bc}$ for distinct $b, c \neq i$, the product equals $t_{\gamma_r\gamma_{r+1}}t_{\gamma_r\gamma_{r+2}}\cdots t_{\gamma_r\gamma_{r+m-1}}$, which in the notation (6.1) is the cycle $(\gamma_{r+m-1}, \dots, \gamma_{r+1}, \gamma_r)$, sending $\gamma_{r+m-1} \mapsto \gamma_{r+m-2} \mapsto \cdots \mapsto \gamma_{r+1} \mapsto \gamma_r \mapsto \gamma_{r+m-1}$.

(ii) Put $c = \psi(i)$ and $v_j = \psi(\gamma_j)$. Fix r and follow R_r. Before the s-th step ($1 \le s \le m$) the position i carries v_{r+s-2} (that is, c when $s = 1$), and the position γ_{r+s-1}, which has not been moved yet, carries v_{r+s-1}; before the last step the position i carries v_{r-1} and the position γ_r carries c. Call a step exchanging the position i with a position g *ascending* if the value at g exceeds the value at i when $g > i$, and is smaller than it when $g < i$. By Lemma 6.2 an ascending step increases the length by at least 1, a non-ascending step decreases it, and a step is a Bruhat cover exactly when it is ascending and increases the length by exactly 1. The total change in length along R_r is $\ell(\psi C) - \ell(\psi)$ by (i), independently of r. Consequently R_r is valid if and only if all of its $m + 1$ steps are ascending and $\ell(\psi C) - \ell(\psi) = m + 1$.

By the description of the values before each step, the steps $2, \dots, m$ of R_r are all ascending if and only if

$$v_j > v_{j-1} \iff \gamma_j > i \qquad \text{for all } j \neq r, \tag{7.2}$$

while the first step is ascending if and only if ($v_r > c \iff \gamma_r > i$) and the last if and only if ($c > v_{r-1} \iff \gamma_r > i$). The last two conditions together say precisely that c lies strictly between v_{r-1} and v_r and that (7.2) holds for $j = r$ as well. Therefore R_r is valid if and only if

$$\ell(\psi C) - \ell(\psi) = m + 1, \qquad \text{(7.2) holds for all } j, \qquad \text{and } c \text{ lies strictly between } v_{r-1} \text{ and } v_r.$$

The first two conditions do not involve r. If they hold, the valid indices r are exactly those at which the cyclic sequence $v_1, \dots, v_m$, none of whose terms equals c, crosses the level c; and consecutive crossings of a cyclic sequence are alternately upward ($v_{r-1} < c < v_r$) and downward ($v_{r-1} > c > v_r$). By (7.2) the upward crossings are those with $\gamma_r > i$ and the downward ones those with $\gamma_r < i$. □

Theorem 7.6. *Let $u, w \in S_\infty$, $i \ge 1$ and $k \ge 0$. Then $\partial_u^w\big(E_{k,k}(x; z_i)\big) = 0$ unless $u \overset{i}{\Rightarrow} w$, in which case*

$$\partial_u^w\big(E_{k,k}(x; z_i)\big) = (-1)^{\ell(u,w)-\sigma_i(u,w)}\, E_{k-\ell(u,w),\,k}\big(x; z_{D_i(u,w)}\big).$$

Proof. Write $E_{k,k}(x; z_i) = f_1 \cdots f_k$ with $f_j = x_j - z_i$. By Lemma 7.1(iii),

$$\partial_u^w\big(E_{k,k}(x; z_i)\big) = \sum_{u = u_0 \le \cdots \le u_k = w} \prod_{j=1}^{k} \partial_{u_{j-1}}^{u_j}(x_j - z_i),$$

and by Lemmas 7.2 and 7.1(ii) the factor $\partial_{u_{j-1}}^{u_j}(x_j - z_i)$ equals $x_j - z_{u_j(i)}$ if $u_j = u_{j-1}$, equals $\epsilon_i(b)$ if $u_j = u_{j-1}t_{ib}$ with $\ell(u_j) = \ell(u_{j-1}) + 1$, and vanishes otherwise. Call a sequence $(u_0, \dots, u_k)$ in which every step is of one of these two kinds an *i-walk* from u to w. Thus the sum is over the i-walks, and a walk contributes the product of $x_j - z_{u_j(i)}$ over its trivial steps and of $\epsilon_i(b)$ over its nontrivial steps.

An *i-chain* from u to w is a sequence $\beta = (\beta_1, \dots, \beta_\ell)$ of integers different from i such that, with $u^{(0)} = u$ and $u^{(s)} = u^{(s-1)}t_{i\beta_s}$, every step $u^{(s-1)} \lessdot u^{(s)}$ is a Bruhat cover and $u^{(\ell)} = w$; repetitions among the β_s are allowed. Put $c_s(\beta) = u^{(s)}(i)$, the value at position i after s steps, and $\epsilon(\beta) = \prod_s \epsilon_i(\beta_s)$. Every i-walk determines the i-chain formed by its nontrivial steps, and conversely an i-chain of length $\ell \le k$ arises from exactly the $\binom{k}{\ell}$ walks obtained by choosing the positions $j_1 < \cdots < j_p$ ($p = k - \ell$) of the trivial steps. In such a walk the trivial step at j_r is preceded by $j_r - r$ nontrivial steps, so it contributes the factor $x_{j_r} - z_{c_{j_r - r}(\beta)}$. Summing over the choices of $j_1 < \cdots < j_p$ and comparing with the explicit formula for $E_{p,k}$ in Definition 3.1, the walks with chain β contribute in total

$$\epsilon(\beta) \sum_{1 \le j_1 < \cdots < j_p \le k} \prod_{r=1}^{p} \big(x_{j_r} - z_{c_{j_r - r}(\beta)}\big) = \epsilon(\beta)\, E_{k-\ell,k}\big(x; z_{c_0(\beta)}, z_{c_1(\beta)}, \dots, z_{c_\ell(\beta)}\big).$$

Hence

$$\partial_u^w\big(E_{k,k}(x; z_i)\big) = \sum_{\beta} \epsilon(\beta)\, E_{k-\ell(\beta),k}\big(x; z_{c_0(\beta)}, \dots, z_{c_{\ell(\beta)}(\beta)}\big), \tag{7.3}$$

the sum being over all i-chains β from u to w of length $\ell(\beta) \le k$. By Lemma 7.3 the summand depends on $c_0(\beta), \dots, c_\ell(\beta)$ only through their multiset.

Chains without repetitions. If $\beta_1, \dots, \beta_\ell$ are distinct, then β is a witness for $u \overset{i}{\Rightarrow} w$, and by Definition 4.4 it is the unique one; moreover $\ell = \ell(u, w)$, $\epsilon(\beta) = (-1)^{\ell(u,w)-\sigma_i(u,w)}$, and since the position β_s is untouched

before the s-th step, $c_s(\beta) = u(\beta_s)$, so that $\{c_0(\beta), \dots, c_\ell(\beta)\} = D_i(u, w)$. So the chains without repetitions contribute exactly the right-hand side of the theorem if $u \overset{i}{\Rightarrow} w$ and $\ell(u, w) \le k$, and nothing otherwise; if $u \overset{i}{\Rightarrow} w$ and $\ell(u, w) > k$ then both sides of the theorem vanish, the left by degree. It remains to show that the chains with a repetition contribute zero in total.

Chains with repetitions. Let β be an i-chain from u to w with a repetition, and let s be the least index for which $\beta_s \in \{\beta_1, \dots, \beta_{s-1}\}$, say $\beta_s = \beta_t$ with $t < s$. Put $m = s - t$ and $(\gamma_1, \dots, \gamma_m) = (\beta_t, \dots, \beta_{s-1})$; these are distinct, and $m \ge 2$ because two consecutive covers cannot use the same transposition. The segment $(\beta_t, \dots, \beta_s)$ of β is the sequence R_1 of Lemma 7.5 for $\psi = u^{(t-1)}$, and it is valid. The alternation in Lemma 7.5(ii) pairs each valid R_r with $\gamma_r > i$ with the next valid sequence in cyclic order, whose entry lies below i, and each valid R_r with $\gamma_r < i$ with the previous one; this is a fixed-point-free involution on the valid sequences which exchanges the two sides of i. Let R_{r^*} be the sequence paired with R_1, and let β^* be obtained from β by replacing the segment $(\beta_t, \dots, \beta_s)$ with R_{r^*}. By Lemma 7.5(i) the new segment again leads from $u^{(t-1)}$ to $u^{(t-1)}C = u^{(s)}$, so β^* is an i-chain from u to w of the same length as β. The entries of β^* preceding the index s form the same set as those of β and are distinct, and $\beta^*_s = \beta^*_t$, so the first repetition of β^* occurs at the same index s with the same t, the same ψ, and the same cyclic sequence γ; hence $\beta^{**} = \beta$, and $\beta \mapsto \beta^*$ is a fixed-point-free involution on the set of i-chains with repetitions. The values $c_0, \dots, c_{t-1}$ and $c_s, \dots, c_\ell$ are the same for β and β^*, since the permutations $u^{(0)}, \dots, u^{(t-1)}$ and $u^{(s)}, \dots, u^{(\ell)}$ are; and $c_t, \dots, c_{s-1}$ are the values $\psi(\gamma_1), \dots, \psi(\gamma_m)$ in some order for both chains. So the multisets of values coincide, and the E-factors in (7.3) agree. Finally $\epsilon(\beta^*) = -\epsilon(\beta)$: the two segments consist of the entries $\gamma_1, \dots, \gamma_m$, each once, together with one additional entry, which is γ_1 for β and γ_{r^*} for β^*, and these lie on opposite sides of i. Hence the chains with repetitions cancel in pairs in (7.3), and the theorem follows. □

7.4. **Conclusion of the proofs.**

Proof of Theorem 4.5. From the definition of the double Schubert polynomials, for every w and a,

$$\partial^a \mathfrak{S}_w(x; z) = \begin{cases} -\mathfrak{S}_{s_a w}(x; z) & \text{if } s_a w < w, \\ 0 & \text{if } s_a w > w\,: \end{cases} \tag{7.4}$$

indeed $\mathfrak{S}_w = (-1)^{\ell(w_0)-\ell(w)} \partial^{ww_0} \mathfrak{S}_{w_0}$, and $\partial^a \partial^{ww_0}$ equals $\partial^{s_a ww_0}$ if $\ell(s_a ww_0) > \ell(ww_0)$, which is the case $s_a w < w$, and equals 0 otherwise. Since $\mu = v^{-1} \cdot y$ is reduced, so is $\mu^{-1} = y^{-1} \cdot v$; writing $y = s_{a_1} \cdots s_{a_r}$ reduced, the permutations $s_{a_j} s_{a_{j+1}} \cdots s_{a_r} \mu^{-1}$, $j = r+1, r, \dots, 1$, therefore decrease in length by one at each step, from μ^{-1} down to $y\mu^{-1} = v$. Applying (7.4) r times and then Lemma 7.4,

$$\mathfrak{S}_v(x; z) = (-1)^{\ell(y)}\, \partial^y\, \mathfrak{S}_{\mu^{-1}}(x; z) = (-1)^{\ell(y)}\, \partial^y \Big(\prod_{i=1}^{k} E_{\theta_i, \theta_i}(x; z_i) \Big).$$

Now expand by the ascending Leibniz formula, Lemma 7.1(iii) with $u = 1$ and $w = y$, and evaluate each factor by Theorem 7.6:

$$\begin{aligned} \partial^y \Big(\prod_{i=1}^{k} E_{\theta_i, \theta_i}(x; z_i) \Big) &= \sum_{1 = v_0 \le \cdots \le v_k = y} \prod_{i=1}^{k} \partial^{v_i}_{v_{i-1}} \big(E_{\theta_i, \theta_i}(x; z_i) \big) \\ &= \sum_{1 = v_0 \overset{1}{\Rightarrow} v_1 \overset{2}{\Rightarrow} \cdots \overset{k}{\Rightarrow} v_k = y} \prod_{i=1}^{k} (-1)^{\ell(v_{i-1}, v_i) - \sigma_i(v_{i-1}, v_i)} E_{\theta_i - \ell(v_{i-1}, v_i),\, \theta_i} \big(x; z_{D_i(v_{i-1}, v_i)} \big). \end{aligned}$$

Since $\sum_i \ell(v_{i-1}, v_i) = \ell(y)$, the signs combine to $\prod_i (-1)^{\sigma_i(v_{i-1}, v_i)}$; and $y = v\mu$. □

Proof of Theorem 4.6. A strict dominant permutation is dominant, so Theorem 4.5 applies to v and μ and expresses $\mathfrak{S}_v(x; z)$ as the sum, over the chains $1 = v_0 \overset{1}{\Rightarrow} \cdots \overset{k}{\Rightarrow} v_k = v\mu$, of $\prod_i (-1)^{\sigma_i} E_{\widetilde{\theta}_i - \ell_i, \widetilde{\theta}_i}(x; z_{D_i})$, where $\ell_i = \ell(v_{i-1}, v_i)$, $\sigma_i = \sigma_i(v_{i-1}, v_i)$ and $D_i = D_i(v_{i-1}, v_i)$. Apply $\mathcal{Q}$ to both sides. The left-hand side becomes $\widetilde{\mathfrak{S}}_v(x; z)$. On the right, by Lemma 7.3 each factor is $E_{\widetilde{\theta}_i - \ell_i, \widetilde{\theta}_i}(x; z_{D_i}) = \sum_m c_{i,m}\, e_m(x_1, \dots, x_{\widetilde{\theta}_i})$ with coefficients $c_{i,m} \in \mathbb{Z}[z]$, and multiplying out, each term $\prod_i e_{m_i}(x_1, \dots, x_{\widetilde{\theta}_i})$ of the product over i is a standard

elementary monomial because $\widetilde{\theta}_1 > \cdots > \widetilde{\theta}_k$. Hence $\mathcal{Q}$ sends the product to $\prod_i \sum_m c_{i,m}\, e_m^{(q)}(x_1, \ldots, x_{\widetilde{\theta}_i}) = \prod_i \widetilde{E}_{\widetilde{\theta}_i - \ell_i, \widetilde{\theta}_i}(x; z_{D_i})$, again by Lemma 7.3. □

We prove Theorem 4.7 by reaching the medium shape from the strict shape by divided differences, one for each retained pair; this also explains the form of the doubled factor. For $k \geq 1$ and $i \geq 1$ write $\widetilde{\mathfrak{S}}_{d_2[k]}(x; z_i, z_{i+1})$ for the quantum double Schubert polynomial of the $k \times 2$ rectangle (Section 8) with (z_1, z_2) replaced by (z_i, z_{i+1}), i.e. $\widetilde{\mathfrak{S}}_{d_2[k]}(x; z_i, z_{i+1}) = \mathcal{Q}\big(E_{k,k}(x; z_i) E_{k,k}(x; z_{i+1})\big) = \widetilde{E}_{k,k;k}(x; z_i, z_{i+1})$.

Corollary 7.7 (Factorization for the medium shape)**.** *Let μ be a dominant permutation of medium shape with $\overline{\theta} = \mathfrak{c}(\mu)$, paired indices $\mathcal{P}$ and single indices $\mathcal{S}$ as in Theorem 4.7, and let $\nu = \mu \prod_{i \in \mathcal{P}} s_i$. Then ν is strict dominant, $\mathfrak{c}(\nu)$ is $\overline{\theta}$ with the first entry of each retained pair increased by one, $\mu = \nu \prod_{i \in \mathcal{P}} s_i$ with $\ell(\mu) = \ell(\nu) - |\mathcal{P}|$, and*

$$\widetilde{\mathfrak{S}}_{\mu^{-1}}(x; z) = (-1)^{|\mathcal{P}|} \Big(\prod_{i \in \mathcal{P}} \partial^i \Big)\, \widetilde{\mathfrak{S}}_{\nu^{-1}}(x; z) = \prod_{i \in \mathcal{S}} \widetilde{E}_{\overline{\theta}_i, \overline{\theta}_i}(x; z_i) \prod_{i \in \mathcal{P}} \widetilde{\mathfrak{S}}_{d_2[\overline{\theta}_i]}(x; z_i, z_{i+1}).$$

In particular, for $\mu = \overline{\mu}(v^{-1})$ one has $\nu = \widetilde{\mu}(v^{-1})$ by Proposition 4.3(iii).

Proof. For $i \in \mathcal{P}$ with $k = \overline{\theta}_i = \overline{\theta}_{i+1}$ we have $\mu(i) < \mu(i+1)$, so $\ell(\mu s_i) = \ell(\mu) + 1$ and $\mathfrak{c}(\mu s_i)$ is $\mathfrak{c}(\mu)$ with $(\mathfrak{c}_i, \mathfrak{c}_{i+1}) = (k, k)$ replaced by $(k+1, k)$; the indices in $\mathcal{P}$ are pairwise at distance ≥ 2, so the s_i commute and act on disjoint pairs of positions, which gives the assertions about ν (its code is strictly decreasing on its nonzero part because the entry preceding a retained pair is $\geq k+2$ and the entry following it is $\leq k-1$). Enumerate $\mathcal{P} = \{i_1 < \cdots < i_r\}$ and put $\mu_0 = \mu$, $\mu_t = \mu_{t-1} s_{i_t}$, so $\mu_r = \nu$. Since $s_{i_t} \mu_t^{-1} = (\mu_t s_{i_t})^{-1} = \mu_{t-1}^{-1}$ is shorter than μ_t^{-1}, (7.4) gives $\mathfrak{S}_{\mu_{t-1}^{-1}}(x; z) = -\partial^{i_t} \mathfrak{S}_{\mu_t^{-1}}(x; z)$, and iterating, $\mathfrak{S}_{\mu^{-1}}(x; z) = (-1)^r \partial^{i_1} \cdots \partial^{i_r} \mathfrak{S}_{\nu^{-1}}(x; z)$. The map $\mathcal{Q}$ acts on the x-variables and is $\mathbb{Z}[q][z]$-linear, while the ∂^a act on the z-variables and are $\mathbb{Z}[q][x]$-linear; hence they commute, and applying $\mathcal{Q}$ gives the first equality. For the second, Lemma 7.4 gives $\widetilde{\mathfrak{S}}_{\nu^{-1}}(x; z) = \prod_j \widetilde{E}_{\mathfrak{c}_j(\nu), \mathfrak{c}_j(\nu)}(x; z_j)$, in which the factor of index $j \notin \mathcal{P} \cup (\mathcal{P}+1)$ is $\widetilde{E}_{\overline{\theta}_j, \overline{\theta}_j}(x; z_j)$ and the factors of indices $i, i+1$ for $i \in \mathcal{P}$, $k = \overline{\theta}_i$, are $\widetilde{E}_{k+1,k+1}(x; z_i) \widetilde{E}_{k,k}(x; z_{i+1})$. The operator ∂^i involves only z_i, z_{i+1}, so it acts on this pair of factors alone, and the operators for distinct $i \in \mathcal{P}$ act on disjoint sets of variables. Finally, with $P_m(t) = \prod_{j \leq m} (x_j - t)$,

$$P_{k+1}(z_i) P_k(z_{i+1}) - P_{k+1}(z_{i+1}) P_k(z_i) = P_k(z_i) P_k(z_{i+1}) \big[(x_{k+1} - z_i) - (x_{k+1} - z_{i+1}) \big] = (z_{i+1} - z_i)\, P_k(z_i) P_k(z_{i+1}),$$

so $-\partial^i \big(P_{k+1}(z_i) P_k(z_{i+1}) \big) = P_k(z_i) P_k(z_{i+1})$. Applying $\mathcal{Q}$, which is multiplicative on $P_{k+1}(z_i) P_k(z_{i+1})$ (distinct alphabets, Lemma 7.4) and commutes with ∂^i, gives $-\partial^i \big(\widetilde{E}_{k+1,k+1}(x; z_i) \widetilde{E}_{k,k}(x; z_{i+1}) \big) = \mathcal{Q}\big(P_k(z_i) P_k(z_{i+1}) \big) = \widetilde{\mathfrak{S}}_{d_2[k]}(x; z_i, z_{i+1})$. □

Proof of Theorem 4.7. Exactly as in the proof of Theorem 4.5, $\widetilde{\mathfrak{S}}_v(x; z) = (-1)^{\ell(y)} \partial^y \widetilde{\mathfrak{S}}_{\mu^{-1}}(x; z)$ (the identity (7.4) holds for the quantum polynomials, $\mathcal{Q}$ commuting with the z-divided differences). Insert the factorization of Corollary 7.7 and expand by the ascending Leibniz formula, Lemma 7.1(iii), treating each doubled factor as a single factor. The skew operators ∂_u^w are $\mathbb{Z}[q][x]$-linear and so commute with $\mathcal{Q}$; hence, by Lemma 7.1(iii) once more and Theorem 7.6,

$$\begin{aligned} \partial_u^w\, \widetilde{\mathfrak{S}}_{d_2[k]}(x; z_i, z_{i+1}) &= \mathcal{Q}\Big(\sum_{u \leq u' \leq w} \partial_u^{u'} \big(E_{k,k}(x; z_i) \big)\, \partial_{u'}^w \big(E_{k,k}(x; z_{i+1}) \big) \Big) \\ &= \sum_{u \overset{i}{\Rightarrow} u' \overset{i+1}{\dashrightarrow} w} (-1)^{\ell_i + \ell_{i+1} - \sigma_i - \sigma_{i+1}}\, \widetilde{E}_{k-\ell_i,\, k-\ell_{i+1};\, k}\big(x; z_{D_i}, z_{D_{i+1}}\big), \end{aligned}$$

with ℓ_i, σ_i, D_i attached to $u \overset{i}{\Rightarrow} u'$ and $\ell_{i+1}, \sigma_{i+1}, D_{i+1}$ to $u' \overset{i+1}{\dashrightarrow} w$, while the single factors are evaluated by Theorem 7.6 and Lemma 7.3 as in the proof of Theorem 4.6. Collecting the signs as in the proof of Theorem 4.5 gives the theorem. In this form it is literally Theorem 4.6 for $\nu = \mu \prod_{i \in \mathcal{P}} s_i$ with the divided differences ∂^i, $i \in \mathcal{P}$, applied first: the two Pieri steps at columns $i, i+1$ of the strict shape, in $k+1$ and k variables, are replaced by the doubled step in k variables. □

8. General-degree and doubled Pieri formulas

In this section we record the two forms of the Pieri rule that the quantum multiplication algorithm of Section 4 actually uses: the rule for a quantum factorial elementary polynomial of arbitrary degree $p \le k$, and a "doubled" rule for the quantization of a product of two such polynomials in the *same* k variables. Recall from Definition 3.1 that $n_k(u,w) = k - |\mathrm{fix}_k(u,w)|$ is the number of positions $i \le k$ at which u and w differ, equivalently the number of transpositions in a chain witnessing $u \overset{k}{\rightsquigarrow} w$, and that $E_{p,k}(y_A; z)$ denotes the factorial elementary polynomial in the variables y_a, $a \in A$ (with $|A| = k$), and $z_1, z_2, \ldots$.

Theorem 8.1 (Quantum Pieri formula, general degree)**.** *Suppose $u \in S_\infty$ and $0 \le p \le k$. Then*

$$\widetilde{\mathfrak{S}}_u(x;y)\,\widetilde{E}_{p,k}(x;z) = \sum_{\substack{u \overset{k}{\rightsquigarrow} w \\ n_k(u,w) \le p}} q_k(u,w)\, E_{p-n_k(u,w),\,k-n_k(u,w)}\big(y_{\mathrm{fix}_k(u,w)}; z\big)\ \widetilde{\mathfrak{S}}_w(x;y).$$

For $p = k$ the coefficient is $q_k(u,w)\mathbf{E}(y_{\mathrm{fix}_k(u,w)}; z)$ and this is Theorem 3.4; at $q = 0$ it is the Pieri formula of [33, Theorem 7.1]. This is the form implemented in [34] (the routines `elem_sym_perms_q` and `elem_sym_func_q`). We deduce the general case from the case $p = k$.

Proof. Let $X = (X_1, \ldots, X_k)$ be indeterminates. The factorial elementary polynomial

$$E_{p,k}(X; z) = \sum_{1 \le i_1 < \cdots < i_p \le k}\ \prod_{j=1}^{p} (X_{i_j} - z_{i_j - j + 1})$$

is symmetric in X and of degree at most one in each X_i, hence is a $\mathbb{Z}[z]$-linear combination of the elementary symmetric polynomials:

$$E_{p,k}(X; z) = \sum_{j=0}^{p} c_{p,k,j}(z)\, e_j(X_1, \ldots, X_k), \qquad c_{p,k,j}(z) \in \mathbb{Z}[z]. \tag{8.1}$$

Two consequences. First, applying the quantization map $\mathcal{Q}$ (which is $\mathbb{Z}[z]$-linear and sends $e_j(x_1, \ldots, x_k)$ to $e_j^{(q)}(x_1, \ldots, x_k)$) gives

$$\widetilde{E}_{p,k}(x; z) = \sum_{j=0}^{p} c_{p,k,j}(z)\, e_j^{(q)}(x_1, \ldots, x_k). \tag{8.2}$$

Second, for $0 \le n \le k$ extract from (8.1) the coefficient of the monomial $X_{k-n+1}X_{k-n+2}\cdots X_k$. On the right it is $\sum_j c_{p,k,j}(z)\, e_{j-n}(X_1, \ldots, X_{k-n})$. On the left, a term of $E_{p,k}(X; z)$ contains this monomial exactly when $\{i_{p-n+1}, \ldots, i_p\} = \{k-n+1, \ldots, k\}$, and then the remaining factors $\prod_{j \le p-n}(X_{i_j} - z_{i_j - j + 1})$ with $i_1 < \cdots < i_{p-n} \le k - n$ are precisely the terms of $E_{p-n,k-n}(X_1, \ldots, X_{k-n}; z)$. Hence, for every set A of $k - n$ indices,

$$\sum_{j=0}^{p} c_{p,k,j}(z)\, e_{j-n}(y_A) = E_{p-n,\,k-n}(y_A; z), \tag{8.3}$$

both sides being 0 when $n > p$.

Now put $z_1 = -t$ in Theorem 3.4. By Lemma 3.3 the left side becomes $\sum_{j=0}^{k} t^{k-j}\ \widetilde{\mathfrak{S}}_u(x;y)\, e_j^{(q)}(x_1, \ldots, x_k)$, while on the right $\mathbf{E}(y_{\mathrm{fix}_k(u,w)}; z) = \prod_{a \in \mathrm{fix}_k(u,w)}(y_a + t) = \sum_i t^{\,k - n_k(u,w) - i} e_i(y_{\mathrm{fix}_k(u,w)})$, since $|\mathrm{fix}_k(u,w)| = k - n_k(u,w)$. Comparing the coefficients of t^{k-j} gives, for $0 \le j \le k$,

$$\widetilde{\mathfrak{S}}_u(x;y)\, e_j^{(q)}(x_1, \ldots, x_k) = \sum_{u \overset{k}{\rightsquigarrow} w} q_k(u,w)\, e_{j - n_k(u,w)}\big(y_{\mathrm{fix}_k(u,w)}\big)\ \widetilde{\mathfrak{S}}_w(x;y). \tag{8.4}$$

Multiplying (8.4) by $c_{p,k,j}(z)$, summing over j and using (8.2) and then (8.3) with $A = \mathrm{fix}_k(u,w)$ and $n = n_k(u,w)$ yields the theorem. □

The doubled formula. Recall that the quantization map $\mathcal{Q}$ of Fomin, Gelfand and Postnikov (Lemma 2.1) is linear over the coefficient variables, sends each standard elementary monomial $\prod_j e_{i_j}(x_1,\dots,x_j)$ to $\prod_j e^{(q)}_{i_j}(x_1,\dots,x_j)$, and satisfies $\mathcal{Q}(\mathfrak{S}_w(x;z)) = \widetilde{\mathfrak{S}}_w(x;z)$. In particular $\mathcal{Q}(E_{p,k}(x;z)) = \widetilde{E}_{p,k}(x;z)$, and more generally $\mathcal{Q}$ is multiplicative on a product of factorial elementary polynomials $E_{p_j,k_j}(x;z^{(j)})$, $j=1,2,\dots$, whose numbers of variables k_j are *pairwise distinct*, since such a product expands into standard elementary monomials. It is *not* multiplicative when two factors have the same number of variables: in general

$$\mathcal{Q}\big(E_{p_1,k}(x;z)E_{p_2,k}(x;z')\big) \neq \widetilde{E}_{p_1,k}(x;z)\,\widetilde{E}_{p_2,k}(x;z').$$

The following definition isolates the terms of the iterated Pieri formula (8.5) below which survive in the quantization of the product.

Definition 8.2 (Compatible pairs of chains)**.** Let $u \overset{k}{\rightsquigarrow} v \overset{k}{\rightsquigarrow} w$. We say that the pair of chains is *compatible* if for every nontrivial cycle d of $v^{-1}w$ and every nontrivial cycle c of $u^{-1}v$ with $\max c = \max d$, one has $c \cap d = \{\max d\}$. In words: a cycle of the second step which shares its top with a cycle of the first step may meet that cycle only in the top. (A cycle of $v^{-1}w$ whose top is not the top of any cycle of $u^{-1}v$ is unrestricted.)

Throughout this subsection we write $\widetilde{E}_{k,k}(x;z_1)$ and $\widetilde{E}_{k,k}(x;z_2)$ for the quantum factorial elementary polynomial $\widetilde{E}_{k,k}(x;z) = \mathcal{Q}\big(\prod_{i\le k}(x_i - z_1)\big)$ and for the same polynomial with z_1 replaced by z_2; thus $\widetilde{E}_{k,k}(x;z_2)$ is $\widetilde{E}_{k,k}(x;z')$ for the shifted alphabet $z' = (z_2, z_3, \dots)$. For a pair of chains $u \overset{k}{\rightsquigarrow} v \overset{k}{\rightsquigarrow} w$ put

$$\omega_k(u,v,w) = q_k(u,v)\,q_k(v,w) \prod_{a\in \mathrm{fix}_k(u,v)} (y_a - z_1) \prod_{b\in \mathrm{fix}_k(v,w)} (y_b - z_2),$$

so that two applications of Theorem 3.4, the second with z_1 replaced by z_2, give

$$\widetilde{\mathfrak{S}}_u(x;y)\,\widetilde{E}_{k,k}(x;z_1)\,\widetilde{E}_{k,k}(x;z_2) = \sum_{u\overset{k}{\rightsquigarrow}v\overset{k}{\rightsquigarrow}w} \omega_k(u,v,w)\;\widetilde{\mathfrak{S}}_w(x;y). \tag{8.5}$$

Recall that $d_2[k]$ denotes the dominant permutation with $\mathfrak{c}(d_2[k]) = (2^k)$, whose diagram is a $k\times 2$ rectangle, so that

$$\mathfrak{S}_{d_2[k]}(x;z) = \prod_{i\le k}(x_i - z_1)(x_i - z_2) = E_{k,k}(x;z_1)E_{k,k}(x;z_2), \qquad \widetilde{\mathfrak{S}}_{d_2[k]}(x;z) = \mathcal{Q}\big(E_{k,k}(x;z_1)E_{k,k}(x;z_2)\big);$$

we set $d_2[0] = \mathrm{id}$.

Theorem 8.3 (Doubled quantum Pieri formula)**.** *For $u \in S_\infty$ and $k \ge 1$,*

$$\begin{aligned}\widetilde{\mathfrak{S}}_u(x;y)\;\widetilde{\mathfrak{S}}_{d_2[k]}(x;z) &= \sum_{\substack{u\overset{k}{\rightsquigarrow}v\overset{k}{\rightsquigarrow}w\\ \textit{compatible}}} \omega_k(u,v,w)\;\widetilde{\mathfrak{S}}_w(x;y)\\ &= \sum_{\substack{u\overset{k}{\rightsquigarrow}v\overset{k}{\rightsquigarrow}w\\ \textit{compatible}}} q_k(u,v)q_k(v,w) \prod_{a\in \mathrm{fix}_k(u,v)} (y_a - z_1) \prod_{b\in \mathrm{fix}_k(v,w)} (y_b - z_2)\;\widetilde{\mathfrak{S}}_w(x;y).\end{aligned}$$

Thus the doubled formula is obtained from the iterated formula (8.5) by *omitting* the incompatible pairs, exactly as the quantization $\widetilde{\mathfrak{S}}_{d_2[k]}$ of the product $E_{k,k}(x;z_1)E_{k,k}(x;z_2)$ omits terms from the product $\widetilde{E}_{k,k}(x;z_1)\widetilde{E}_{k,k}(x;z_2)$ of the quantizations. This is the rule implemented by the routine `double_elem_sym_q` of [34], and it is the "paired step" of the quantum algorithm of Section 4. We have verified the theorem by computer for all $u \in S_4$ and all $k \le 3$; in these cases the unrestricted sum (8.5) fails to equal the left-hand side in roughly half of the instances, so the compatibility condition is essential.

Example 8.4 (Compatible and incompatible pairs)**.** (a) $k = 1$. A step $u \overset{1}{\rightsquigarrow} v$ is either trivial or a single cover t_{1b}, so a cycle of $u^{-1}v$ and a cycle of $v^{-1}w$ with a common top B sharing the element 1 are both equal to $(1,B)$, i.e. $v = ut_{1B}$ and $w = u$; and t_{1B} can be a cover of both u and ut_{1B} only when $B = 2$ (if $B \ge 3$, the classical cover among the two requires no intermediate value in the interval between $u(1)$ and $u(B)$, while the quantum cover requires every intermediate value to lie in it). Thus the

incompatible pairs are exactly the round trips $u \overset{1}{\rightsquigarrow} us_1 \overset{1}{\rightsquigarrow} u$, in which one step is a classical and the other a quantum cover, both fixed sets are empty, and $\omega_1(u, us_1, u) = q_1$. Theorem 8.3 for $k = 1$ therefore says that $\widetilde{\mathfrak{S}}_u\,\widetilde{\mathfrak{S}}_{312}(x;z)$ is obtained from $\widetilde{\mathfrak{S}}_u\,\widetilde{E}_{1,1}(x;z_1)\,\widetilde{E}_{1,1}(x;z_2)$ by deleting the term $q_1\,\widetilde{\mathfrak{S}}_u$, in agreement with $\widetilde{\mathfrak{S}}_{312}(x;z) = \mathcal{Q}\big((x_1-z_1)(x_1-z_2)\big) = (x_1-z_1)(x_1-z_2) - q_1$, which is the case $k=1$ of the recurrence of Lemma 8.6 below. For $u = 21$ the seven compatible pairs and the rejected round trip $21 \overset{1}{\rightsquigarrow} 12 \overset{1}{\rightsquigarrow} 21$ are listed in Example 4.18.

(b) $k = 2$, $u = 1324$. The steps $u \overset{2}{\rightsquigarrow} v$ are: the trivial one; the single covers t_{13} (classical, $v = 2314$), t_{23} (quantum, weight q_2, $v = 1234$) and t_{24} (classical, $v = 1423$); and the two-step chains $t_{24}t_{13}$ ($v = 2413$) and $t_{13}t_{23}$ (weight q_2, $v = 2134$). Consider first $w = 3124$. Exactly two pairs $u \overset{2}{\rightsquigarrow} v \overset{2}{\rightsquigarrow} w$ exist, namely

$$1324 \xrightarrow{t_{13}} 2314 \xrightarrow{t_{23}t_{13}} 3124 \qquad \text{and} \qquad 1324 \xrightarrow{t_{13}t_{23}} 2134 \xrightarrow{t_{13}} 3124,$$

with $\omega_2 = q_2(y_3 - z_1)$ and $\omega_2 = q_2(y_1 - z_2)$ respectively ($\mathrm{fix}_2(1324, 2314) = \{3\}$, $\mathrm{fix}_2(2134, 3124) = \{1\}$, the other fixed sets being empty). In the first, the cycles $(1,3)$ of $u^{-1}v$ and $(1,2,3)$ of $v^{-1}w$ share the top $3 = k+1$ and the element 1; in the second, the cycles $(2,1,3)$ and $(1,3)$ do. Both pairs are incompatible, so the coefficient of $\widetilde{\mathfrak{S}}_{3124}$ in $\widetilde{\mathfrak{S}}_{1324}\,\widetilde{\mathfrak{S}}_{3412}(x;z)$ is 0, whereas the coefficient in $\widetilde{\mathfrak{S}}_{1324}\,\widetilde{E}_{2,2}(x;z_1)\widetilde{E}_{2,2}(x;z_2)$ is $q_2\big[(y_3-z_1)+(y_1-z_2)\big]$. The recurrence of Lemma 8.6 accounts for this difference by q_2 times the coefficient of $\widetilde{\mathfrak{S}}_{3124}$ in $\widetilde{\mathfrak{S}}_{1324}\,\widetilde{\mathfrak{S}}_{d_2[1]}(x;z)$, which by (a) is the sum over the two (compatible) level-1 pairs $1324 \overset{1}{\rightsquigarrow} 1324 \overset{1}{\rightsquigarrow} 3124$ and $1324 \overset{1}{\rightsquigarrow} 3124 \overset{1}{\rightsquigarrow} 3124$, namely $(y_1-z_1)+(y_3-z_2)$; and indeed $(y_3-z_1)+(y_1-z_2) = (y_1-z_1)+(y_3-z_2)$, although the two sums do not agree term by term. For $w = 2413$, by contrast, all four pairs

$$1324 \overset{2}{\rightsquigarrow} 1324 \xrightarrow{t_{24}t_{13}} 2413, \quad 1324 \xrightarrow{t_{13}} 2314 \xrightarrow{t_{24}} 2413, \quad 1324 \xrightarrow{t_{24}} 1423 \xrightarrow{t_{13}} 2413, \quad 1324 \xrightarrow{t_{24}t_{13}} 2413 \overset{2}{\rightsquigarrow} 2413$$

are compatible (the cycles of the two steps have distinct tops, or one step is trivial), and the coefficient of $\widetilde{\mathfrak{S}}_{2413}$ in both products is

$$(y_1-z_1)(y_3-z_1) + (y_3-z_1)(y_2-z_2) + (y_1-z_1)(y_4-z_2) + (y_2-z_2)(y_4-z_2).$$

Both coefficients were checked against the output of [34].

Corollary 8.5 (Doubled formula in lower degrees)**.** *For $u \in S_\infty$, $0 \le p_1, p_2 \le k$ and two alphabets z, z', with $\widetilde{E}_{p_1,p_2;k}(x;z,z') = \mathcal{Q}\big(E_{p_1,k}(x;z)E_{p_2,k}(x;z')\big)$ as in* (4.1)*, and writing $n_1 = n_k(u,v)$, $n_2 = n_k(v,w)$ for a pair $u \overset{k}{\rightsquigarrow} v \overset{k}{\rightsquigarrow} w$,*

$$\widetilde{\mathfrak{S}}_u(x;y)\,\widetilde{E}_{p_1,p_2;k}(x;z,z') = \sum_{\substack{u \overset{k}{\rightsquigarrow} v \overset{k}{\rightsquigarrow} w \ \text{compatible} \\ n_1 \le p_1,\ n_2 \le p_2}} q_k(u,v)\,q_k(v,w)\,E_{p_1-n_1,\,k-n_1}\big(y_{\mathrm{fix}_k(u,v)};z\big)\,E_{p_2-n_2,\,k-n_2}\big(y_{\mathrm{fix}_k(v,w)};z'\big)\,\widetilde{\mathfrak{S}}_w(x;y).$$

For $p_1 = p_2 = k$ this is Theorem 8.3. This is the form implemented in the routine `double_elem_sym_q` *of* [34] *and used by the paired step of Method 4.16.*

Proof. Theorem 8.3 is an identity of polynomials in z_1, z_2, in which $\widetilde{\mathfrak{S}}_{d_2[k]}(x;z) = \mathcal{Q}\big(\prod_{i\le k}(x_i - z_1)\prod_{i\le k}(x_i - z_2)\big)$ and $\mathcal{Q}$ is linear over $\mathbb{Z}[z_1, z_2]$. Substitute $z_1 = -t$, $z_2 = -t'$ for independent indeterminates t, t'. On the left, $\prod_{i\le k}(x_i + t)\prod_{i\le k}(x_i + t') = \sum_{a,b=0}^{k} t^{k-a}t'^{\,k-b}e_a(x_1,\dots,x_k)e_b(x_1,\dots,x_k)$. On the right, for a pair $u \overset{k}{\rightsquigarrow} v \overset{k}{\rightsquigarrow} w$ with $n_1 = n_k(u,v)$ and $n_2 = n_k(v,w)$ we have $\prod_{a\in\mathrm{fix}_k(u,v)}(y_a + t) = \sum_i t^{k-n_1-i}e_i(y_{\mathrm{fix}_k(u,v)})$ and similarly for the second step, since $|\mathrm{fix}_k(u,v)| = k - n_1$. Comparing the coefficients of $t^{k-a}t'^{\,k-b}$ gives, for $0 \le a, b \le k$,

(8.6)
$$\widetilde{\mathfrak{S}}_u(x;y)\,\mathcal{Q}\big(e_a(x_{\le k})e_b(x_{\le k})\big) = \sum_{\substack{u \overset{k}{\rightsquigarrow} v \overset{k}{\rightsquigarrow} w \\ \text{compatible}}} q_k(u,v)q_k(v,w)\,e_{a-n_k(u,v)}\big(y_{\mathrm{fix}_k(u,v)}\big)\,e_{b-n_k(v,w)}\big(y_{\mathrm{fix}_k(v,w)}\big)\,\widetilde{\mathfrak{S}}_w(x;y).$$

Now expand both factors by (8.1): $E_{p_1,k}(x;z)E_{p_2,k}(x;z') = \sum_{a,b} c_{p_1,k,a}(z)c_{p_2,k,b}(z')\,e_a(x_{\le k})e_b(x_{\le k})$. Apply $\mathcal{Q}$, which is linear over $\mathbb{Z}[z,z']$, multiply by $\widetilde{\mathfrak{S}}_u(x;y)$, insert (8.6), and sum first over a and then over b

using (8.3) with $(A,n) = (\mathrm{fix}_k(u,v), n_k(u,v))$ and $(\mathrm{fix}_k(v,w), n_k(v,w))$ respectively; this gives the asserted coefficient, which vanishes unless $n_k(u,v) \le p_1$ and $n_k(v,w) \le p_2$. □

We first reduce Theorem 8.3 to a single combinatorial identity about pairs of chains, (J_k) below (Proposition 8.7). To prove (J_k) we pass to a termwise reformulation (B_k) based on the quantum Giambelli formula, prove that the pairs of chains organize into blocks along which (B_k) telescopes (Lemma 8.12), reduce (B_k) to an identity between the ends of the blocks and the mixed chains that move the position $k+1$ (Proposition 8.13), and prove that identity by an explicit injection, built from a single local lemma which generalizes the up/down pairing of Lemma 6.16 (Theorem 8.23). This completes the proof of Theorem 8.3 for all k.

The recurrence. The quantization of two equal columns is governed by the following recurrence, in which only powers of q appear.

Lemma 8.6 (Quantization of two equal columns)**.** *For all $k \ge 1$,*

$$\widetilde{E}_{k,k}(x;z_1)\,\widetilde{E}_{k,k}(x;z_2) = \widetilde{\mathfrak{S}}_{d_2[k]}(x;z) + q_k\;\widetilde{\mathfrak{S}}_{d_2[k-1]}(x;z).$$

Proof. Write $P_k(z_1) = \prod_{i\le k}(x_i - z_1) = E_{k,k}(x;z)$ and $Q_k(z_1) = \mathcal{Q}(P_k(z_1)) = \widetilde{E}_{k,k}(x;z)$, and similarly with z_2, with $P_0 = Q_0 = 1$ and $P_{-1} = Q_{-1} = 0$; thus $\widetilde{\mathfrak{S}}_{d_2[k]}(x;z)$ equals $\mathcal{Q}(P_k(z_1)P_k(z_2))$. Since $P_{k+1}(z_2) = (x_{k+1} - z_2)P_k(z_2)$, we have

$$P_k(z_1)P_{k+1}(z_2) - P_{k+1}(z_1)P_k(z_2) = (z_1 - z_2)\,P_k(z_1)P_k(z_2).$$

The products $P_k(z_1)P_{k+1}(z_2)$ and $P_{k+1}(z_1)P_k(z_2)$ are $\mathbb{Z}[z_1,z_2]$-linear combinations of the standard elementary monomials $e_a(x_{\le k})e_b(x_{\le k+1})$, so $\mathcal{Q}$ acts on them multiplicatively, and

$$(z_1 - z_2)\;\widetilde{\mathfrak{S}}_{d_2[k]}(x;z) = W_k, \qquad W_k := Q_k(z_1)Q_{k+1}(z_2) - Q_{k+1}(z_1)Q_k(z_2).$$

The three-term recursion $Q_{k+1}(z_2) = (x_{k+1} - z_2)Q_k(z_2) + q_kQ_{k-1}(z_2)$ of Lemma 6.22 gives

$$\begin{aligned} W_k = Q_k(z_1)\big[(x_{k+1} - z_2)Q_k(z_2) + q_kQ_{k-1}(z_2)\big] - \big[(x_{k+1} - z_1)Q_k(z_1) + q_kQ_{k-1}(z_1)\big]Q_k(z_2) \\ = (z_1 - z_2)Q_k(z_1)Q_k(z_2) - q_kW_{k-1}. \end{aligned}$$

Dividing by $z_1 - z_2$ (all terms are divisible, W_k being antisymmetric in z_1, z_2) gives $\widetilde{\mathfrak{S}}_{d_2[k]} = Q_kQ_k - q_k\,\widetilde{\mathfrak{S}}_{d_2[k-1]}$. □

Lemma 8.6 reduces Theorem 8.3 to a recurrence in k. For $u, w \in S_\infty$ let $\mathrm{All}_k(u,w)$, $\mathrm{Comp}_k(u,w)$ and $\mathrm{Inc}_k(u,w)$ denote the sum of $\omega_k(u,v,w)$ over all pairs $u \overset{k}{\rightsquigarrow} v \overset{k}{\rightsquigarrow} w$, over the compatible pairs, and over the incompatible pairs, respectively, so that $\mathrm{All}_k = \mathrm{Comp}_k + \mathrm{Inc}_k$ and, by (8.5), $\mathrm{All}_k(u,w) = [\widetilde{\mathfrak{S}}_w]\big(\widetilde{\mathfrak{S}}_u\;\widetilde{E}_{k,k}(x;z_1)\widetilde{E}_{k,k}(x;z_2)\big)$; here $\mathrm{Comp}_0(u,w) = \mathrm{All}_0(u,w) = \delta_{u,w}$.

Proposition 8.7 (Reduction)**.** *Suppose Theorem 8.3 holds with k replaced by $k-1$ (for $k = 1$ this is the trivial statement $\widetilde{\mathfrak{S}}_u\,\widetilde{\mathfrak{S}}_{d_2[0]} = \widetilde{\mathfrak{S}}_u$). Then Theorem 8.3 holds for k if and only if*

$$\mathrm{Inc}_k(u,w) = q_k\,\mathrm{Comp}_{k-1}(u,w) \qquad \textit{for all } u,w \in S_\infty. \tag{J_k}$$

Consequently Theorem 8.3 holds for all $k \le K$ if and only if (J_k) holds for all $k \le K$.

Proof. By Lemma 8.6, (8.5) and the hypothesis,

$$[\widetilde{\mathfrak{S}}_w]\big(\widetilde{\mathfrak{S}}_u\;\widetilde{\mathfrak{S}}_{d_2[k]}\big) = \mathrm{All}_k(u,w) - q_k[\widetilde{\mathfrak{S}}_w]\big(\widetilde{\mathfrak{S}}_u\;\widetilde{\mathfrak{S}}_{d_2[k-1]}\big) = \mathrm{All}_k(u,w) - q_k\,\mathrm{Comp}_{k-1}(u,w),$$

while $\mathrm{Comp}_k = \mathrm{All}_k - \mathrm{Inc}_k$. The two sides of Theorem 8.3 therefore differ by $\mathrm{Inc}_k(u,w) - q_k\mathrm{Comp}_{k-1}(u,w)$. The last assertion follows by induction on k. □

The identity (J_k) says that the terms omitted from the iterated Pieri formula at level k are exactly accounted for, with an extra factor q_k, by the doubled formula one level down. Its left-hand side is a sum over pairs of chains whose cycles overlap, and the following lemma locates the overlap.

Lemma 8.8 (Overlapping cycles have top $k+1$)**.** *Let $u \overset{k}{\rightsquigarrow} v \overset{k}{\rightsquigarrow} w$, and suppose that a cycle c of $u^{-1}v$ and a cycle d of $v^{-1}w$ have the same top B and share an element $a \le k$. Then $B = k+1$, and moreover $k \in c \cup d$, i.e. one of the two cycles contains the transposition $t_{k,k+1}$. Consequently a pair of chains in which either step contains no transposition $t_{\cdot,k+1}$ is compatible, and in an incompatible pair both steps contain a cycle with top $k+1$, these two cycles overlap, no other pair of cycles does, and the letter $t_{k,k+1}$ occurs in one of the two chains.*

Proof. We show more generally that there is no position $r \notin c \cup d$ with $a' < r < B$ for every $a' \in (c \cup d) \setminus \{B\}$; applied to $r = k+1$ this gives $B = k+1$ (as $k+1 \notin c \cup d$ when $B \ge k+2$), and applied to $r = k$ it gives $k \in c \cup d$. Suppose such an r exists. By Lemma 6.7(iv) and Lemma 6.8 the fixed sets and weights of the two chains do not depend on the order in which their cycles are processed, and every order gives a chain of covers; we process c last in the first step and d first in the second. Write u' for the permutation obtained from u by applying the other cycles of $u^{-1}v$, so that $v = u'c$ and c is u'-tight, and write $c = (a_p, \dots, a_1, B)$, $\beta_0 = u'(B) = u(B)$, $\beta_l = u'(a_l) = u(a_l)$, and $d = (b_r, \dots, b_1, B)$, $\gamma_0 = v(B)$, $\gamma_m = v(b_m)$. By Lemma 6.7(i), $v(B) = \beta_p$ and $v(a_l) = \beta_{l-1}$ for $1 \le l \le p$; as $a = a_i = b_j$ with $i, j \ge 1$, we get

$$\gamma_0 = \beta_p, \qquad \gamma_j = \beta_{i-1}, \qquad \text{and} \qquad \beta_{i-1} \ne \beta_p.$$

Let η be the value at the position r when the letters of c are applied. Since $r \notin c \cup d$ and nothing happens between the last letter of c and the first letter of d, η is also the value at position r when the letters of d are applied. Moreover $a_l < r < B$ for every l and $b_m < r < B$ for every m, so the conditions (6.2) and (6.3) apply to the position r at every step of c and of d; for c they read: $\eta \notin (\beta_l, \beta_{l-1})$ if $\beta_l < \beta_{l-1}$, and $\eta \in (\beta_{l-1}, \beta_l)$ if $\beta_l > \beta_{l-1}$, for $1 \le l \le p$. By Corollary 6.13 the cyclic sequence $\Gamma = (\beta_0, \dots, \beta_p)$ is unimodal, and the displayed conditions say precisely that η lies in the gap (β_p, β_0) of Γ: if Γ has no descent then $\eta \notin (\beta_p, \beta_0)$, i.e. $\eta < \beta_p = \min \Gamma$ or $\eta > \beta_0 = \max \Gamma$; and if the ascent of Γ is (β_{q-1}, β_q) then $\eta \in (\min \Gamma, \max \Gamma)$ and η avoids all the descent gaps (β_l, β_{l-1}) with $l \ne q$, which together with (β_0, β_p) cover $(\min \Gamma, \max \Gamma) \setminus \Gamma$, so $\beta_0 < \eta < \beta_p$. In other words, in the cyclically decreasing arrangement of $\Gamma \cup \{\eta\}$ the value η sits between β_p and β_0. The same argument for d shows that in the cyclically decreasing arrangement of $\Gamma' \cup \{\eta\}$, $\Gamma' = (\gamma_0, \dots, \gamma_r)$, the value η sits between γ_r and $\gamma_0 = \beta_p$.

If $\eta > \beta_p$, the first statement forces $\beta_p = \min \Gamma$ and $\eta > \max \Gamma$, while the second forces $\Gamma' \cap (\beta_p, \eta) = \emptyset$; but $\beta_{i-1} \in \Gamma \setminus \{\beta_p\}$ lies in (β_p, η) and belongs to Γ', a contradiction. If $\eta < \beta_p$, the second statement forces $\eta = \min(\Gamma' \cup \{\eta\})$ and $\beta_p = \max \Gamma'$, while the first forces $\Gamma \cap (\eta, \beta_p) = \emptyset$; but $\beta_{i-1} \in \Gamma' \setminus \{\beta_p\}$ lies in (η, β_p) and belongs to Γ, again a contradiction. This proves the claim, hence $B = k+1$ and $k \in c \cup d$. The consequences follow because a step contains at most one cycle with a given top. □

The identity (J_k) will be established for all k in Theorem 8.23. Before that we record two features of it which any proof must account for, and which explain the detour through (B_k). First, by Lemma 8.8, in every incompatible pair the overlapping cycles have top $B = k+1$ and one of them contains k, so that both steps contain a transposition $t_{a,k+1}$ and one of them contains $t_{k,k+1}$; the incompatibility is a "detour" of the level-$(k-1)$ transposition t_{ak} through the position $k+1$, in accordance with $t_{a,k+1}t_{k,k+1}t_{a,k+1} = t_{ak}$. Second, ($\mathrm{J}_k$) is not a termwise identity: already for $u = 1324$, $k = 2$, $w = 3124$ (Example 8.4(b)) the left-hand side is $q_2\big[(y_3 - z_1) + (y_1 - z_2)\big]$ and the right-hand side is $q_2\big[(y_1 - z_1) + (y_3 - z_2)\big]$. In all examples the pairs with given (u, w) and q-weight organize into chains $v_1, \dots, v_{m+1}$ whose fixed sets are nested, with contributions $\prod_{j<i}(y_{c_j} - z_1) \prod_{j>i}(y_{c_j} - z_2)$, which sum to the divided difference $\big(\prod_j (y_{c_j} - z_1) - \prod_j (y_{c_j} - z_2)\big)/(z_2 - z_1)$ and so depend only on the set $\{c_j\}$; the identity (J_k) then amounts to a matching of these sets between the two levels. We now explain the origin of this divided-difference structure and use it to recast (J_k) as a termwise identity, which we then prove.

A termwise reformulation via the Giambelli formula. The proof of Lemma 8.6 shows that

$$(z_1 - z_2)\, \widetilde{\mathfrak{S}}_{d_2[k]}(x; z) = \widetilde{E}_{k,k}(x; z_1)\, \widetilde{E}_{k+1,k+1}(x; z_2) - \widetilde{E}_{k+1,k+1}(x; z_1)\, \widetilde{E}_{k,k}(x; z_2),$$

which is the quantum Giambelli formula [13] for the $k \times 2$ rectangle written as a divided difference. Expanding both products on the right by Theorem 3.4 at the two levels k and $k+1$ gives the following reformulation. For $u, w \in S_\infty$ put

$$M_k(u, w) = \sum_{u \overset{k}{\rightsquigarrow} v \overset{k+1}{\rightsquigarrow} w} q_k(u, v)\, q_{k+1}(v, w) \prod_{a \in \mathrm{fix}_k(u,v)} (y_a - z_1) \prod_{b \in \mathrm{fix}_{k+1}(v,w)} (y_b - z_2),$$

$$M'_k(u,w)=\sum_{u\overset{k+1}{\rightsquigarrow}v\overset{k}{\rightsquigarrow}w} q_{k+1}(u,v)\,q_k(v,w)\prod_{a\in \mathrm{fix}_{k+1}(u,v)}(y_a-z_1)\prod_{b\in \mathrm{fix}_k(v,w)}(y_b-z_2).$$

Proposition 8.9 (Giambelli form)**.** *For every $k\geq 1$ and all $u,w\in S_\infty$,*

$$(z_1-z_2)\,[\widetilde{\mathfrak{S}}_w]\big(\,\widetilde{\mathfrak{S}}_u(x;y)\;\widetilde{\mathfrak{S}}_{d_2[k]}(x;z)\big)=M_k(u,w)-M'_k(u,w).$$

Consequently Theorem 8.3 holds for k if and only if $(z_1-z_2)\,\mathrm{Comp}_k(u,w)=M_k(u,w)-M'_k(u,w)$ for all u,w.

Proof. Multiply the displayed Giambelli identity by $\widetilde{\mathfrak{S}}_u(x;y)$ and expand $\widetilde{\mathfrak{S}}_u\;\widetilde{E}_{k,k}(x;z_1)\widetilde{E}_{k+1,k+1}(x;z_2)$ by two applications of Theorem 3.4, first at level k with the variable z_1 and then at level $k+1$ with the variable z_2; this gives M_k. The product $\widetilde{\mathfrak{S}}_u\;\widetilde{E}_{k+1,k+1}(x;z_1)\widetilde{E}_{k,k}(x;z_2)$, expanded first at level $k+1$ and then at level k, gives M'_k. □

The right-hand side of Proposition 8.9 is a sum of products in which the levels of the two steps differ, and the factor z_1-z_2 on the left can be absorbed into the compatible pairs as follows. For a pair $u\overset{k}{\rightsquigarrow}v\overset{k}{\rightsquigarrow}w$ let $c(v)=v(k+1)$. If the first step does not use the position $k+1$ then $c(v)=u(k+1)$, and otherwise $c(v)=u(a)$ for a position $a\leq k$ moved by the first step; in either case $c(v)\notin \mathrm{fix}_k(u,v)$, and $c(v)\notin \mathrm{fix}_k(v,w)\subseteq\{v(1),\dots,v(k)\}$. Hence, writing $z_1-z_2=(y_{c(v)}-z_2)-(y_{c(v)}-z_1)$,

$$\begin{aligned}&(z_1-z_2)\,\omega_k(u,v,w)\\&=q_k(u,v)q_k(v,w)\Big[\prod_{a\in \mathrm{fix}_k(u,v)}(y_a-z_1)\prod_{b\in \mathrm{fix}_k(v,w)\cup\{c(v)\}}(y_b-z_2)-\prod_{a\in \mathrm{fix}_k(u,v)\cup\{c(v)\}}(y_a-z_1)\prod_{b\in \mathrm{fix}_k(v,w)}(y_b-z_2)\Big],\end{aligned}$$

and each of the two products on the right has the same shape as a term of M_k or M'_k. Writing $\mathcal{T}(u,w)$ for the multiset of triples $\big(q_k(u,v)q_k(v,w),\ \mathrm{fix}_k(u,v),\ \mathrm{fix}_k(v,w)\cup\{c(v)\}\big)$ over the compatible pairs $u\overset{k}{\rightsquigarrow}v\overset{k}{\rightsquigarrow}w$, $\mathcal{S}(u,w)$ for the multiset of $\big(q_k(u,v)q_k(v,w),\ \mathrm{fix}_k(u,v)\cup\{c(v)\},\ \mathrm{fix}_k(v,w)\big)$ over the same pairs, $\mathcal{M}_k(u,w)$ for the multiset of $\big(q_k(u,v)q_{k+1}(v,w),\mathrm{fix}_k(u,v),\mathrm{fix}_{k+1}(v,w)\big)$ over the chains $u\overset{k}{\rightsquigarrow}v\overset{k+1}{\rightsquigarrow}w$ and $\mathcal{M}'_k(u,w)$ for the corresponding multiset over the chains $u\overset{k+1}{\rightsquigarrow}v\overset{k}{\rightsquigarrow}w$, we are led to the following termwise statement, from which Theorem 8.3 follows immediately by Proposition 8.9.

Theorem 8.10 (Termwise form)**.** *For all $k\geq 1$ and $u,w\in S_\infty$ there is an equality of multisets*

$$\mathcal{T}(u,w)\uplus\mathcal{M}'_k(u,w)=\mathcal{S}(u,w)\uplus\mathcal{M}_k(u,w). \tag{B_k}$$

Proof. Proposition 8.13 and Theorem 8.23 below. □

Unlike (J_k), the identity (B_k) is an equality of multisets, so it calls for a bijection. We first show that the greater part of this bijection is forced: the compatible pairs with given (u,w) fall into explicit *blocks*, inside which the terms of $\mathcal{T}$ and $\mathcal{S}$ cancel telescopically, and the unblocked ends of the blocks are matched with the chains of $\mathcal{M}_k$ and $\mathcal{M}'_k$ that do not move the position $k+1$. This reduces (B_k) to an identity between the blocked ends of the blocks and the mixed chains that do move $k+1$ (Proposition 8.13); in the simplest situation that identity is exactly the up/down pairing of Lemma 6.16, and we then prove it in general (Theorem 8.23).

Blocks, and the reduction of (B_k) to its junctions. For a pair $u\overset{k}{\rightsquigarrow}v\overset{k}{\rightsquigarrow}w$ we write $[A\mid B]=\prod_{a\in A}(y_a-z_1)\prod_{b\in B}(y_b-z_2)$ and

$$T(v)=q_k(u,v)q_k(v,w)\,[\mathrm{fix}_k(u,v)\mid \mathrm{fix}_k(v,w)\cup\{c(v)\}],\qquad S(v)=q_k(u,v)q_k(v,w)\,[\mathrm{fix}_k(u,v)\cup\{c(v)\}\mid \mathrm{fix}_k(v,w)],$$

so that $(z_1-z_2)\,\omega_k(u,v,w)=T(v)-S(v)$ by the display above, and $\mathcal{T}(u,w)$, $\mathcal{S}(u,w)$ are the multisets of the $T(v)$, $S(v)$ over the compatible pairs (we identify the monomial $q[A\mid B]$ with the triple (q,A,B)). Recall from Lemma 6.7(iv) and Lemma 6.8 that the cycles of a step may be processed in any order.

Definition 8.11 (Block data, cut, block)**.** Let $u\overset{k}{\rightsquigarrow}v\overset{k}{\rightsquigarrow}w$ be compatible, $\pi_1=u^{-1}v$, $\pi_2=v^{-1}w$. Let c be the cycle of π_1 with top $k+1$ and d the cycle of π_2 with top $k+1$ (either may be trivial). By compatibility $c\cap d\subseteq\{k+1\}$, so writing $c=(x_i,\dots,x_1,k+1)$ and $d=(x_n,\dots,x_{i+1},k+1)$ we have, by (6.1),

$$X:=cd=t_{x_1,k+1}\cdots t_{x_n,k+1}=(x_n,\dots,x_1,k+1),$$

a cycle with top $k+1$ which we call the *merged cycle* of the pair. Let $\rho_1 = \pi_1 c^{-1}$ and $\rho_2 = d^{-1}\pi_2$ be the products of the remaining cycles of π_1 and of π_2; all of these have tops $\geq k+2$, and $u^{-1}w = \rho_1 X \rho_2$. We call (ρ_1, X, ρ_2) the *block data* and i the *cut* of the pair, and we put

$$i_{\min} = \max\big(\{j : x_j \in \mathrm{supp}\rho_2\} \cup \{0\}\big), \qquad i_{\max} = \min\big(\{j : x_j \in \mathrm{supp}\rho_1\} \cup \{n+1\}\big) - 1.$$

Since the cycles of π_1 are disjoint, $x_1, \ldots, x_i \notin \mathrm{supp}\rho_1$, and likewise $x_{i+1}, \ldots, x_n \notin \mathrm{supp}\rho_2$; hence $i_{\min} \leq i \leq i_{\max}$. The *block* of the pair is the set of compatible pairs $u \overset{k}{\rightsquigarrow} v' \overset{k}{\rightsquigarrow} w$ with the same block data; the blocks partition the compatible pairs with given (u, w).

Lemma 8.12 (Blocks)**.** *Let $u \overset{k}{\rightsquigarrow} v \overset{k}{\rightsquigarrow} w$ be compatible with block data (ρ_1, X, ρ_2), $X = (x_n, \ldots, x_1, k+1)$, and put $u' = u\rho_1$, $c_j = (x_j, \ldots, x_1, k+1)$, $d_j = (x_n, \ldots, x_{j+1}, k+1)$ and $v_j = u'c_j$ for $0 \leq j \leq n$, so that $v = v_i$ where i is the cut. Then the block of the pair is $\{v_j : i_{\min} \leq j \leq i_{\max}\}$, and for every j in this range:*

(i) *$u \overset{k}{\rightsquigarrow} v_j \overset{k}{\rightsquigarrow} w$ is a compatible pair with block data (ρ_1, X, ρ_2) and cut j, and $q_k(u, v_j)\, q_k(v_j, w) = q_k(u, v)\, q_k(v, w)$;*

(ii) *$\mathrm{fix}_k(u, v_j) = F_1 \setminus \{u(x_1), \ldots, u(x_j)\}$ and $\mathrm{fix}_k(v_j, w) = F_2 \setminus \{w(x_{j+1}), \ldots, w(x_n)\}$, where $F_1 = \{u(a) : a \leq k,\ a \notin \mathrm{supp}\rho_1\}$ and $F_2 = \{w(a) : a \leq k,\ a \notin \mathrm{supp}\rho_2\}$;*

(iii) *$c(v_j) = u(x_j)$ if $j \geq 1$ and $c(v_0) = u(k+1)$; moreover $w(x_j) = c(v_{j-1})$ for $i_{\min} < j \leq i_{\max}$.*

Consequently $S(v_j) = T(v_{j-1})$ for $i_{\min} < j \leq i_{\max}$, and the contribution of the block telescopes:

$$(z_1 - z_2) \sum_{j=i_{\min}}^{i_{\max}} \omega_k(u, v_j, w) = T(v_{i_{\max}}) - S(v_{i_{\min}}).$$

Proof. (i) For $j \leq i_{\max}$ the positions $x_1, \ldots, x_j$ avoid $\mathrm{supp}\rho_1$, so $\rho_1 c_j$ is a product of disjoint cycles, each with exactly one element greater than k; similarly $d_j\rho_2$ for $j \geq i_{\min}$. By Lemma 6.8 and Lemma 6.7(iv), $u \overset{k}{\rightsquigarrow} v_j \overset{k}{\rightsquigarrow} w$ holds if and only if ρ_1 is u-tight, c_j is u'-tight, d_j is v_j-tight, and ρ_2 is $v_j d_j$-tight, where $v_j d_j = u'X$ does not depend on j. The first and the last condition do not involve j, and they hold because they hold for $j = i$ (as $\pi_1 = \rho_1 c$ is u-tight and $\pi_2 = d\rho_2$ is v-tight). For the two middle conditions write $\beta_0 = u'(k+1)$ and $\beta_l = u'(x_l)$ for $1 \leq l \leq n$. The pivot factorization of X is the concatenation of those of c_j and d_j, so by Lemma 6.7(i) applied to X and u', the l-th letter $t_{x_l,k+1}$ exchanges the value β_l at x_l with the value β_{l-1} at $k+1$, for every l and independently of j; in particular $v_j(k+1) = \beta_j$ and $v_j(x_l) = \beta_l$ for $l > j$. Hence

$$\mathrm{desc}_{u'}(c_j) = \{x_l : l \leq j,\ \beta_l > \beta_{l-1}\}, \qquad \mathrm{desc}_{v_j}(d_j) = \{x_l : l > j,\ \beta_l > \beta_{l-1}\},$$

and by Definition 6.6

$$L_{u'}(c_j) + L_{v_j}(d_j) = n - 2 \sum_{l:\, \beta_l > \beta_{l-1}} (k + 1 - x_l) = L_{u'}(X), \qquad \mathbf{q}_{u'}(c_j)\, \mathbf{q}_{v_j}(d_j) = \prod_{l:\, \beta_l > \beta_{l-1}} q_{x_l,k+1} = \mathbf{q}_{u'}(X),$$

both independent of j. By Lemma 6.7(ii),

$$\ell(u'X) - \ell(u') = \big(\ell(v_j) - \ell(u')\big) + \big(\ell(u'X) - \ell(v_j)\big) \geq L_{u'}(c_j) + L_{v_j}(d_j) = L_{u'}(X),$$

with equality if and only if c_j is u'-tight and d_j is v_j-tight. Equality holds for $j = i$, hence for every j. By Corollary 6.9 and Lemma 6.7(iv), $q_k(u, v_j) q_k(v_j, w) = \mathbf{q}_u(\rho_1)\, \mathbf{q}_{u'}(c_j)\, \mathbf{q}_{v_j}(d_j)\, \mathbf{q}_{u'X}(\rho_2) = \mathbf{q}_u(\rho_1) \mathbf{q}_{u'}(X) \mathbf{q}_{u'X}(\rho_2)$, independent of j. The pair is compatible, since its cycles with tops $\geq k+2$ are those of ρ_1 and ρ_2, as for the given pair, and $c_j \cap d_j = \{k+1\}$; its block data is (ρ_1, X, ρ_2) and its cut is j by construction. Conversely every pair in the block has cut in $[i_{\min}, i_{\max}]$ and is therefore one of the v_j, since the block data and the cut determine $\pi_1 = \rho_1 c_j$.

(ii) The positions $a \leq k$ moved by $\pi_1^{(j)} = \rho_1 c_j$ are those of $\mathrm{supp}\rho_1$ together with $x_1, \ldots, x_j$, which gives the first formula (note $u(x_l) \in F_1$ for $l \leq i_{\max}$). A position $a \leq k$ is fixed by $d_j\rho_2$ if and only if $a \notin \mathrm{supp}\rho_2 \cup \{x_{j+1}, \ldots, x_n\}$, and then $v_j(a) = w(a)$; this gives the second formula (note $w(x_l) \in F_2$ for $l > i_{\min}$).

(iii) We saw that $v_j(k+1) = \beta_j$, which is $u'(x_j) = u(x_j)$ for $1 \leq j \leq i_{\max}$ (as $x_j \notin \mathrm{supp}\rho_1$) and $u'(k+1) = u(k+1)$ for $j = 0$ (as $k+1 \notin \mathrm{supp}\rho_1$). Now let $i_{\min} < j \leq i_{\max}$ and consider the pair v_{j-1}. The position x_j is not moved by its first step $\rho_1 c_{j-1}$. Its second step is $d_{j-1}\rho_2$, whose first letter $t_{x_j,k+1}$ moves

the value $v_{j-1}(k+1) = c(v_{j-1})$ to the position x_j, and x_j is moved by no later letter: not by the remaining letters of d_{j-1}, and not by ρ_2 because $j > i_{\min}$. Hence $w(x_j) = c(v_{j-1})$.

For the consequence write $A_j = \mathrm{fix}_k(u, v_j)$ and $B_j = \mathrm{fix}_k(v_j, w)$. For $i_{\min} < j \le i_{\max}$, (ii) and (iii) give $A_j \cup \{c(v_j)\} = A_j \cup \{u(x_j)\} = A_{j-1}$ and $B_j = B_{j-1} \cup \{w(x_j)\} = B_{j-1} \cup \{c(v_{j-1})\}$, while the q-weights agree by (i); thus $S(v_j) = T(v_{j-1})$. Summing $(z_1 - z_2)\omega_k(u, v_j, w) = T(v_j) - S(v_j)$ over the block, all terms cancel except $T(v_{i_{\max}})$ and $S(v_{i_{\min}})$. □

We call $v_{i_{\max}}$ the *head* and $v_{i_{\min}}$ the *tail* of the block; the head is *blocked* if $i_{\max} < n$ (some x_j lies in $\mathrm{supp}\rho_1$) and the tail is *blocked* if $i_{\min} > 0$ (some x_j lies in $\mathrm{supp}\rho_2$). Thus a block is the set of all cuts of the merged transposition word $t_{x_1,k+1} \cdots t_{x_n,k+1}$ that are not obstructed by a spectator cycle of ρ_1 or ρ_2 sharing a position with X, and the transit value $c(v_j)$ runs through the values of u at $x_n, x_{n-1}, \ldots, x_1, k+1$ as the cut moves from right to left.

Proposition 8.13 (Reduction of (B_k) to junctions)**.** *Let $u, w \in S_\infty$. Let $\mathcal{M}_k^{\mathrm{mov}}(u, w) \subseteq \mathcal{M}_k(u, w)$ be the submultiset of the chains $u \overset{k}{\rightsquigarrow} v \overset{k+1}{\rightsquigarrow} w$ whose second step moves the position $k+1$, let $\mathcal{M}_k'^{\mathrm{mov}}(u, w) \subseteq \mathcal{M}_k'(u, w)$ consist of the chains $u \overset{k+1}{\rightsquigarrow} v \overset{k}{\rightsquigarrow} w$ whose first step moves $k+1$, and let $\mathcal{T}^{\mathrm{bl}}(u, w)$, respectively $\mathcal{S}^{\mathrm{bl}}(u, w)$, be the multiset of $T(v)$ over the blocked heads v, respectively of $S(v)$ over the blocked tails v, of the blocks of compatible pairs $u \overset{k}{\rightsquigarrow} v \overset{k}{\rightsquigarrow} w$. Then ($\mathrm{B}_k$) holds for (u, w) if and only if*

$$\mathcal{T}^{\mathrm{bl}}(u, w) \uplus \mathcal{M}_k'^{\mathrm{mov}}(u, w) = \mathcal{S}^{\mathrm{bl}}(u, w) \uplus \mathcal{M}_k^{\mathrm{mov}}(u, w), \tag{$\mathrm{J}_k^{\mathrm{B}}$}$$

and in that case $(z_1 - z_2)\,\mathrm{Comp}_k(u, w) = M_k(u, w) - M_k'(u, w)$. In particular, Theorem 8.3 holds for k as soon as ($\mathrm{J}_k^{\mathrm{B}}$) holds for all $u, w \in S_\infty$.

Proof. By Lemma 8.12 the compatible pairs with given (u, w) are partitioned into blocks, and inside each block the elements $S(v_j)$, $i_{\min} < j \le i_{\max}$, of $\mathcal{S}(u, w)$ coincide with the elements $T(v_{j-1})$ of $\mathcal{T}(u, w)$. Cancelling them, (B_k) is equivalent to

$$\{T(v) : v \text{ a head}\} \uplus \mathcal{M}_k'(u, w) = \{S(v) : v \text{ a tail}\} \uplus \mathcal{M}_k(u, w).$$

Now let $v = v_n$ be an unblocked head, so $d = 1$ and $v(k+1) = w(k+1)$. By Lemma 6.14(i), applied with $k+1$ in place of k, $v \overset{k+1}{\rightsquigarrow} w$ with $q_{k+1}(v, w) = q_k(v, w)$ and $\mathrm{fix}_{k+1}(v, w) = \mathrm{fix}_k(v, w) \cup \{v(k+1)\}$; thus $u \overset{k}{\rightsquigarrow} v \overset{k+1}{\rightsquigarrow} w$ is a chain of $\mathcal{M}_k(u, w) \setminus \mathcal{M}_k^{\mathrm{mov}}(u, w)$, and its triple is $T(v)$. Conversely, let $u \overset{k}{\rightsquigarrow} v \overset{k+1}{\rightsquigarrow} w$ be a chain not moving $k+1$. By the same lemma $v \overset{k}{\rightsquigarrow} w$ with the same weight, the second step has no cycle with top $k+1$, so the pair $u \overset{k}{\rightsquigarrow} v \overset{k}{\rightsquigarrow} w$ is compatible with cut $i = n$; hence it is the head of its block, this head is unblocked ($x_1, \ldots, x_n$ avoid $\mathrm{supp}\rho_1$ as the cycles of π_1 are disjoint), and its triple is $T(v)$. Since distinct blocks have distinct heads, $v \mapsto v$ is a bijection between the unblocked heads and the chains of $\mathcal{M}_k(u, w)$ not moving $k+1$, preserving the triples. Symmetrically, the unblocked tails v_0 (for which $c = 1$ and $u(k+1) = v_0(k+1)$) correspond bijectively to the chains $u \overset{k+1}{\rightsquigarrow} v \overset{k}{\rightsquigarrow} w$ not moving $k+1$, with $S(v_0) = \big(q, \mathrm{fix}_{k+1}(u, v_0), \mathrm{fix}_k(v_0, w)\big)$. Cancelling these two families gives the equivalence with ($\mathrm{J}_k^{\mathrm{B}}$). The remaining assertions follow from Proposition 8.9. □

The identity ($\mathrm{J}_k^{\mathrm{B}}$) concerns only the *junctions* of the blocks: a blocked head $v_{i_{\max}}$, whose next cut is obstructed by a cycle e of ρ_1 with top $B \ge k+2$ through $x_{i_{\max}+1}$, and a blocked tail $v_{i_{\min}}$, obstructed by a cycle of ρ_2 through $x_{i_{\min}}$, together with the mixed chains in which the position $k+1$ is moved by a cycle with top $B \ge k+2$. The correspondence realising ($\mathrm{J}_k^{\mathrm{B}}$) *transports* the obstructing cycle e across X: in the simplest case a blocked tail with block data (ρ_1, X, ρ_2) and $e \subseteq \rho_2$ through x_j, $j = i_{\min}$, is matched with the blocked head of the block with data $(\rho_1 \cdot XeX^{-1}, X, \rho_2 e^{-1})$, in which e has been conjugated into the first step and is attached to x_{j-1}, so that the cut $j-1$ is skipped; in general pivots of X migrate into the transported cycle or out of it, exactly as in the transposition words F_m of Lemma 6.16, and a chain of $\mathcal{M}_k^{\mathrm{mov}}$ is matched in the same way with the head obtained by transporting the cycle through $k+1$ into the first step. The simplest junction, in which the obstructing cycle is a single transposition $t_{k+1,B}$, is exactly Lemma 6.16 (applied with $k+1, B$ in the roles of k, B): its transposition words F_{up}, F_{std} and F_m are a chain of $\mathcal{M}_k^{\mathrm{mov}}$, a chain of $\mathcal{M}_k'^{\mathrm{mov}}$ with trivial second step, and the blocked heads with block data $\big((x_n, \ldots, x_m, B), (x_m, \ldots, x_1, k+1), 1\big)$, and they reappear below as the junction word $J = F_{\mathrm{up}}$ and its resolutions $J_0 = F_{\mathrm{std}}$, $J_m \sim F_m$.

Proof of the junction identity. We now prove ($\mathrm{J}_k^{\mathrm{B}}$), and hence ($\mathrm{B}_k$) and Theorem 8.3, for all k. Throughout we write $K = k+1$. The proof constructs, for every u, w and every q-monomial q, an injective map from the *sources* (the blocked tails and the chains of $\mathcal{M}_k^{\mathrm{mov}}$ of weight q) to the *targets* (the blocked heads and the chains of $\mathcal{M}_k'^{\mathrm{mov}}$ of weight q) which preserves the triples, and a symmetry which exchanges sources and targets; the two together give ($\mathrm{J}_k^{\mathrm{B}}$). The map is a sequence of local rewritings of the pair of chains, each governed by a single lemma (Lemma 8.18) which generalizes Lemma 6.16; the lemma is proved not by tracking covers, but by comparing the formal q-weights of transposition words.

Overview. For orientation we summarize the four kinds of configurations in ($\mathrm{J}_k^{\mathrm{B}}$) and the triples (q, A, B), $A, B \subseteq \{1, 2, \ldots\}$ sets of values, that they carry; here M_1, M_2 denote the sets of positions $\le k$ moved by the first and by the second step, $\overline{S} = [1,k] \setminus S$, and q is the product of the q-weights of the two steps.

configuration	steps	role	triple
blocked tail v	$u \overset{k}{\rightsquigarrow} v \overset{k}{\rightsquigarrow} w$	source	$S(v) = \big(q,\ u(\overline{M_1}) \cup \{v(K)\},\ w(\overline{M_2})\big)$
chain of $\mathcal{M}_k^{\mathrm{mov}}$	$u \overset{k}{\rightsquigarrow} v \overset{k+1}{\rightsquigarrow} w$, K moved by step 2	source	$\big(q,\ u(\overline{M_1}),\ w(\overline{M_2})\big)$
blocked head v	$u \overset{k}{\rightsquigarrow} v \overset{k}{\rightsquigarrow} w$	target	$T(v) = \big(q,\ u(\overline{M_1}),\ w(\overline{M_2}) \cup \{v(K)\}\big)$
chain of $\mathcal{M}_k'^{\mathrm{mov}}$	$u \overset{k+1}{\rightsquigarrow} v \overset{k}{\rightsquigarrow} w$, K moved by step 1	target	$\big(q,\ u(\overline{M_1}),\ w(\overline{M_2})\big)$

(For a blocked tail, $v(K) = u(x)$ where $x \le k$ is the position whose next cut is obstructed, so $u(\overline{M_1}) \cup \{v(K)\} = u(\overline{M_1 \setminus \{x\}})$; for a blocked head, $v(K) = w(x_g)$ for the obstructed position x_g, so the last entry is $w(\overline{M_2 \setminus \{x_g\}})$. In the mixed chains one of the two fixed sets is taken at level $k+1$, which accounts for the missing extra element.) In every case the triple is thus determined by the weight and by the two sets of positions $\le k$ moved by the two steps, with the obstructed position removed from the appropriate set. The junction move of Proposition 8.19 rewrites a source as a transposition word with a cut and transports the obstructing cycle from the second step to the first across the top-K letters; it preserves the product $u^{-1}w$, the weight (Lemma 8.18), and the two sets of moved positions up to the obstructed one (the invariants N_1 and M_2 below), and therefore the triple. After finitely many moves the result is a target, and the symmetry $v \mapsto w_0 v$ of Lemma 8.16, which exchanges sources and targets, gives the reverse inclusion. A complete instance is worked out in Example 8.20.

Transposition words, cover paths and formal weights. A *transposition word* is a finite sequence $W = (t_1, \ldots, t_N)$ of transpositions $t_i = t_{a_i b_i}$, $a_i < b_i$ (not of simple reflections: the pivot words and junction words below are transposition words, and the only words in the simple reflections used in this paper are the reduced words of Section 7); its product is $t_1 \cdots t_N$ and its length is N. Given a permutation ψ put $\psi_0 = \psi$ and $\psi_i = \psi_{i-1} t_i$. The i-th letter *exchanges* the values $\psi_{i-1}(a_i)$ and $\psi_{i-1}(b_i)$, and we call it a *descent letter* (from ψ) if $\psi_{i-1}(a_i) > \psi_{i-1}(b_i)$. The *formal weight* of W from ψ is

$$\mathrm{wt}_\psi(W) = \prod_{i \text{ a descent letter}} q_{a_i b_i},$$

and W is a *cover path* from ψ if $\psi_{i-1} \lhd \psi_i$ for every i; in that case $\mathrm{wt}_\psi(W) = \prod_i \kappa(\psi_{i-1}, t_i)$. Two transposition words are *equivalent* if they have the same product and the same length.

Lemma 8.14 (Degree criterion)**.** *Let W be a transposition word of length N with product π. Then $\ell(\psi\pi) - \ell(\psi) \ge N - \deg \mathrm{wt}_\psi(W)$, with equality if and only if W is a cover path from ψ. Consequently, if W is a cover path from ψ and W' is equivalent to W, then W' is a cover path from ψ if and only if $\deg \mathrm{wt}_\psi(W') = \deg \mathrm{wt}_\psi(W)$; in particular this is the case whenever $\mathrm{wt}_\psi(W') = \mathrm{wt}_\psi(W)$.*

Proof. By Lemma 6.2, the i-th step changes the length by at least 1 if the letter is not a descent letter and by at least $1 - 2(b_i - a_i) = 1 - \deg q_{a_i b_i}$ if it is, with equality in both cases exactly when $\psi_{i-1} \lhd \psi_i$. Summing over i gives the inequality and the characterization of equality, and the rest follows because $\ell(\psi\pi) - \ell(\psi)$ and N are the same for W and W'. □

Lemma 8.15 (Inert reordering)**.** *Let $W = W_1 U V W_2$ be a concatenation of transposition words such that no position is moved both by a letter of U and by a letter of V. Then $W' = W_1 V U W_2$ is equivalent to W, each letter of W' exchanges the same two values from ψ as the corresponding letter of W, $\mathrm{wt}_\psi(W') = \mathrm{wt}_\psi(W)$, and W' is a cover path from ψ if and only if W is.*

Proof. The letters of U and V commute, so the products agree. A letter of U only moves positions untouched by V, so it finds the same values at its two positions whether or not V has been applied before it, and symmetrically for V; the letters of W_1 and W_2 see the same permutations in both transposition words. The last assertion is Lemma 8.14. □

Lemma 8.16 (Reversal)**.** *Work in S_n and let w_0 be the longest element. Let $W = (t_1, \dots, t_N)$ be a transposition word with product π, and let $W^{\mathrm{rev}} = (t_N, \dots, t_1)$. Then the i-th letter of W is a descent letter from ψ if and only if the $(N+1-i)$-th letter of W^{rev} is a descent letter from $w_0\psi\pi$; hence $\mathrm{wt}_{w_0\psi\pi}(W^{\mathrm{rev}}) = \mathrm{wt}_\psi(W)$, and W is a cover path from ψ if and only if W^{rev} is a cover path from $w_0\psi\pi$.*

Proof. The $(N+1-i)$-th step of W^{rev} goes from $w_0\psi_i$ to $w_0\psi_{i-1} = (w_0\psi_i)t_i$ and exchanges the values $w_0\psi_i(a_i) = n+1-\psi_{i-1}(b_i)$ and $w_0\psi_i(b_i) = n+1-\psi_{i-1}(a_i)$; the first is larger than the second if and only if $\psi_{i-1}(a_i) > \psi_{i-1}(b_i)$. Since $\ell(w_0 v) = \ell(w_0) - \ell(v)$, the length change of this step is $\ell(\psi_i) - \ell(\psi_{i-1})$, the length change of the i-th step of W, so the two steps are covers of the same kind simultaneously (Definition 6.3). □

The token lemma. Fix $C \geq k+2$ and write $T_x = t_{xK}$, $S_x = t_{xC}$ for $x \leq k$, and $Z = t_{KC}$; thus for every $x \leq k$ the three products $T_xS_x = S_xZ = ZT_x$ coincide (they are the 3-cycle (x, K, C) in some order), and likewise $T_xZ = ZS_x = S_xT_x$. Let $x_1, \dots, x_{j-1}$ be distinct positions $\leq k$, let x_j be either a further position $\leq k$ (the *tail case*) or $x_j = K$ (the $\mathcal{M}$ *case*), and let $e_1, \dots, e_{s-1}$ be distinct positions $\leq k$ different from x_j (they may coincide with some of $x_1, \dots, x_{j-1}$). The *junction word* attached to these data is the transposition word

$$J = T_{x_1} \cdots T_{x_j} \cdot S_{e_1} \cdots S_{e_{s-1}} \cdot S_{x_j},$$

where in the $\mathcal{M}$ case the letter $T_{x_j} = T_K$ is omitted and $S_{x_j} = Z$. For $0 \leq m \leq j-1$ its *m-th resolution* is

$$J_m = S_{x_m} S_{x_{m+1}} \cdots S_{x_{j-1}} \cdot T_{x_1} \cdots T_{x_{m-1}} \cdot T_{x_m} \cdot T_{e_1} \cdots T_{e_{s-1}} \cdot T_{x_j} \qquad (1 \leq m \leq j-1),$$
$$J_0 = Z\, S_{x_1} \cdots S_{x_{j-1}} \cdot T_{e_1} \cdots T_{e_{s-1}} \cdot T_{x_j},$$

again with T_{x_j} omitted in the $\mathcal{M}$ case. (In the language of the detours: the letter S_{x_j} is converted into the token Z, which travels to the left through the letters $S_{e_{s-1}}, \dots, S_{e_1}$, turning them into $T_{e_{s-1}}, \dots, T_{e_1}$, and then through $T_{x_{j-1}}, \dots, T_{x_{m+1}}$, turning them into S's, until it is absorbed by T_{x_m}, which it turns into $S_{x_m}T_{x_m}$; for $m = 0$ it is not absorbed.) Lemma 6.16 is the case $s = 1$ of the $\mathcal{M}$ case: there $J = F_{\mathrm{up}}$, $J_0 = F_{\mathrm{std}}$ and J_m is F_m with its first two blocks interchanged.

Lemma 8.17. *Each J_m is equivalent to J.*

Proof. The lengths agree. In the tail case, T_{x_j} commutes with $S_{e_1}, \dots, S_{e_{s-1}}$, and $T_{x_j}S_{x_j} = ZT_{x_j}$; in the $\mathcal{M}$ case the last letter is Z already. In both cases $J = T_{x_1} \cdots T_{x_{j-1}}\, S_{e_1} \cdots S_{e_{s-1}}\, Z\, T_{x_j}$ (with T_{x_j} omitted in the $\mathcal{M}$ case). Using $S_eZ = ZT_e$ repeatedly, $J = T_{x_1} \cdots T_{x_{j-1}}\, Z\, T_{e_1} \cdots T_{e_{s-1}} T_{x_j}$. Using $T_xZ = ZS_x$ for $x = x_{j-1}, \dots, x_{m+1}$ and then $T_{x_m}Z = S_{x_m}T_{x_m}$, we get $T_{x_1} \cdots T_{x_{j-1}}Z = T_{x_1} \cdots T_{x_{m-1}}\, S_{x_m}T_{x_m}\, S_{x_{m+1}} \cdots S_{x_{j-1}}$ for $m \geq 1$, and $= Z\, S_{x_1} \cdots S_{x_{j-1}}$ for $m = 0$. Finally the letters $S_{x_{m+1}}, \dots, S_{x_{j-1}}$ commute with $T_{x_1}, \dots, T_{x_m}$ and may be moved to the front; this gives J_m. □

Lemma 8.18 (Token lemma)**.** *Let $\psi \in S_\infty$ and suppose that the junction word J is a cover path from ψ. Put $\beta_0 = \psi(K)$, $\beta_l = \psi(x_l)$ for $1 \leq l \leq j-1$, and $\gamma = \psi(C)$. Then the cyclic sequence $\Gamma = (\beta_0, \beta_1, \dots, \beta_{j-1})$ is unimodal; let $g \in \{0, 1, \dots, j-1\}$ be the insertion gap of γ in Γ (gap $g \geq 1$ is (β_{g-1}, β_g), gap 0 is the wrap (β_{j-1}, β_0); $g = 0$ if $j = 1$).*

(i) *For $0 \leq m \leq j-1$, the resolution J_m is a cover path from ψ with $\mathrm{wt}_\psi(J_m) = \mathrm{wt}_\psi(J)$ if and only if $m = g$.*

(ii) *If $e_t = x_l$ for some $t \leq s-1$ and $l \leq j-1$, then $l > g$. If moreover the transposition word $J\, S_{e_{s+1}} \cdots S_{e_r}$ is a cover path from ψ for some further distinct positions $e_{s+1}, \dots, e_r \leq k$ (different from x_j and from $e_1, \dots, e_{s-1}$), and $e_t = x_l$ for some $t \geq s+1$ and $l \leq j-1$, then $l \leq g$.*

Proof. Write $T = T_{x_1} \cdots T_{x_{j'}}$, where $j' = j$ in the tail case and $j' = j-1$ in the $\mathcal{M}$ case, and $\beta_j = \psi(x_j)$ in the tail case. The transposition word T is the pivot chain of the cycle $(x_{j'}, \dots, x_1, K)$, so by Lemma 6.7(i) its letter T_{x_l} exchanges β_l (at x_l) with β_{l-1} (at K), and after T the position x_l carries β_{l-1} and K carries $\beta_{j'}$. The remaining letters $S_{e_1}, \dots, S_{e_{s-1}}, S_{x_j}$ of J form the pivot chain of the cycle $(x_j, e_{s-1}, \dots, e_1, C)$ (in the $\mathcal{M}$ case, of $(K, e_{s-1}, \dots, e_1, C)$) from $\psi' = \psi T$; put $\varepsilon_0 = \gamma$, $\varepsilon_t = \psi'(e_t)$ for $t \leq s-1$, and $\varepsilon_s = \psi'(x_j) = \beta_{j-1}$ (in both cases). By Lemma 6.7(i) the letter S_{e_t} exchanges ε_t with ε_{t-1} and S_{x_j} exchanges ε_s with ε_{s-1}.

Since J is a cover path, both cycles are tight (Lemma 6.7(iii)), so by Corollary 6.13 the cyclic sequences $(\beta_0,\dots,\beta_{j'})$ and $E=(\varepsilon_0,\dots,\varepsilon_s)$ are unimodal; in particular Γ, a block of the first, is unimodal. We record

$$\mathrm{wt}_\psi(J)=\prod_{l=1}^{j'} q_{x_lK}^{[\beta_l>\beta_{l-1}]}\ \prod_{t=1}^{s} q_{e_tC}^{[\varepsilon_t>\varepsilon_{t-1}]}, \tag{8.7}$$

where $[P]\in\{0,1\}$ is the truth value of P and e_s stands for x_j (tail case) or K ($\mathcal{M}$ case).

Now track J_m for $m\ge 1$. Its first block is the pivot chain of $(x_{j-1},\dots,x_m,C)$ from ψ: S_{x_m} exchanges β_m with γ, and S_{x_l} ($m<l<j$) exchanges β_l with β_{l-1}; afterwards x_m carries γ, x_l carries β_{l-1} for $m<l<j$, and C carries β_{j-1}. The block $T_{x_1}\cdots T_{x_{m-1}}$ exchanges β_l with β_{l-1} and leaves β_{m-1} at K. Then T_{x_m} exchanges γ (at x_m) with β_{m-1} (at K), after which K carries γ and x_m carries β_{m-1}. Next, T_{e_t} ($t\le s-1$) exchanges the value at e_t with the value at K. The value at e_t is ε_t: if e_t is none of the x_l it is $\psi(e_t)=\psi'(e_t)$, and if $e_t=x_l$ ($l\le j-1$) it is β_{l-1} in the present transposition word (by the above, in each of the cases $l<m$, $l=m$, $l>m$) and also $\psi'(x_l)=\beta_{l-1}$. The value at K is ε_{t-1} by induction. Finally, in the tail case, T_{x_j} exchanges β_j (the position x_j has not been touched) with ε_{s-1}. For $m=0$ the tracking is the same except that Z exchanges β_0 with γ and then S_{x_l} ($1\le l\le j-1$) exchanges β_l with β_{l-1}, after which K carries γ. Hence, with $q_{aC}=q_{aK}q_{KC}$ for $a\le k$ and (8.7),

$$\mathrm{wt}_\psi(J_m)=\mathrm{wt}_\psi(J)\cdot q_{x_mK}^{P_m}\,q_{x_jK}^{Q}\,q_{KC}^{a_m-a_E}, \tag{8.8}$$

where $P_0=0$, $P_m=[\beta_m>\gamma]+[\gamma>\beta_{m-1}]-[\beta_m>\beta_{m-1}]$ for $m\ge 1$, $Q=[\beta_j>\varepsilon_{s-1}]-[\beta_j>\beta_{j-1}]-[\beta_{j-1}>\varepsilon_{s-1}]$ in the tail case and $Q=0$ in the $\mathcal{M}$ case, a_m is the number of ascents of the sequence $(\gamma,\beta_m,\beta_{m+1},\dots,\beta_{j-1})$ (for $m=0$: of $(\gamma,\beta_0,\dots,\beta_{j-1})$), and a_E is the number of ascents of $E=(\varepsilon_0,\dots,\varepsilon_s)$.

$Q=0$. In the tail case, the letter S_{x_j} of J is a cover of the permutation reached before it, in which x_j carries β_{j-1}, C carries ε_{s-1}, and the position K, which lies strictly between x_j and C, carries β_j. By Definition 6.3, if $\beta_{j-1}<\varepsilon_{s-1}$ then $\beta_j\notin(\beta_{j-1},\varepsilon_{s-1})$, and if $\beta_{j-1}>\varepsilon_{s-1}$ then $\beta_j\in(\varepsilon_{s-1},\beta_{j-1})$; in each case $[\beta_j>\varepsilon_{s-1}]=[\beta_j>\beta_{j-1}]+[\beta_{j-1}>\varepsilon_{s-1}]$.

$a_E=[\beta_{j-1}>\gamma]$. The cyclic sequence E is unimodal, so as a linear sequence it has at most one ascent; if it has none then $\varepsilon_s<\varepsilon_0$, and if its ascent is $(\varepsilon_{t-1},\varepsilon_t)$ then by Corollary 6.13 $\varepsilon_t>\cdots>\varepsilon_s>\varepsilon_0>\cdots>\varepsilon_{t-1}$, so $\varepsilon_s>\varepsilon_0$. Thus $a_E=[\varepsilon_s>\varepsilon_0]=[\beta_{j-1}>\gamma]$. The same argument applies to any linear block of a unimodal cyclic sequence: *the block has an ascent if and only if its last entry exceeds its first.*

$P_m\ge 0$, *with equality if and only if* $m=g$ ($1\le m\le j-1$). If $\beta_m<\beta_{m-1}$ then $P_m=0$ exactly when $\beta_m<\gamma<\beta_{m-1}$, and $P_m=1$ otherwise. If $\beta_m>\beta_{m-1}$, then (β_{m-1},β_m) is the ascent of Γ, so $\beta_{m-1}=\min\Gamma$ and $\beta_m=\max\Gamma$, and $P_m=0$ exactly when $\gamma\notin(\beta_{m-1},\beta_m)$, i.e. when γ lies outside $[\min\Gamma,\max\Gamma]$; again $P_m=1$ otherwise. By the definition of the insertion gap, in both cases $P_m=0$ if and only if gap m is the insertion gap of γ in Γ.

(i) Since the monomials q_{x_mK}, q_{x_jK} involve only q_i with $i<K$ and q_{KC} only q_i with $i\ge K$, and $Q=0$, (8.8) shows that $\mathrm{wt}_\psi(J_m)=\mathrm{wt}_\psi(J)$ if and only if $P_m=0$ and $a_m=a_E$; by Lemmas 8.17 and 8.14 this is also equivalent to J_m being a cover path of weight $\mathrm{wt}_\psi(J)$. By the previous step, $P_m=0$ holds for $m=0$ and for $m=g$ (if $g\ge 1$) only. Let Σ be the unimodal cyclic sequence obtained by inserting γ into its insertion gap of Γ.

If $g\ge 1$: the sequence $(\gamma,\beta_g,\dots,\beta_{j-1})$ is a block of Σ, so $a_g=[\beta_{j-1}>\gamma]=a_E$, and J_g qualifies. For $m=0$ we show $a_0\ne a_E$. Let $d\in\{0,1\}$ be the number of linear ascents of $(\beta_0,\dots,\beta_{j-1})$, so that $a_0=[\beta_0>\gamma]+d$. If $d=0$ then $\beta_0>\cdots>\beta_{j-1}$ and the wrap is the ascent of Γ; as $g\ne 0$, γ lies in (β_{j-1},β_0), so $a_0=1$ while $a_E=[\beta_{j-1}>\gamma]=0$. If $d=1$, the wrap (β_{j-1},β_0) is a descent gap, and $g\ne 0$ means $\gamma\notin(\beta_0,\beta_{j-1})$: if $\gamma<\beta_0$ then $a_0=2\ne a_E$, and if $\gamma>\beta_{j-1}$ then $a_0=1$ while $a_E=0$. So J_0 does not qualify, and J_g is the unique resolution that does.

If $g=0$: only $m=0$ is possible, and $(\gamma,\beta_0,\dots,\beta_{j-1})$ is the whole of Σ read from γ, a linear block of a unimodal cyclic sequence, so $a_0=[\beta_{j-1}>\gamma]=a_E$ and J_0 qualifies.

(ii) Recall that a cyclic subsequence of a unimodal cyclic sequence is unimodal, and that a unimodal cyclic sequence of three distinct numbers is cyclically decreasing, i.e. reads $(\max,\mathrm{mid},\min)$ up to rotation; so two unimodal cyclic sequences containing the same three distinct numbers list them in the same cyclic order. Let $e_t=x_l$ with $t\le s-1$, $l\le j-1$, so that $\varepsilon_t=\beta_{l-1}$, and suppose $l\le g$ (so $g\ge 1$). In $\Sigma=(\beta_0,\dots,\beta_{g-1},\gamma,\beta_g,\dots,\beta_{j-1})$ the three distinct numbers $\beta_{l-1},\gamma,\beta_{j-1}$ occur in the cyclic order $(\beta_{l-1},\gamma,\beta_{j-1})$, because $l-1\le g-1$ and $l-1<j-1$. In the unimodal cyclic sequence E they occur as $\varepsilon_0=\gamma$, $\varepsilon_t=\beta_{l-1}$,

$\varepsilon_s = \beta_{j-1}$ with $0 < t < s$, i.e. in the cyclic order $(\gamma, \beta_{l-1}, \beta_{j-1})$, which is the opposite one; this contradiction proves $l > g$. For the second assertion, the whole cycle $(e_r, \dots, e_1, C)$ is tight from ψ' and its cyclic value sequence $(\varepsilon_0, \dots, \varepsilon_r)$, with $\varepsilon_t = \psi'(e_t) = \beta_{l-1}$ for $e_t = x_l$, is unimodal. Suppose $e_t = x_l$ with $t \geq s+1$ and $g < l \leq j-1$. In Σ the numbers $\gamma, \beta_{l-1}, \beta_{j-1}$ occur in the cyclic order $(\gamma, \beta_{l-1}, \beta_{j-1})$, because $g \leq l-1 < j-1$; in the e-sequence they occur as $\varepsilon_0, \varepsilon_s, \varepsilon_t$ with $0 < s < t$, i.e. in the order $(\gamma, \beta_{j-1}, \beta_{l-1})$, the opposite one. Hence $l \leq g$. □

States and the junction process. Fix $u, w \in S_\infty$ and a monomial q. By Lemma 6.7(iii), a level-k step $v \overset{k}{\rightsquigarrow} v'$ may be represented by any *pivot word* of $v^{-1}v'$, i.e. any transposition word obtained by interleaving the pivot factorizations of its cycles; all of them are cover paths from v of weight $q_k(v, v')$, and the same holds for level-$(k+1)$ steps. A *state* is a transposition word $U = W_1 \,|\, W_2$ with a cut, which is a cover path from u of weight q with product $u^{-1}w$, together with an *active position* x, such that:

- (S1) $W_1 = R_1\, T_{x_1} \cdots T_{x_j}$, where R_1 is a pivot word of a product of pairwise disjoint cycles, each consisting of its top, which is $\geq k+2$, together with positions $\leq k$, and $x_1, \dots, x_j$ are distinct positions $\leq k$ with $x_j = x$ and $x_1, \dots, x_{j-1} \notin \mathrm{supp} R_1$ (the active position may or may not lie in $\mathrm{supp} R_1$); or, in the $\mathcal{M}$ *case*, $x = K$ and $W_1 = R_1 T_{x_1} \cdots T_{x_{j-1}}$ with $x_1, \dots, x_{j-1}$ as before.
- (S2) $W_2 = E\, W_2'$, where $E = S_{e_1} \cdots S_{e_r}$ is the pivot factorization of a cycle $e = (e_r, \dots, e_1, C)$ with top $C \geq k+2$ containing x, say $x = e_s$, whose other elements e_t ($t \neq s$) are positions $\leq k$; and W_2' is a pivot word of a product of pairwise disjoint cycles, each consisting of its top, which is $\geq K$, together with positions $\leq k$, disjoint from e and not containing x.
- (S3) The cycle of W_2' with top K, if there is one, is disjoint from $\{x_1, \dots, x_{j-1}\}$.

Here $T_y = t_{yK}$ and $S_y = t_{yC}$ as before (with C the top of e). Conditions (S1) and (S2) say that every cycle of R_1 and of W_2', and in the tail case also e, has exactly one element greater than k, as the cycles of a level-k step do, while in the $\mathcal{M}$ case e has exactly one element greater than K; the requirement on the tops alone would not suffice, since a cycle such as (K, C') with $C' \geq k+2$ has top $\geq K$ but is not a cycle of any level-k or level-$(k+1)$ step. Two states are identified if they differ by reorderings as in Lemma 8.15, i.e. we regard a state as the data (cycles of R_1, the sequence $x_1, \dots, x_j$, the cycle e, the cycles of W_2') rather than as a particular interleaving. A state is a *source* if $x \notin \mathrm{supp} R_1$ (and the active position is in $[1, k]$ or K accordingly); the sources are exactly the blocked tails and the chains of $\mathcal{M}_k^{\mathrm{mov}}$: a blocked tail $u \overset{k}{\rightsquigarrow} v \overset{k}{\rightsquigarrow} w$ with block data (ρ_1, X, ρ_2) and cut $i = i_{\min} > 0$ is the state with R_1 a pivot word of ρ_1, $(x_1, \dots, x_j) = (x_1, \dots, x_i)$, e the cycle of ρ_2 through x_i, and W_2' a pivot word of $d_i \rho_2 e^{-1}$; a chain $u \overset{k}{\rightsquigarrow} v \overset{k+1}{\rightsquigarrow} w$ of $\mathcal{M}_k^{\mathrm{mov}}$ with $X = (x_n, \dots, x_1, K)$ is the state with $(x_1, \dots, x_{j-1}) = (x_1, \dots, x_n)$, $x = K$, e the cycle of $v^{-1}w$ through K, and W_2' a pivot word of the other cycles. In both cases (S1) and (S2) hold because the two steps are k-shaped, respectively k- and $(k+1)$-shaped (the cycles of ρ_1, ρ_2 and $v^{-1}w$ other than those through K have tops $\geq k+2$); (S3) holds for a blocked tail because its top-K cycle $d_i = (x_n, \dots, x_{i+1}, K)$ of W_2' avoids $x_1, \dots, x_{i-1}$, and vacuously for a chain of $\mathcal{M}_k^{\mathrm{mov}}$, whose W_2' contains no cycle through K. Conversely such a state is a blocked tail (resp. a chain of $\mathcal{M}_k^{\mathrm{mov}}$) of weight q: since $x \notin \mathrm{supp} R_1$, (S1) and (S2) make W_1 and W_2 pivot words of k-shaped permutations (in the $\mathcal{M}$ case, of a k-shaped and a $(k+1)$-shaped permutation), which are tight because U is a cover path (Lemma 6.7(iii)), so $u \overset{k}{\rightsquigarrow} v \overset{k}{\rightsquigarrow} w$ (resp. $u \overset{k}{\rightsquigarrow} v \overset{k+1}{\rightsquigarrow} w$ with K moved) for $v = uW_1$; in the tail case the pair is compatible, its two cycles with top K, namely $(x_j, \dots, x_1, K)$ and the top-K cycle of W_2', meeting only in K by (S3) and $x \notin \mathrm{supp} W_2'$, and cycles with a common top $\geq k+2$ sharing no element $\leq k$ by Lemma 8.8; and its cut j is $i_{\min}$, which is positive, because $x_j = x \in \mathrm{supp} e \subseteq \mathrm{supp} \rho_2$. For a state U with active position x put

$$N_1(U) = \mathrm{supp} R_1 \cup \{x_1, \dots, x_{j-1}\}, \qquad M_2(U) = \{a \leq k : a \text{ is moved by } W_2\},$$

the set of positions $\leq k$ moved by W_1 other than (possibly) the active one, and the set of positions $\leq k$ moved by W_2.

The *junction move* transforms a state U as follows. The transposition word $J = T_{x_1} \cdots T_{x_j}\, S_{e_1} \cdots S_{e_{s-1}}\, S_x$ formed by the K-letters of W_1 and the first s letters of E is a junction word in the sense of the token lemma (tail case if $x \leq k$, $\mathcal{M}$ case if $x = K$), it occupies a contiguous segment of U, and it is a cover path from $\psi = uR_1$. Let g be the index provided by Lemma 8.18, and replace J by J_g; by Lemmas 8.14, 8.17 and 8.18,

the resulting transposition word

$$U^+ = \begin{cases} R_1\, S_{x_g} \cdots S_{x_{j-1}}\, T_{x_1} \cdots T_{x_{g-1}} \;\big|\; T_{x_g}\, T_{e_1} \cdots T_{e_{s-1}}\, T_x\, S_{e_{s+1}} \cdots S_{e_r}\, W_2' & (g \geq 1), \\ R_1\, Z\, S_{x_1} \cdots S_{x_{j-1}} \;\big|\; T_{e_1} \cdots T_{e_{s-1}}\, T_x\, S_{e_{s+1}} \cdots S_{e_r}\, W_2' & (g = 0), \end{cases}$$

with T_x omitted when $x = K$ and with the cut placed as indicated, is a cover path from u of weight q with product $u^{-1}w$.

Proposition 8.19 (The junction process)**.** *Let U be a state with active position x and let U^+ be obtained by the junction move, with index g.*

(a) *If $g = 0$, then U^+ is a chain $u \overset{k+1}{\rightsquigarrow} v^+ \overset{k}{\rightsquigarrow} w$ of $\mathcal{M}_k'^{\mathrm{mov}}$ of weight q, with triple $\big(q,\ u(\overline{N_1(U)}),\ w(\overline{M_2(U)})\big)$, where $\overline{S} = [1,k] \setminus S$.*

(b) *If $g \geq 1$ and x_g is not moved by $S_{e_{s+1}} \cdots S_{e_r} W_2'$, then U^+ is a compatible pair $u \overset{k}{\rightsquigarrow} v^+ \overset{k}{\rightsquigarrow} w$ of weight q which is the blocked head of its block, with $T(v^+) = \big(q,\ u(\overline{N_1(U)}),\ w(\overline{M_2(U)})\big)$.*

(c) *If $g \geq 1$ and x_g is moved by $S_{e_{s+1}} \cdots S_{e_r} W_2'$, let f be the cycle of $S_{e_{s+1}} \cdots S_{e_r} W_2'$ through x_g (so f is either $(e_r, \ldots, e_{s+1}, C)$ or a cycle of W_2'). Then the transposition word U' obtained from U^+ by moving the cut one letter to the right, past T_{x_g}, and moving the letters of f to the front of the suffix, is a state with active position x_g, with $N_1(U') = N_1(U)$ and $M_2(U') = M_2(U)$, and the index j of its junction word is $g < j$.*

Consequently, starting from a source and iterating the junction move, one reaches after at most j moves a target (a blocked head or a chain of $\mathcal{M}_k'^{\mathrm{mov}}$) of weight q whose triple is the triple $S(v)$ of the source if the source is a blocked tail v, and the triple of the source if the source is a chain of $\mathcal{M}_k^{\mathrm{mov}}$. We write $\Phi(U)$ for this target.

Proof. We first collect what Lemma 8.18(ii) says about the positions e_t which are among $x_1, \ldots, x_{j-1}$: if $t < s$ then $e_t = x_l$ with $l > g$, and if $t > s$ then $e_t = x_l$ with $l \leq g$. (The second assertion of (ii) applies because $JS_{e_{s+1}} \cdots S_{e_r}$ is a segment of the cover path U.) In particular $x_g \notin \{e_1, \ldots, e_{s-1}\}$, and $\{e_{s+1}, \ldots, e_r\} \cap \{x_{g+1}, \ldots, x_{j-1}\} = \emptyset$.

The prefix of U^+. For $g \geq 1$ it is R_1 followed by the pivot factorization of the cycle $(x_{j-1}, \ldots, x_g, C)$ and the pivot factorization of $(x_{g-1}, \ldots, x_1, K)$. The positions $x_1, \ldots, x_{j-1}$ avoid $\mathrm{supp} R_1$ by (S1), so if R_1 has no cycle with top C these are three pairwise disjoint pieces, and if R_1 contains a cycle $(z_t, \ldots, z_1, C)$, then its letters followed by $S_{x_g} \cdots S_{x_{j-1}}$ form the pivot factorization of the cycle $(x_{j-1}, \ldots, x_g, z_t, \ldots, z_1, C)$, disjoint from the other cycles. Thus the prefix is a pivot word of a product π_1^+ of pairwise disjoint cycles, each with exactly one element $> k$ (by (S1), the cycles of R_1 have this property, and the two new cycles have elements $x_l \leq k$ besides their tops), i.e. π_1^+ is k-shaped; being an initial segment of a cover path it is u-tight (Lemma 6.7(iii)), so $u \overset{k}{\rightsquigarrow} v^+ := u\pi_1^+$. The positions $\leq k$ it moves form $N_1(U)$. For $g = 0$ the prefix is R_1 followed by $ZS_{x_1} \cdots S_{x_{j-1}}$, which merges with the cycle of R_1 with top C (if any) into the cycle $(x_{j-1}, \ldots, x_1, K, z_t, \ldots, z_1, C)$; so the prefix is a pivot word of a $(k+1)$-shaped, u-tight permutation moving K, i.e. $u \overset{k+1}{\rightsquigarrow} v^+$ with K moved, and it moves the positions $N_1(U) \cup \{K\}$.

The suffix of U^+. Its letters T are T_{x_g} (if $g \geq 1$), $T_{e_1}, \ldots, T_{e_{s-1}}$, T_x (if $x \leq k$), and the letters T of W_2', which form the pivot factorization of a cycle D with top K (trivial if $x = K$, since then $K \in e$ and W_2' is disjoint from e by (S2)). The positions $x_g, e_1, \ldots, e_{s-1}, x$ and those of D are pairwise distinct ($x_g \notin \{e_t\}_{t<s}$ by the first paragraph; $x \notin \{e_t\}_{t \neq s}$; D is disjoint from e and does not contain x by (S2), and does not contain x_g by (S3)), so these letters form, in the order in which they occur, the pivot factorization of the cycle $D^+ = (D,\ x,\ e_{s-1}, \ldots, e_1,\ x_g,\ K)$ (we list the elements following the cycle from the top), respectively $(D, x, e_{s-1}, \ldots, e_1, K)$ when $g = 0$ and $(D, e_{s-1}, \ldots, e_1, x_g, K)$ in the $\mathcal{M}$ case; by (S2) all its elements other than K are $\leq k$. The remaining letters of the suffix are those of $(e_r, \ldots, e_{s+1}, C)$ and of the other cycles of W_2', which are pairwise disjoint, disjoint from D, x and $e_1, \ldots, e_{s-1}$, have, by (S2), exactly one element greater than k, and, under the hypothesis of (a) or (b), do not contain x_g. Hence in cases (a) and (b) the suffix is a pivot word of a k-shaped permutation π_2^+, which is v^+-tight because the suffix is a terminal segment of a cover path; so $v^+ \overset{k}{\rightsquigarrow} w$. The positions $\leq k$ moved by the suffix are $M_2(U)$ in case (a) and $M_2(U) \sqcup \{x_g\}$ in case (b).

(a) By the above, U^+ is a chain $u \overset{k+1}{\rightsquigarrow} v^+ \overset{k}{\rightsquigarrow} w$ whose first step moves K; its weight is q and its triple is $(q, \mathrm{fix}_{k+1}(u, v^+), \mathrm{fix}_k(v^+, w)) = (q, u(\overline{N_1(U)}), w(\overline{M_2(U)}))$.

(b) We have $u \overset{k}{\rightsquigarrow} v^+ \overset{k}{\rightsquigarrow} w$. The two steps are compatible: the cycles with top K are $(x_{g-1},\dots,x_1,K)$ and D^+, which meet only in K (as $e_t = x_l$ with $t<s$ forces $l>g$); and by Lemma 8.8 two cycles of the two steps with a common top $\geq k+2$ cannot share an element $\leq k$. The block data of the pair are $\big(\rho_1^+,\ X^+ = (x_{g-1},\dots,x_1,K)D^+,\ \rho_2^+\big)$ with cut $g-1$, where ρ_1^+ contains the cycle through x_g with top C; since x_g is the g-th pivot of X^+, $i_{\max} = g-1$, so the pair is the head of its block and this head is blocked. Finally $\mathrm{fix}_k(u,v^+) = u(\overline{N_1(U)})$, $\mathrm{fix}_k(v^+,w) = w(\overline{M_2(U)\sqcup\{x_g\}})$ and $c(v^+) = v^+(K) = \beta_{g-1}$, the value at K after the prefix; the letter T_{x_g} moves this value to x_g, which is not moved by any later letter, so $w(x_g) = c(v^+)$ and $T(v^+) = (q, u(\overline{N_1(U)}), w(\overline{M_2(U)}))$.

(c) Since $x_g \notin D$ by (S3), $f \neq D$: so f is either $(e_r,\dots,e_{s+1},C)$ or a cycle of W_2' other than D, and in either case, by (S2), f consists of its top, which is $\geq k+2$, together with positions $\leq k$, one of them x_g. The prefix of U' is the prefix of U^+ followed by T_{x_g}: it has the form (S1) with R_1' the prefix of U^+ minus its letters T (a pivot word of the cycles of R_1 other than the one with top C, if any, together with $(x_{j-1},\dots,x_g,z_t,\dots,z_1,C)$; each of these consists of a top $\geq k+2$ and positions $\leq k$, by (S1) for R_1), with K-letters $T_{x_1},\dots,T_{x_{g-1}},T_{x_g}$, and active position $x_g \in \mathrm{supp}R_1'$; the positions $x_1,\dots,x_{g-1}$ avoid $\mathrm{supp}R_1'$. The letters of f move no position moved by the letters of the suffix of U^+ which they pass (these letters are T_{x_g}—not passed—$T_{e_1},\dots,T_{e_{s-1}},T_x$, the letters of $(e_r,\dots,e_{s+1},C)$ if $f \neq$ this cycle, and letters of cycles of W_2' other than f; all are disjoint from f because the cycles of the suffix are pairwise disjoint apart from the common element x_g of T_{x_g} and f), so by Lemma 8.15 U' is a cover path of weight q. Its suffix is f followed by a pivot word of pairwise disjoint cycles: the cycle $D' = (D,x,e_{s-1},\dots,e_1,K)$ formed by the letters $T_{e_1},\dots,T_{e_{s-1}},T_x$ and those of D (the letter T_{x_g} having moved to the prefix), the cycle $(e_r,\dots,e_{s+1},C)$ if f is not this cycle, and the cycles of W_2' other than f; all of these are disjoint from f, none of them contains x_g, and each consists of a top $\geq K$ together with positions $\leq k$. This is (S2) for U', with $e' = f$ and $x' = x_g$. For (S3), the top-K cycle of the new W_2' is D', and it avoids $x_1,\dots,x_{g-1}$: D does by (S3) for U, $x = x_j$ does, and $e_t = x_l$ with $t<s$ forces $l>g$ by the first paragraph. So U' is a state with active position $x_g \leq k$, whose junction word has g letters T and index $g<j$. Its sets are $N_1(U') = \mathrm{supp}R_1' \cup \{x_1,\dots,x_{g-1}\} = \mathrm{supp}R_1 \cup \{x_1,\dots,x_{j-1}\} = N_1(U)$ and $M_2(U') = M_2(U)\cup\{x_g\} = M_2(U)$, since $x_g \in \mathrm{supp}f \subseteq M_2(U)$.

For the consequence, note that a source U has $N_1(U) = M_1 \setminus \{x\}$ where M_1 is the set of positions $\leq k$ moved by its first step, and $M_2(U) = M_2$ is the set moved by its second step. If U is a blocked tail v, then $S(v) = (q, \mathrm{fix}_k(u,v)\cup\{c(v)\}, \mathrm{fix}_k(v,w))$ with $c(v) = u(x)$ (Lemma 8.12(iii)), which is $(q, u(\overline{M_1\setminus\{x\}}), w(\overline{M_2}))$; if U is a chain of $\mathcal{M}_k^{\mathrm{mov}}$ its triple is $(q, \mathrm{fix}_k(u,v), \mathrm{fix}_{k+1}(v,w)) = (q, u(\overline{M_1}), w(\overline{M_2}))$, and $N_1(U) = M_1$ as $x = K$. Since the junction move strictly decreases the index j of the junction word, the iteration terminates, by (a) or (b), after at most j moves, and the invariance of N_1 and M_2 in (c) gives the triple. □

Example 8.20 (A junction move)**.** Let $k=1$, $K=2$, $u=123$ and $w=312$, so that $u^{-1}w = 312 = t_{12}t_{13}$. The steps $u \overset{1}{\rightsquigarrow} v$ are the trivial one and t_{12} ($v=213$; t_{13} is not a cover of 123 because of the intermediate value 2), and the only compatible pair $u \overset{1}{\rightsquigarrow} v \overset{1}{\rightsquigarrow} w$ is

$$123 \xrightarrow{t_{12}} 213 \xrightarrow{t_{13}} 312,$$

with weight $q=1$ and $\omega_1 = [\emptyset \mid \emptyset] = 1$. Its block data are $\rho_1 = 1$, $X = (x_1,K) = (1,2)$, $\rho_2 = (1,3)$, with cut $i = 1 = n$; since $x_1 = 1 \in \mathrm{supp}\rho_2$ we have $i_{\min} = 1$, so $v = 213$ is a *blocked tail*, and since $\mathrm{supp}\rho_1 = \emptyset$, $i_{\max} = 1$ and v is also an unblocked head. Its triples are $S(v) = (1,\{u(1)\},\emptyset) = (1,\{1\},\emptyset)$ and $T(v) = (1,\emptyset,\{v(2)\}) = (1,\emptyset,\{1\})$.

As a state, v is the transposition word $U = T_1 \,|\, S_1 = t_{12} \,|\, t_{13}$ with R_1 empty, $(x_1) = (1)$, active position $x=1$, $e = (1,3)$ with $C=3$, $r=s=1$, and W_2' empty; its junction word is $J = T_1S_1$ with $j=1$, so the only resolution is $J_0 = Z\,T_1 = t_{23}\,t_{12}$ (indeed $T_xS_x = ZT_x$), and $g=0$. From $\psi = u = 123$ the word J exchanges $1,2$ and then $2,3$, both ascents, and J_0 exchanges $2,3$ and then $1,3$, both ascents; so $\mathrm{wt}_u(J_0) = \mathrm{wt}_u(J) = 1$, as Lemma 8.18(i) asserts, and J_0 is a cover path by Lemma 8.14. The junction move gives

$$U^+ = Z \;\big|\; T_1 = t_{23} \;\big|\; t_{12}, \qquad \text{i.e.} \qquad 123 \xrightarrow{t_{23}} 132 \xrightarrow{t_{12}} 312,$$

a chain $u \overset{2}{\rightsquigarrow} v^+ \overset{1}{\rightsquigarrow} w$ whose first step moves $K=2$: a chain of $\mathcal{M}_1'^{\mathrm{mov}}$, as in Proposition 8.19(a). Its triple is $\big(1, \mathrm{fix}_2(123,132), \mathrm{fix}_1(132,312)\big) = (1,\{1\},\emptyset) = S(v)$, as required. Here $N_1(U) = \emptyset$ and $M_2(U) = \{1\}$, and the triple is $(1, u(\overline{N_1}), w(\overline{M_2})) = (1,\{1\},\emptyset)$ in the notation of the proposition.

For completeness, the other side of (B_k) for this (u,w): $\mathcal{M}_1(u,w)$ consists of the single chain $123 \xrightarrow{t_{12}} 213 \xrightarrow{t_{13}} 312$ read with the second step at level 2 (it does not move K, so it is not in $\mathcal{M}_1^{\mathrm{mov}}$), with triple $(1,\emptyset,\mathrm{fix}_2(213,312)) = (1,\emptyset,\{1\}) = T(v)$, matching the unblocked head as in the proof of Proposition 8.13. Thus ($\mathrm{J}_k^{\mathrm{B}}$) reads $\{S(v)\} = \{(1,\{1\},\emptyset)\}$ on both sides, (B_k) reads $\{T(v)\}\uplus\{(1,\{1\},\emptyset)\} = \{S(v)\}\uplus\{(1,\emptyset,\{1\})\}$, and Proposition 8.9 reads $(z_1-z_2)\cdot 1 = (y_1-z_2)-(y_1-z_1)$. The reversal of Lemma 8.16 sends this configuration to the pair $w_0w = 132 \overset{1}{\rightsquigarrow} w_0v = 231 \overset{1}{\rightsquigarrow} w_0u = 321$ for $(u,w) = (132,321)$, which is a blocked head, and the chain U^+ to the chain $132 \xrightarrow{t_{12}} 312 \xrightarrow{t_{23}} 321$ of $\mathcal{M}_1^{\mathrm{mov}}$ for that pair.

Injectivity.

Lemma 8.21. *Let U be a state and U^+ the result of the junction move, and work in S_n. Then the reversed transposition word $(U^+)^{\mathrm{rev}}$, read from w_0w with the reversed cut, is (after reorderings as in Lemma 8.15) a state for the pair (w_0w, w_0u) and the weight q, with active position x_g if $g \geq 1$ and K if $g = 0$, and the junction move applied to it produces U^{rev}.*

Proof. By Lemma 8.16, $(U^+)^{\mathrm{rev}}$ and U^{rev} are cover paths from w_0w of weight q with product $w^{-1}u$. Let $g \geq 1$. The prefix of $(U^+)^{\mathrm{rev}}$ is the reversed suffix of U^+: the reversal of the pivot word of W_2' and of $S_{e_{s+1}}\cdots S_{e_r}$, followed by $T_x\,T_{e_{s-1}}\cdots T_{e_1}\,T_{x_g}$. Its letters T are those of D (reversed), then $T_x, T_{e_{s-1}}, \ldots, T_{e_1}, T_{x_g}$; the other letters move positions disjoint from those of D, x and $e_1,\ldots,e_{s-1}$, so by Lemma 8.15 they may be moved to the front, giving the form (S1) with R_1^{rev} consisting of the reversed pivot words of the cycles of W_2' other than D and of $(e_r,\ldots,e_{s+1},C)$, the K-letters being $T_{y_1},\ldots,T_{y_p}$ with $(y_1,\ldots,y_p) = (D \text{ reversed}, x, e_{s-1},\ldots,e_1,x_g)$, and active position x_g (which may lie in $\mathrm{supp}R_1^{\mathrm{rev}}$). The suffix of $(U^+)^{\mathrm{rev}}$ is $T_{x_{g-1}}\cdots T_{x_1}\,S_{x_{j-1}}\cdots S_{x_g}\,R_1^{\mathrm{rev}}$; here $S_{x_{j-1}}\cdots S_{x_g}$ followed by the reversed letters of the cycle of R_1 with top C (if any) is the pivot factorization of a cycle e^{rev} with top C through x_g, in which x_g is preceded by $x_{j-1},\ldots,x_{g+1}$; moving $T_{x_{g-1}}\cdots T_{x_1}$ past it (Lemma 8.15) gives the form (S2), the remaining cycles being $(x_1,\ldots,x_{g-1},K)$ and the other cycles of R_1, which do not contain x_g. The shape requirements of (S1) and (S2) are inherited from those of U: the cycles of R_1^{rev} are cycles of W_2' other than D, which by (S2) for U have tops $\geq k+2$ (their tops are $\geq K$ and differ from K) and other elements $\leq k$, together with $(e_r,\ldots,e_{s+1},C)$; the K-letters $y_1,\ldots,y_{p-1}$ are positions $\leq k$ not in $\mathrm{supp}R_1^{\mathrm{rev}}$ (the cycles of W_2' are pairwise disjoint and disjoint from e, and $x \notin \mathrm{supp}W_2'$); the cycle e^{rev} has top $C \geq k+2$ and other elements $x_{j-1},\ldots,x_{g+1}$ and the z's of the top-C cycle of R_1, all $\leq k$; and the remaining cycles $(x_1,\ldots,x_{g-1},K)$ and those of R_1 consist of a top $\geq K$ and positions $\leq k$ by (S1) for U. Finally (S3) holds for $(U^+)^{\mathrm{rev}}$: its top-K cycle $(x_1,\ldots,x_{g-1},K)$ avoids $y_1,\ldots,y_{p-1}$, because D avoids $x_1,\ldots,x_{j-1}$ by (S3) for U, $x = x_j$, and $e_t = x_l$ with $t < s$ forces $l > g$ (Lemma 8.18(ii)). So $(U^+)^{\mathrm{rev}}$ is a state. Its junction word is $T_{y_1}\cdots T_{y_p}\,S_{x_{j-1}}\cdots S_{x_{g+1}}\,S_{x_g}$. Suppose first that $x \leq k$. The resolution with stop at the letter T_x (the letter y_m with $y_m = x$) is

$$S_x\,S_{e_{s-1}}\cdots S_{e_1}\ \cdot\ (D \text{ reversed})\ \big|\ T_x\,T_{x_{j-1}}\cdots T_{x_{g+1}}\,T_{x_g}\ \cdot\ (\text{rest of the suffix}),$$

which, after moving the block $S_xS_{e_{s-1}}\cdots S_{e_1}$ past the letters of D and the reversed letters of the cycle of R_1 with top C past $T_{x_{g-1}}\cdots T_{x_1}$ (both inert), is exactly

$$U^{\mathrm{rev}} = \big(W_2'\big)^{\mathrm{rev}}\,S_{e_r}\cdots S_{e_{s+1}}\,S_xS_{e_{s-1}}\cdots S_{e_1}\ \big|\ T_xT_{x_{j-1}}\cdots T_{x_1}\,R_1^{\mathrm{rev}}.$$

Since U^{rev} is a cover path of weight q, Lemma 8.18(i) shows that this resolution is the one selected by the junction move. If $x = K$, there is no letter T_x, the sequence $(y_1,\ldots,y_p)$ is $(e_{s-1},\ldots,e_1,x_g)$, and the same comparison shows that the resolution $J_0 = Z\,S_{e_{s-1}}\cdots S_{e_1}\cdot T_{x_{j-1}}\cdots T_{x_g}$ of the junction word yields $U^{\mathrm{rev}} = (W_2')^{\mathrm{rev}}S_{e_r}\cdots S_{e_{s+1}}ZS_{e_{s-1}}\cdots S_{e_1}\,|\,T_{x_{j-1}}\cdots T_{x_1}R_1^{\mathrm{rev}}$ up to inert reorderings. The case $g = 0$ is identical, with the prefix of $(U^+)^{\mathrm{rev}}$ having no active letter after T_{e_1}, the suffix containing the cycle $(x_{j-1},\ldots,x_1,K,\ldots,C)$ through K, so that the reversed state is of $\mathcal{M}$ type (its e^{rev} has top $C \geq k+2$, contains K, and has all other elements $\leq k$; its W_2' consists of the cycles of R_1 other than the one with top C, which by (S1) have tops $\geq k+2$ and positions $\leq k$, and (S3) is vacuous), and the resolution with stop at T_x (resp. J_0 if $x = K$) again giving U^{rev}. □

Proposition 8.22. *The map Φ from the sources of weight q for (u,w) to the targets of weight q for (u,w) is injective.*

Proof. Let U_0 be a source and $U_0, U_1, \ldots, U_m$ the states traversed by the process, U_{i+1} being obtained from U_i by a junction move followed, in case (c), by the normalization of Proposition 8.19(c), and $U_m^+ = \Phi(U_0)$ the

final target. We show that U_i is determined by U_{i+1} (resp. by U_m^+ for $i = m$). The transposition word U_i^+ (the result of the move, before normalization) is determined by U_{i+1}: one moves the active letter T of U_{i+1}, the last letter of its prefix, back to the front of the suffix (the normalization of (c) only reorders inert letters otherwise). By Lemma 8.21, $(U_i^+)^{\mathrm{rev}}$ is a state whose junction move produces U_i^{rev}; since the junction move is deterministic (Lemma 8.18(i) selects a unique resolution), U_i^{rev}, hence U_i, is determined by U_i^+. Finally, whether U_i is the source or an intermediate state is read off from U_i itself: a state is a source if and only if its active position does not lie in $\mathrm{supp} R_1$. Hence the whole sequence, and in particular U_0, is determined by $\Phi(U_0)$. □

Theorem 8.23 (The junction identity)**.** *The identity* ($\mathrm{J}_k^{\mathrm{B}}$) *holds for all $k \geq 1$ and all $u, w \in S_\infty$. Consequently* (B_k)*, Theorem 8.3 and* (J_k) *hold for all $k \geq 1$.*

Proof. Fix k, u, w, a monomial q, and n so large that everything takes place in S_n. Let $\mathrm{Src}(u, w)$ be the multiset of triples of the sources of weight q for (u, w), i.e. $S(v)$ for the blocked tails v and the triples of the chains of $\mathcal{M}_k^{\mathrm{mov}}$, and $\mathrm{Tgt}(u, w)$ the multiset of $T(v)$ for the blocked heads and of the triples of the chains of $\mathcal{M}_k'^{\mathrm{mov}}$, all of weight q. By Propositions 8.19 and 8.22, Φ is an injective, triple-preserving map from sources to targets; hence $\mathrm{Src}(u, w) \subseteq \mathrm{Tgt}(u, w)$ as multisets.

Now apply the reversal of Lemma 8.16 to whole configurations: it sends a compatible pair $u \overset{k}{\rightsquigarrow} v \overset{k}{\rightsquigarrow} w$ to the pair $w_0 w \overset{k}{\rightsquigarrow} w_0 v \overset{k}{\rightsquigarrow} w_0 u$ (the reversed pivot words are pivot words of the inverse permutations, which are tight by Lemma 6.7(iii)), with the same weight, with block data $(\rho_2^{-1}, X^{-1}, \rho_1^{-1})$ and cut $n - i$, so that $i_{\min}$ and $i_{\max}$ are exchanged with $n - i_{\max}$ and $n - i_{\min}$: blocked tails go to blocked heads and conversely. Likewise it exchanges the chains of $\mathcal{M}_k^{\mathrm{mov}}$ for (u, w) with those of $\mathcal{M}_k'^{\mathrm{mov}}$ for $(w_0 w, w_0 u)$. On triples it acts by $\sigma(q, A, B) = (q, w_0 B, w_0 A)$, where w_0 acts on values: indeed $\mathrm{fix}_k(w_0 w, w_0 v) = w_0 \mathrm{fix}_k(v, w)$, $\mathrm{fix}_k(w_0 v, w_0 u) = w_0 \mathrm{fix}_k(u, v)$ and $(w_0 v)(K) = w_0(v(K))$, so that $S(w_0 v) = \sigma(T(v))$ and $T(w_0 v) = \sigma(S(v))$, and similarly for the mixed chains. Hence $\sigma(\mathrm{Tgt}(u, w)) = \mathrm{Src}(w_0 w, w_0 u)$ and $\sigma(\mathrm{Src}(u, w)) = \mathrm{Tgt}(w_0 w, w_0 u)$. Applying the first paragraph to $(w_0 w, w_0 u)$ gives $\mathrm{Src}(w_0 w, w_0 u) \subseteq \mathrm{Tgt}(w_0 w, w_0 u)$, i.e. $\mathrm{Tgt}(u, w) \subseteq \mathrm{Src}(u, w)$. Therefore $\mathrm{Src}(u, w) = \mathrm{Tgt}(u, w)$ for every q, which is ($\mathrm{J}_k^{\mathrm{B}}$). The consequences follow from Propositions 8.13 and 8.7. □

9. Proof of the parabolic reduction

In this section we prove Proposition 2.9 in finite rank. Peterson's comparison formula expresses the Gromov–Witten invariants of G/P in terms of those of G/B; it was proved by Woodward [37] and extended to equivariant quantum cohomology by Huang and Li [17], whose form we use. We use the notation of Section 2: W_P is a parabolic subgroup of S_N generated by $P \subseteq \{s_1, \ldots, s_{N-1}\}$, $\omega_P = w_P$ is its longest element, and for $\lambda_P \in Q^\vee / Q_P^\vee$ we write $\lambda_B \in Q^\vee$ for its Peterson–Woodward lift. In the language of roots, λ_B is the unique lift with $\langle \alpha, \lambda_B \rangle \in \{0, -1\}$ for all positive roots α of the Levi subgroup of P; in type A these are the roots $\alpha_i + \cdots + \alpha_{j-1}$ with $s_i, \ldots, s_{j-1} \in P$, and $\langle \alpha_i + \cdots + \alpha_{j-1}, d \rangle = \langle d \rangle_{ij}$, so this is the condition of Definition 2.6.

Proposition 9.1 ([17, Proposition 2.10])**.** *Let W_P be a parabolic subgroup of W and $u, v, w \in W^P$. Then for any $\lambda_P \in Q^\vee / Q_P^\vee$, we have that*

$$c_{u,v}^{w,\lambda_P}(y; y) = c_{u,v}^{w\omega_P \omega_{P'}, \lambda_B}(y; y),$$

where ω_P is the longest element in W_P, $\omega_{P'}$ is the longest element in $W_{P'}$, where P' is the parabolic subgroup generated by $\{s_\alpha \mid \alpha \in \Delta_P, \langle \alpha, \lambda_B \rangle = 0\}$, and λ_B is the Peterson-Woodward lift of λ_P.

Here the left-hand side is a structure constant of $QH_T^*(G/P)$, i.e. of the parabolic quantum double Schubert polynomials $\widetilde{\mathfrak{S}}_w^P(x; y)$ of Definition 2.4, which represent the Schubert classes of $QH_T^*(G/P)$ [21], and the right-hand side is a structure constant of $QH_T^*(G/B)$.

Corollary 9.2. *With the notation of Proposition 9.1, for all $u, v, w \in W^P$,*

$$c_{u,v}^{w,\lambda_P}(y; z) = c_{u,v}^{w\omega_P \omega_{P'}, \lambda_B}(y; z).$$

Proof. The parabolic polynomials satisfy the same quantum Cauchy identity as in Lemma 2.1. Indeed, by [21, Lemma 18], $\widetilde{\mathfrak{S}}_w^P(x; y) = \theta^P(\mathfrak{S}_w(x; y))$ for $w \in W^P$, where θ^P is the parabolic quantization map of Ciocan-Fontanine [11], extended linearly over the coefficient variables; applying θ^P to the classical identity

$\mathfrak{S}_v(x;z) = \sum_{v_1\cdot v_2=v} \mathfrak{S}_{v_1}(y;z)\,\mathfrak{S}_{v_2}(x;y)$ gives $\widetilde{\mathfrak{S}}^P_v(x;z) = \sum_{v_1\cdot v_2=v} \mathfrak{S}_{v_1}(y;z)\,\widetilde{\mathfrak{S}}^P_{v_2}(x;y)$, where we note that $v_2 \in W^P$ for every reduced factorization (see below), so that $\theta^P(\mathfrak{S}_{v_2}(x;y))$ is indeed $\widetilde{\mathfrak{S}}^P_{v_2}(x;y)$. Hence

$$c^{w,\lambda_P}_{u,v}(y;z) = \sum_{v_1\cdot v_2=v} c^{w,\lambda_P}_{u,v_2}(y;y)\ \mathfrak{S}_{v_1}(y;z).$$

Here, if $v = v_1\cdot v_2$ is reduced and $v \in W^P$, then $v_2 \in W^P$: for $s_i \in P$, $v_2 s_i < v_2$ would give $\ell(vs_i) \le \ell(v_1)+\ell(v_2 s_i) < \ell(v)$, contradicting $v \in W^P$. Thus Proposition 9.1 applies to each term, and resumming by Lemma 2.1 gives

$$\sum_{v_1\cdot v_2=v} c^{w\omega_P\omega_{P'},\lambda_B}_{u,v_2}(y;y)\ \mathfrak{S}_{v_1}(y;z) = c^{w\omega_P\omega_{P'},\lambda_B}_{u,v}(y;z). \quad \square$$

Proof of Proposition 2.9. Let $d_P \in Q^\vee/Q^\vee_P$ with Peterson–Woodward lift $d = \lambda_B$. In type A, $\langle \alpha_i, d\rangle = 2d_i - d_{i-1} - d_{i+1} = \langle d\rangle_i$, so the parabolic subgroup P' of Proposition 9.1 is generated by $\{s_i \in P \mid \langle d\rangle_i = 0\}$, that is, by $\{s_i \mid i \in z_P(d)\}$, and $\omega_P\omega_{P'} = w_P w_{z_P(d)}$. The claim is therefore Corollary 9.2. Its application to stabilized polynomial coefficients is explained immediately after Proposition 2.9. $\square$

Appendix A. Proof of the positivity theorem

by Tongren Xiao and Yihua Yue[1]

This appendix proves Theorem 2.3. The argument combines the geometric argument of Mihalcea [25] with the Graham-positivity theorem of Gao and Xiong [15] for B^--stable effective cycles.

Proof of Theorem 2.3. Our product is written with the output basis carrying the parameters of the second factor, so commutativity of multiplication identifies the coefficient $C^{w,d}_{v,u}(z,y)$ below with $c^{w,d}_{u,v}(y;z)$. It is therefore enough to prove that $C^{w,d}_{u,v}(y,t)$ is in $\mathbb{Z}_{\ge 0}[t_i - y_j]$.

Fix $u, v, w \in S_n$, increasing n if necessary so that all permutations and parameters occurring in the coefficient under consideration are included. Write $\mathbf{a} = (t_1,\dots,t_n,y_1,\dots,y_n)$, and let $\tau \in S_{2n}$ interchange the two blocks, so that $\tau\mathbf{a} = (y_1,\dots,y_n,t_1,\dots,t_n)$. Regard the product expansion in S_{2n}:

$$\widetilde{\mathfrak{S}}_u(x;\tau\mathbf{a})\,\widetilde{\mathfrak{S}}_v(x;\mathbf{a}) = \sum_{w\in S_{2n}} C^{w,\tau}_{u,v}(\mathbf{a};q)\,\widetilde{\mathfrak{S}}_w(x;\mathbf{a}). \tag{A.1}$$

We recall the geometric interpretation of these coefficients. Let $e_i(x)$ and $E_i(x,q)$ denote, respectively, the elementary symmetric polynomials and their quantum analogues in $2n$ variables, and set

$$J^{2n}_{\mathbf{a}} = \langle e_i(x) - e_i(\mathbf{a}) : 1 \le i \le 2n\rangle, \qquad J^{2n}_{q\mathbf{a}} = \langle E_i(x,q) - e_i(\mathbf{a}) : 1 \le i \le 2n\rangle.$$

The Fomin–Gelfand–Postnikov deformation map [13], extended $\mathbb{Z}[\mathbf{a}]$-linearly, induces

$$\overline{\mathcal{Q}} : \mathbb{Z}[x,\mathbf{a}]/J^{2n}_{\mathbf{a}} \longrightarrow \mathbb{Z}[x,\mathbf{a},q]/J^{2n}_{q\mathbf{a}}.$$

The classical Borel presentation identifies the source with $H^*_T(X)$, and Kim's presentation [19] identifies the target with $QH^*_T(X)$, where $X = \mathrm{Fl}_{2n}$. Under the Lam–Shimozono identification [21] of quantum double Schubert polynomials with quantum Schubert classes, (A.1) is the quantum equivariant Littlewood–Richardson expansion for the classes obtained by applying $\overline{\mathcal{Q}}$ to the first factor. Consequently,

$$C^{w,\tau}_{u,v}(\mathbf{a};q) = \sum_{d\ge 0} q^d (\pi_{M_d})^T_* \left(\mathrm{ev}_1^*[\tau X_u]^T\ \mathrm{ev}_2^*[X_v]^T\ \mathrm{ev}_3^*[Y_w]^T\right). \tag{A.2}$$

Here $M_d = \overline{M}_{0,3}(X,d)$, $\mathrm{ev}_i : M_d \to X$ are the evaluation maps, $\pi_{M_d} : M_d \to \mathrm{pt}$ is the structure map, and X_r and Y_r denote the Schubert and opposite Schubert varieties indexed by r.

For each effective degree d, put

$$F = (\mathrm{ev}_1, \mathrm{ev}_2) : M_d \longrightarrow X\times X, \qquad I^d_{u,v} = F^{-1}(\tau X_u \times X_v), \qquad \Gamma^d_{u,v} = (\mathrm{ev}_3)_*[I^d_{u,v}].$$

The incidence scheme $I^d_{u,v}$ is reduced and locally irreducible, and all its irreducible components have codimension $\ell(u)+\ell(v)$. Indeed, the moduli space M_d is irreducible by Thomsen's theorem [36], and Mihalcea's

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incidence lemma [25, Lemma 6.3] applies after diagonal translation by $a_0 = 1 \times w_0 \in S_{2n}$, where w_0 is the longest element in the second block: it identifies $\tau X_u \times X_v$ with $w_0^{(2n)} X_u \times X_v$. The codimension is therefore

$$\operatorname{codim}_X(w_0^{(2n)} X_u) + \ell(v) = \ell(u) + \ell(v).$$

By the incidence formula of Mihalcea [25, Section 3.1], this gives

$$\operatorname{ev}_1^*[\tau X_u]^T \operatorname{ev}_2^*[X_v]^T \cap [M_d]^T = [I^d_{u,v}]_T. \tag{A.3}$$

The cycle $\Gamma^d_{u,v}$ is effective and $B^-(\tau)$-invariant. To see invariance, write $N^-(\tau) = N^- \cap \tau N^- \tau^{-1}$ and $B^-(\tau) = TN^-(\tau)$. The varieties X_v and τX_u are stable under the corresponding actions, so their product, its inverse image under the equivariant map F, and its proper push-forward under ev_3 are $B^-(\tau)$-invariant. Effectivity follows from the reducedness of $I^d_{u,v}$.

There is a unique expansion

$$[\Gamma^d_{u,v}]_T = \sum_{r \in S_{2n}} p^{r,d}_{u,v}(\mathbf{a})[X_r]_T \quad \text{in } H^*_T(X).$$

By Gao and Xiong's Graham–positivity theorem for $B^-(\tau)$-invariant effective cycles [15, Theorem 2.4 and Corollary 2.5],

$$p^{r,d}_{u,v}(\mathbf{a}) \in \mathbb{Z}_{\geq 0}[t_i - y_j : 1 \leq i, j \leq n].$$

Finally, by equivariant Poincaré duality, (A.3), and the projection formula,

$$(\operatorname{ev}_3)^T_* \left(\operatorname{ev}_1^*[\tau X_u]^T \operatorname{ev}_2^*[X_v]^T \right) = [\Gamma^d_{u,v}]_T,$$

and hence the degree-d summand of (A.2) is

$$(\pi_X)^T_* \left([\Gamma^d_{u,v}]_T [Y_w]^T \right) = \sum_{r \in S_{2n}} p^{r,d}_{u,v}(\mathbf{a}) (\pi_X)^T_* \left([X_r]^T [Y_w]^T \right) = p^{w,d}_{u,v}(\mathbf{a}).$$

Thus every coefficient $C^{w,d}_{u,v}(y,t)$ is in $\mathbb{Z}_{\geq 0}[t_i - y_j]$. Replacing (u, v, y, t) by (v, u, z, y) proves the asserted positivity of $c^{w,d}_{u,v}(y; z)$. □

References

[1] Benedetti, C., Bergeron, N., Colmenarejo, L., Saliola, F., and Sottile, F. A quantum Murnaghan–Nakayama rule for the flag manifold. *Algebraic Combinatorics 8*, 3 (2025), 619–653.

[2] Björner, A., and Brenti, F. *Combinatorics of Coxeter Groups*. Springer, 2005.

[3] Björner, A., and Wachs, M. Shellable nonpure complexes and posets. II. *Transactions of the American Mathematical Society 349*, 10 (1997), 3945–3975.

[4] Buch, A. S. A direct proof of the quantum version of Monk's formula. *Proceedings of the American Mathematical Society 131*, 7 (2003), 2037–2042.

[5] Buch, A. S. Quantum cohomology of partial flag manifolds. *Transactions of the American Mathematical Society 357* (2005), 443–458.

[6] Buch, A. S. Mutations of puzzles and equivariant cohomology of two-step flag varieties. *Ann. of Math.* (2015), 173–220.

[7] Buch, A. S., Kresch, A., Purbhoo, K., and Tamvakis, H. The puzzle conjecture for the cohomology of two-step flag manifolds. *Journal of algebraic combinatorics 44* (2016), 973–1007.

[8] Buch, A. S., Kresch, A., and Tamvakis, H. Gromov–Witten invariants on Grassmannians. *Journal of the American Mathematical Society 16* (2003), 901–915.

[9] Buch, A. S., and Mihalcea, L. C. Quantum K-theory of Grassmannians. *Duke Mathematical Journal 156*, 3 (2011), 501–538.

[10] Buch, A. S., and Mihalcea, L. C. Curve neighborhoods of Schubert varieties. *Journal of Differential Geometry 99* (2015), 255–283.

[11] Ciocan-Fontanine, I. On quantum cohomology rings of partial flag varieties. *Duke Mathematical Journal 98*, 3 (1999), 485–524.

[12] Colmenarejo, L., and Mayers, N. The quantum k-Bruhat order. *arXiv preprint arXiv:2601.03437* (2026).

[13] Fomin, S., Gelfand, S., and Postnikov, A. Quantum Schubert polynomials. *Journal of the American Mathematical Society* (1997), 565–596.

[14] Fulton, W., and Woodward, C. On the quantum product of Schubert classes. *Journal of Algebraic Geometry 13*, 4 (2004), 641–661.

[15] Gao, Y., and Xiong, R. Positivity in triple Schubert calculus. *Séminaire Lotharingien de Combinatoire 95B* (2026). Proceedings of FPSAC 2026, Art. 37, 11 pp.

[16] Gu, W., and Kalashnikov, E. A rim-hook rule for quiver flag varieties. *arXiv preprint arXiv:2009.02810* (2020).

[17] Huang, Y., and Li, C. On equivariant quantum Schubert calculus for G/P. *Journal of Algebra 441* (2015), 21–56.

[18] Humphreys, J. E. *Reflection groups and Coxeter groups*. Cambridge university press, 1992.

[19] Kim, B. On equivariant quantum cohomology. *International Mathematics Research Notices*, 17 (1996), 841–851.
[20] Kirillov, A. N., and Maeno, T. Quantum double Schubert polynomials, quantum Schubert polynomials and Vafa–Intriligator formula. *Discrete Mathematics 217*, 1-3 (2000), 191–223.
[21] Lam, T., and Shimozono, M. Quantum double Schubert polynomials represent Schubert classes. *Proceedings of the American Mathematical Society 142*, 3 (2014), 835–850.
[22] Leung, N. C., and Li, C. Classical aspects of quantum cohomology of generalized flag varieties. *International Mathematics Research Notices 2012*, 16 (2012), 3706–3722.
[23] Macdonald, I. G. *Notes on Schubert Polynomials*. Université Du Québec à Montréal, Dép. de mathématiques et d'informatique, 1991.
[24] Mihalcea, L. C. Equivariant quantum Schubert calculus. *Advances in Mathematics 203*, 1 (2006), 1–33.
[25] Mihalcea, L. C. Positivity in equivariant quantum Schubert calculus. *American Journal of Mathematics 128*, 3 (2006), 787–803.
[26] Mihalcea, L. C. On equivariant quantum cohomology of homogeneous spaces: Chevalley formulae and algorithms. *Duke Mathematical Journal 140*, 2 (2007), 321–350.
[27] Molev, A., and Sagan, B. A Littlewood-Richardson rule for factorial Schur functions. *Trans. Amer. Math. Soc. 351*, 11 (1999), 4429–4443.
[28] Postnikov, A. On a quantum version of Pieri's formula. In *Advances in geometry*. Springer, 1999, pp. 371–383.
[29] Reading, N. Lattice congruences of the weak order. *Order 21*, 4 (2004), 315–344.
[30] Reading, N. Cambrian lattices. *Advances in Mathematics 205*, 2 (2006), 313–353.
[31] Robinson, S. A Pieri-Type Formula for $H^*T(SL_n(C)/B)$. *J. Algebra 249*, 1 (2002), 38–58.
[32] Samuel, M. J. *The Leibniz formula for divided difference operators associated to Kac-Moody root systems*. PhD thesis, Rutgers, The State University of New Jersey, 2014.
[33] Samuel, M. J. A Molev-Sagan type formula for double Schubert polynomials. *J. Pure Appl. Algebra 228*, 7 (2024), 107636.
[34] Samuel, M. J. `schubmult`: a Python package for multiplying Schubert polynomials. Available from the Python Package Index, `https://pypi.org/project/schubmult/`, with sources at `https://github.com/matthematics/schubmult`, 2026.
[35] Sottile, F. Pieri's rule for flag manifolds and Schubert polynomials. *Ann. Inst. Fourier 46*, 1 (1996), 89–110.
[36] Thomsen, J. F. Irreducibility of $m_{0,n}(g/p, \beta)$. *International Journal of Mathematics 9*, 3 (1998), 367–376.
[37] Woodward, C. T. On D. Peterson's comparison formula for Gromov-Witten invariants of G/P. *Proceedings of the American Mathematical Society* (2005), 1601–1609.